

voloshin-hypergraph — formula report

2555 inline formulas (0 shown: the ones that did not render), 266 display equations, 2 tables, 176 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0124	159	■ 0.310 ● N +0	$a_{ij}=\begin{cases}1&\text{ if }\mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \emptyset \\0&\text{ otherwise. }\end{cases}$	$a_{ij}=\begin{cases}1&\text{ if }\mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \emptyset \\0&\text{ otherwise. }\end{cases}$	$a_{ij}=\begin{cases}1&\text{ if }\mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \emptyset, \\0&\text{ otherwise. }\end{cases}$
voloshin-hypergraph_ EQ0133	170	■ 0.340 ● N +0	$\left(\mathcal{H}\right)_2=(X,\mathcal{E})\text{ where }\left\{x_i,x_j\right\}\in \mathcal{E}\Leftrightarrow \mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \mathbb{0}.$	$\left(\mathcal{H}\right)_2=(X,\mathcal{E})\text{ where }\left\{x_i,x_j\right\}\in \mathcal{E}\Leftrightarrow \mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \mathbb{0}.$	$\left(\mathcal{H}\right)_2=(X,\mathcal{E})\text{ where }\left\{x_i,x_j\right\}\in \mathcal{E}\Leftrightarrow \mathcal{D}\left(x_i\right)\cap \mathcal{D}\left(x_j\right)\neq \emptyset.$
voloshin-hypergraph_ EQ0141 (8.1)	185	■ 0.473 ● N -1	$\tau(\mathcal{H})=\nu(\mathcal{H})\text{ .}$	$\tau(\mathcal{H})=\nu(\mathcal{H}).$	$\mathfrak{T}(\mathcal{H})=\mathfrak{V}(\mathcal{H}).$
voloshin-hypergraph_ EQ0197	220	■ 0.500 ● N -1	$\chi(\mathcal{H})=\overline{\chi}(\mathcal{H})=0,\quad R(\mathcal{H})=(0,0,\ldots,0),\quad P(\mathcal{H},\lambda)=0.$	$\chi(\mathcal{H})=\overline{\chi}(\mathcal{H})=0,\quad R(\mathcal{H})=(0,0,\ldots,0),\quad P(\mathcal{H},\lambda)=0.$	$\chi(\mathcal{H})=\overline{\chi}(\mathcal{H})=0,\quad R(\mathcal{H})=(0,0,\ldots,0),\quad P(\mathcal{H},\lambda)=0.$
voloshin-hypergraph_ EQ0244	258	■ 0.530 ● C -3	$\overline{\chi}(\mathcal{H})=\alpha_C(\mathcal{H})= X -\tau(\mathcal{H}_C)= X -\nu(\mathcal{H}_C).$	$\overline{\chi}(\mathcal{H})=\alpha_C(\mathcal{H})= X -\tau(\mathcal{H}_C)= X -\nu(\mathcal{H}_C).$	$\overline{\chi}(\mathcal{H})=\alpha_C(\mathcal{H})= X -\mathfrak{T}(\mathcal{H}_C)= X -\mathfrak{V}(\mathcal{H}_C).$
voloshin-hypergraph_ EQ0207	230	■ 0.531 ● K +0	$P(\mathcal{H},\lambda)=P(\mathcal{H}_1,\lambda),\quad R(\mathcal{H})=R(\mathcal{H}_1),\text{ where }$	$P(\mathcal{H},\lambda)=P(\mathcal{H}_1,\lambda),\quad R(\mathcal{H})=R(\mathcal{H}_1),\text{ where }$	$P(\mathcal{H},\lambda)=P(\mathcal{H}_1,\lambda),\quad R(\mathcal{H})=R(\mathcal{H}_1),\text{ where }$
voloshin-hypergraph_ EQ0233	253	■ 0.543 ● N +0	$\boldsymbol{\alpha}_c(\mathcal{H})-\overline{\chi}(\mathcal{H})>k.$	$\alpha_c(\mathcal{H})-\overline{\chi}(\mathcal{H})>k.$	$\alpha_c(\mathcal{H})-\overline{\chi}(\mathcal{H})>k.$
voloshin-hypergraph_ EQ0130 (7.3)	168	■ 0.556 ● N -1	$\tau(\mathcal{H})=v(\mathcal{H})\text{ .}$	$\tau(\mathcal{H})=v(\mathcal{H}).$	$\mathfrak{T}(\mathcal{H})=\mathfrak{V}(\mathcal{H}).$
voloshin-hypergraph_ EQ0190	216	■ 0.566 ● N +0	$2,3,1,4,5,$	$2,3,1,4,5,$	$2,3,1,4,5,$
voloshin-hypergraph_ EQ0029	085	■ 0.576 ● N +0	$n-m+f=2\text{ .}$	$n-m+f=2.$	$n-m+f=2.$
voloshin-hypergraph_ EQ0051	107	■ 0.608 ● N +1	$P(G\cdot e,\lambda)=\sum_{i=0}^{n-1}(-1)^ib_i\lambda^{n-1-i}$	$P(G\cdot e,\lambda)=\sum_{i=0}^{n-1}(-1)^ib_i\lambda^{n-1-i}$	$P(G\cdot e,\lambda)=\sum_{i=0}^{n-1}(-1)^ib_i\lambda^{n-1-i}.$
voloshin-hypergraph_ EQ0120	156	■ 0.613 ● W +1	$\begin{gathered}\mathcal{H}=(X,\mathcal{D}),\text{ }X=\{1,2,3,4,5,6\},\text{ }\mathcal{D}=\{D_1,D_2,D_3,D_4,D_5\},\\D_1=\{1\},D_2=\{1,2\},D_3=\{1,2,4\},\\D_4=\{2,3,5\},D_5=\{3,4,5\}.\end{gathered}$	$\mathcal{H}=(X,\mathcal{D}),\text{ }X=\{1,2,3,4,5,6\},\text{ }\mathcal{D}=\{D_1,D_2,D_3,D_4,D_5\},\\D_1=\{1\},D_2=\{1,2\},D_3=\{1,2,4\}\\D_4=\{2,3,5\},D_5=\{3,4,5\}.$	$\mathcal{H}=(X,\mathcal{D}),\text{ }X=\{1,2,3,4,5,6\},\text{ }\mathcal{D}=\{D_1,D_2,D_3,D_4,D_5\},\\D_1=\{1\},D_2=\{1,2\},D_3=\{1,2,4\},\\D_4=\{2,3,5\},D_5=\{3,4,5\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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voloshin-hypergraph_ EQ0056	109	■ 0.653 ●	$\begin{gathered} P(G, \lambda) = \frac{P(\text{left}(G_1), \lambda)}{P(\text{left}(G_2), \lambda)} = \\ \frac{P(\text{left}(\frac{P(\text{left}(\lambda(\lambda-1)\text{left}(\lambda(\lambda-1)\text{left}(\lambda^2-3\lambda+3)\text{right})\text{right})\text{right})}{P(\text{left}(\lambda(\lambda-1)\text{left}(\lambda^2-3\lambda+3)\text{right})\text{right})} = \\ \frac{P(\text{left}(\lambda(\lambda-1)\text{left}(\lambda^2-3\lambda+3)\text{right})\text{right})}{P(\text{left}(\lambda(\lambda-1)\text{left}(\lambda^2-3\lambda+3)\text{right})\text{right})} \end{gathered}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(2)}} = \frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)][\lambda(\lambda-1)(\lambda-2)]}{\lambda(\lambda-1)} = \lambda(\lambda-1)(\lambda-2)(\lambda^2-3\lambda+3).$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(2)}} = \frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)][\lambda(\lambda-1)(\lambda-2)]}{\lambda(\lambda-1)} = \lambda(\lambda-1)(\lambda-2)(\lambda^2-3\lambda+3).$
voloshin-hypergraph_ EQ0125	160	■ 0.655 ● W +0	$\left \mathcal{D}(\mathcal{K}_n^r) \right = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$	$ \mathcal{D}(K_n^r) = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$	$ \mathcal{D}(K_n^r) = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$
voloshin-hypergraph_ EQ0189	216	■ 0.679 ● K +0	$5, 4, 1, 3, 2.$	$5, 4, 1, 3, 2.$	$5, 4, 1, 3, 2.$
voloshin-hypergraph_ EQ0198 (10.3)	221	■ 0.699 ● N -2	$1 \leq \chi(\mathcal{H}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$	$1 \leq \chi(\mathcal{H}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$	$1 \leq \chi(\mathcal{H}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$
voloshin-hypergraph_ EQ0160	195	■ 0.709 ● N -1	$\wedge(\mathcal{H}) = \min_T \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$	$\wedge(\mathcal{H}) = \min_T \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$	$\Lambda(\mathcal{H}) = \min_T \Lambda(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$
voloshin-hypergraph_ EQ0040	102	■ 0.712 ● W +1	$\begin{gathered} P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = \lambda^{(3)} + \lambda^{(2)} = \\ \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2 \end{gathered}$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = \lambda^{(3)} + \lambda^{(2)} = \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = \lambda^{(3)} + \lambda^{(2)} = \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2.$
voloshin-hypergraph_ EQ0216	239	■ 0.715 ● K +0	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$
voloshin-hypergraph_ EQ0266	287	■ 0.722 ● K +0	$I(\text{left}(G_1)\text{right}) = \text{left}(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix})$	$I(G_1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$	$I(G_1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$
voloshin-hypergraph_ EQ0009	033	■ 0.738 ● K +0	$A(G) = \text{left}(\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix})$	$A(G) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix};$	$A(G) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix};$
voloshin-hypergraph_ EQ0137 (7.4)	173	■ 0.743 ● K +0	$\left(D_1 \cap D_2 \right) \cup \left(D_1 \cap D_3 \right) \cup \left(D_2 \cap D_3 \right) \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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voloshin-hypergraph_E00119	156	0.755 K +0	$\begin{gathered} \mathcal{Z}=\left\{X_{_1},\, X_{_2},\, \ldots,\, X_{_n}\right\},\, \, \, \backslash\backslash\, X_{_i}=\left\{d_{_j}: \, x_{_i}\, \text{in } D_{_j}\right\} \\ \text{in } \mathcal{H}\right\} . \end{gathered}$	$\mathcal{Z}=\left\{X_{_1},X_{_2},\ldots,X_n\right\},$ $X_i=\left\{d_j: x_i\in D_j \text{ in } \mathcal{H}\right\}.$	$\mathcal{Z}=\left\{X_{_1},X_{_2},\ldots,X_n\right\},$ $X_i=\left\{d_j: x_i\in D_j \text{ in } \mathcal{H}\right\}.$
voloshin-hypergraph_E00052	107	0.759 K +0	$\begin{gathered} P(G,\lambda)=P(G-e,\lambda)-P(G\cdot e,\lambda) \\ =\lambda^n-(m-1)\lambda^{n-1}+a_2\lambda^{n-2}-\cdots+(-1)^ia_i\lambda^{n-i}\cdots \\ -(\lambda^{n-1}-b_1\lambda^{n-2}+b_2\lambda^{n-2}-\cdots+(-1)^ib_{i-1}\lambda^{n-i}\cdots) \\ =\lambda^n-m(G)\lambda^{n-1}+(a_2+b_1)\lambda^{n-2}\cdots(-1)^i(a_i+b_{i-1})\lambda^{n-i}\cdots \end{gathered}$	$P(G,\lambda)=P(G-e,\lambda)-P(G\cdot e,\lambda)$ $=\lambda^n-(m-1)\lambda^{n-1}+a_2\lambda^{n-2}-\cdots+(-1)^ia_i\lambda^{n-i}\cdots$ $-(\lambda^{n-1}-b_1\lambda^{n-2}+b_2\lambda^{n-2}-\cdots+(-1)^ib_{i-1}\lambda^{n-i}\cdots)$ $=\lambda^n-m(G)\lambda^{n-1}+(a_2+b_1)\lambda^{n-2}\cdots(-1)^i(a_i+b_{i-1})\lambda^{n-i}\cdots$	$P(G,\lambda)=P(G-e,\lambda)-P(G\cdot e,\lambda)$ $=\lambda^n-(m-1)\lambda^{n-1}+a_2\lambda^{n-2}-\cdots+(-1)^ia_i\lambda^{n-i}\cdots$ $-(\lambda^{n-1}-b_1\lambda^{n-2}+b_2\lambda^{n-2}-\cdots+(-1)^ib_{i-1}\lambda^{n-i}\cdots)$ $=\lambda^n-m(G)\lambda^{n-1}+(a_2+b_1)\lambda^{n-2}\cdots(-1)^i(a_i+b_{i-1})\lambda^{n-i}\cdots.$
voloshin-hypergraph_E00064	111	0.763 W +0	$\begin{gathered} P\left(G_{_1},\lambda\right)=\left(\lambda-k_{_1}\right)P\left(G_{_2},\lambda\right)=\left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)P\left(G_{_3},\lambda\right)\cdots= \\ \left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)\left(\lambda-k_{_3}\right)\cdots\left(\lambda-k_{_n}\right). \end{gathered}$	$P\left(G_{_1},\lambda\right)=\left(\lambda-k_{_1}\right)P\left(G_{_2},\lambda\right)=\left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)P\left(G_{_3},\lambda\right)\cdots=$ $\left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)\left(\lambda-k_{_3}\right)\cdots\left(\lambda-k_{_n}\right)$	$P\left(G_{_1},\lambda\right)=\left(\lambda-k_{_1}\right)P\left(G_{_2},\lambda\right)=\left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)P\left(G_{_3},\lambda\right)\cdots=$ $\left(\lambda-k_{_1}\right)\left(\lambda-k_{_2}\right)\left(\lambda-k_{_3}\right)\cdots\left(\lambda-k_{_n}\right).$
voloshin-hypergraph_E00206	230	0.771 W -1	$\begin{gathered} \mathcal{H}_{_1}=\left(X,\mathcal{C}\right),\, \mathcal{D}_{_1}=\mathcal{D}\cup\left\{x_k,x_l\right\}, \\ \mathcal{H}_{_2}=\left(X,\mathcal{C}_1,\mathcal{D}\right),\, \mathcal{C}_1=\mathcal{C}\cup\left\{x_k,x_l\right\} \text{ (splitting)}. \end{gathered}$	$\mathcal{H}_{_1}=\left(X,\mathcal{C},\mathcal{D}_{_1}\right),\, \mathcal{D}_{_1}=\mathcal{D}\cup\left\{x_k,x_l\right\},$ $\mathcal{H}_{_2}=\left(X,\mathcal{C}_1,\mathcal{D}\right),\, \mathcal{C}_1=\mathcal{C}\cup\left\{x_k,x_l\right\} \text{ (splitting)}.$	$\mathcal{H}_{_1}=\left(X,\mathcal{C},\mathcal{D}_{_1}\right),\, \mathcal{D}_{_1}=\mathcal{D}\cup\left\{x_k,x_l\right\},$ $\mathcal{H}_{_2}=\left(X,\mathcal{C}_1,\mathcal{D}\right),\, \mathcal{C}_1=\mathcal{C}\cup\left\{x_k,x_l\right\} \text{ (splitting)}.$
voloshin-hypergraph_E00007	030	0.775 K +0	$\begin{gathered} J(G)=\left\{\left\{x_{_1},x_{_2}\right\},\left\{x_{_2},x_{_3}\right\},\left\{x_{_3},x_{_4}\right\},\right. \\ \left.\left\{x_{_4},x_{_5}\right\},\left\{x_{_1},x_{_5}\right\},\left\{x_{_2},x_{_5}\right\},\left\{x_{_2},x_{_4}\right\}\right\}. \end{gathered}$	$J(G)=\left\{\left\{x_{_1},x_{_2}\right\},\left\{x_{_2},x_{_3}\right\},\left\{x_{_3},x_{_4}\right\},\right. \\ \left.\left\{x_{_4},x_{_5}\right\},\left\{x_{_1},x_{_5}\right\},\left\{x_{_2},x_{_5}\right\},\left\{x_{_2},x_{_4}\right\}\right\}.$	$J(G)=\left\{\left\{x_{_1},x_{_2}\right\},\left\{x_{_2},x_{_3}\right\},\left\{x_{_3},x_{_4}\right\},\right. \\ \left.\left\{x_{_4},x_{_5}\right\},\left\{x_{_1},x_{_5}\right\},\left\{x_{_2},x_{_5}\right\},\left\{x_{_2},x_{_4}\right\}\right\}.$
voloshin-hypergraph_E00143 (8.3)	190	0.779 N +0	$\wedge(\mathcal{H},T)=\sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T).$	$\wedge(\mathcal{H},T)=\sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T).$	$\wedge(\mathcal{H},T)=\sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T).$
voloshin-hypergraph_E00245	258	0.790 C -3	$\nu\left(\mathcal{H}_{_C}\right)=\tau\left(\mathcal{H}_{_C}\right).$	$\nu\left(\mathcal{H}_{_C}\right)=\tau\left(\mathcal{H}_{_C}\right).$	$\mathfrak{v}\left(\mathcal{H}_{_C}\right)=\mathfrak{t}\left(\mathcal{H}_{_C}\right).$
voloshin-hypergraph_E00161	195	0.792 K +0	$\wedge(\mathcal{H})=\wedge\left(\mathcal{H}_{_1}\right).$	$\wedge(\mathcal{H})=\wedge\left(\mathcal{H}_{_1}\right).$	$\wedge(\mathcal{H})=\wedge\left(\mathcal{H}_{_1}\right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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voloshin-hypergraph_ EQ0118	155	■ 0.808 ● W +0	$(i, j) \text{ -entry} = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$	$(i, j)\text{-entry} = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$	$(i, j)\text{-entry} = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$
voloshin-hypergraph_ EQ0138	173	■ 0.808 ● K +0	$\left(D_{\{1\}} \cap D_{\{2\}}\right) \cup \left(D_{\{1\}} \cap D_{\{3\}}\right) \cup \left(D_{\{2\}} \cap D_{\{3\}}\right) \subseteq D$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$
voloshin-hypergraph_ EQ0162	195	■ 0.828 ● N +1	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$
voloshin-hypergraph_ EQ0030	085	■ 0.840 ● N +0	$m(G) \leq 3n(G) - 6.$	$m(G) \leq 3n(G) - 6.$	$m(G) \leq 3n(G) - 6.$
voloshin-hypergraph_ EQ0199	221	■ 0.852 ●	$1 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$	$1 \leq \chi(\mathcal{H}_D) = 2 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$	$1 \leq \chi(\mathcal{H}_D) = 2 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$
voloshin-hypergraph_ EQ0087	116	■ 0.863 ● W -1	$\begin{gathered} P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda - 1)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda) \end{gathered}$	$P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda - 1)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda)$	$P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda - 1)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda).$
voloshin-hypergraph_ EQ0153	193	■ 0.873 ● N -1	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$
voloshin-hypergraph_ EQ0214	233	■ 0.876 ● C -5	$\begin{gathered} \chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda). \end{gathered}$	$\chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda).$	$\chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda).$
voloshin-hypergraph_ EQ0099	129	■ 0.890 ● N +0	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$
voloshin-hypergraph_ EQ0235 (10.5)	253	■ 0.901 ● N +0	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$
voloshin-hypergraph_ EQ0201	224	■ 0.913 ● N +1	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$
voloshin-hypergraph_ EQ0232 (10.4)	252	■ 0.932 ● K +0	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$
voloshin-hypergraph_ EQ0259	273	■ 0.935 ● K +0	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0226	248	■ 0.938 ● K +0	$ C \geq n - \chi$.	$ C \geq n - \chi$.	$ C \geq n - \chi$.
voloshin-hypergraph_ EQ0231	250	■ 0.941 ● K +0	$\boldsymbol{\sigma} = (x_5, x_3, x_4, x_2, x_1)$.	$\boldsymbol{\sigma} = (x_5, x_3, x_4, x_2, x_1)$.	$\boldsymbol{\sigma} = (x_5, x_3, x_4, x_2, x_1)$.
voloshin-hypergraph_ EQ0149	192	■ 0.945 ●	$\begin{gathered} \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ \text{\texttt{\textbackslash in \texttt{\textbackslash mathcal\{D\}}(D -1)-w(T)+l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}} =} \\ \text{\texttt{\textbackslash \{ \texttt{\textbackslash text \{ ignore the singletons \}} \}} = \sum_{ D \geq 2} (D - 1) - w(T) + l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}} = \text{\texttt{\textbackslash \{ \texttt{\textbackslash text \{ insert the} \\ \texttt{\textbackslash weight of the forest \}} \}} \texttt{\textbackslash \}} = \sum_{ D \geq 2} (D - 1) - \\ \text{\texttt{\textbackslash sum_{ D \geq 2} \texttt{\textbackslash left E_{D}\texttt{\textbackslash right }+l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}}}} \\ = \text{\texttt{\textbackslash \{ \texttt{\textbackslash text \{ unite the sum \}} \}} = \sum_{ D \geq 2} (D - \\ \text{\texttt{\textbackslash left E_{D}\texttt{\textbackslash right -1\texttt{\textbackslash right }+l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}}}} = \text{\texttt{\textbackslash \{ \\ \texttt{\textbackslash text \{ use (8.5) \}} \}} = \sum_{ D \geq 2} (c(T, D) - 1) + l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}} = \text{\texttt{\textbackslash \{ \texttt{\textbackslash text \{ bring in singletons back \}} \\ \texttt{\textbackslash \}} = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\texttt{\textbackslash mathcal\{H\}}, T) \\ \texttt{\textbackslash . \texttt{\textbackslash end\{gathered\}}} \end{gathered}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \{ \text{ignore the singletons} \} = \sum_{ D \geq 2} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \{ \text{insert the weight of the forest} \} \\ &= \sum_{ D \geq 2} (D - 1) - \sum_{ D \geq 2} E_D + l(\mathcal{H}, T) \\ &= \{ \text{unite the sum} \} = \sum_{ D \geq 2} (D - E_D - 1) + l(\mathcal{H}, T) \\ &= \{ \text{use (8.5)} \} = \sum_{ D \geq 2} (c(T, D) - 1) + l(\mathcal{H}, T) \\ &= \{ \text{bring in singletons back} \} = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\mathcal{H}, T). \end{aligned}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \{ \text{ignore the singletons} \} = \sum_{ D \geq 2} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \{ \text{insert the weight of the forest} \} \\ &= \sum_{ D \geq 2} (D - 1) - \sum_{ D \geq 2} E_D + l(\mathcal{H}, T) \\ &= \{ \text{unite the sum} \} = \sum_{ D \geq 2} (D - E_D - 1) + l(\mathcal{H}, T) \\ &= \{ \text{use (8.5)} \} = \sum_{ D \geq 2} (c(T, D) - 1) + l(\mathcal{H}, T) \\ &= \{ \text{bring in singletons back} \} = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\mathcal{H}, T). \end{aligned}$
voloshin-hypergraph_ EQ0144	191	■ 0.946 ● N +0	$\begin{gathered} \wedge(\mathcal{H}, T) = \sum_{ D =2} (D - 1) - w(T) + l(\mathcal{H}, T) \texttt{\textbackslash \}} = \\ \texttt{\textbackslash mathcal\{D\}} - (X - 1) = m - n + 1 \texttt{\textbackslash . \texttt{\textbackslash end\{gathered\}}} \end{gathered}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{ D =2} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \mathcal{D} - (X - 1) = m - n + 1. \end{aligned}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{ D =2} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= \mathcal{D} - (X - 1) = m - n + 1. \end{aligned}$
voloshin-hypergraph_ EQ0202	224	■ 0.952 ● N -1	$0(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}_Y)$	$O(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}_Y)$	$O(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}_Y)$
voloshin-hypergraph_ EQ0145	191	■ 0.952 ● S +0	$\begin{gathered} \Lambda(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ \text{\texttt{\textbackslash in \texttt{\textbackslash mathcal\{D\}}(D -1)-w(T)+l(\texttt{\textbackslash mathcal\{H\}}, T) \texttt{\textbackslash \}}} \\ = 2 + 2 + 0 + 0 - 3 + 1 = 2 \texttt{\textbackslash . \texttt{\textbackslash end\{gathered\}}} \end{gathered}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= 2 + 2 + 0 + 0 - 3 + 1 = 2. \end{aligned}$	$\begin{aligned} \Lambda(\mathcal{H}, T) &= \sum_{D \in \mathcal{D}} (D - 1) - w(T) + l(\mathcal{H}, T) \\ &= 2 + 2 + 0 + 0 - 3 + 1 = 2. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0041 (5.4)	103	■0.953 ●W -1	$\begin{gathered} P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ P\left(K_1, \lambda\right)+P\left(K_2, \lambda\right)+P\left(K_3, \lambda\right)+P\left(K_4, \lambda\right)= \\ \lambda\left(\lambda-1\right)\left[1+2\left(\lambda-2\right)+\left(\lambda-2\right)\left(\lambda-3\right)\right]=\lambda\left(\lambda-1\right)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda . \end{gathered}$	$P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ \lambda^{(2)}+2 \lambda^{(3)}+\lambda^{(4)}=\lambda(\lambda-1)+2 \lambda(\lambda-1)(\lambda-2)+\lambda(\lambda-1)(\lambda-2)(\lambda-3)= \\ \lambda(\lambda-1)\left[1+2(\lambda-2)+(\lambda-2)(\lambda-3)\right]=\lambda(\lambda-1)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda .$	$P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ \lambda^{(2)}+2 \lambda^{(3)}+\lambda^{(4)}=\lambda(\lambda-1)+2 \lambda(\lambda-1)(\lambda-2)+\lambda(\lambda-1)(\lambda-2)(\lambda-3)= \\ \lambda(\lambda-1)\left[1+2(\lambda-2)+(\lambda-2)(\lambda-3)\right]=\lambda(\lambda-1)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda .$
voloshin-hypergraph_ EQ0265	286	■0.955 ●	$A\left(G_1\right)=\left(\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$	$A\left(G_1\right)=\left(\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$	$A\left(G_1\right)=\left(\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$
voloshin-hypergraph_ EQ0136	171	■0.957 ●N +0	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$
voloshin-hypergraph_ EQ0152 (8.6)	193	■0.970 ●N -1	$\wedge\left(\mathcal{H}, T\right)=\wedge\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$	$\wedge\left(\mathcal{H}, T\right)=\wedge\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$	$\Lambda\left(\mathcal{H}, T\right)=\Lambda\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$
voloshin-hypergraph_ EQ0218	240	■0.971 ●K +0	$(l-1)(m-1) \leq n .$	$(l-1)(m-1) \leq n .$	$(l-1)(m-1) \leq n .$
voloshin-hypergraph_ EQ0243	257	■0.973 ●N +0	$\bigcap_{C \in \mathcal{C}_1} C \neq \emptyset .$	$\bigcap_{C \in \mathcal{C}_1} C \neq \emptyset .$	$\bigcap_{C \in \mathcal{C}_1} C \neq \emptyset .$
voloshin-hypergraph_ EQ0242	256	■0.975 ●N -1	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_{\mathcal{C}}\left(\mathcal{H}^r\right) .$	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_{\mathcal{C}}\left(\mathcal{H}^r\right) .$	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_{\mathcal{C}}\left(\mathcal{H}^r\right) .$
voloshin-hypergraph_ EQ0072	114	■0.983 ●K +0	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'}$	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'}$	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'},$
voloshin-hypergraph_ EQ0015	048	■0.988 ●N -1	$\tau(G) \geq v(G) .$	$\tau(G) \geq v(G) .$	$\tau(G) \geq v(G) .$
voloshin-hypergraph_ EQ0096	118	■0.988 ●N +1	$\begin{gathered} P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda) \end{gathered}$	$P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda)$	$P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda),$
voloshin-hypergraph_ EQ0228	248	■0.988 ●K +0	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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voloshin-hypergraph_ EQ0107	136	■ 0.989 ● N -1	$\bar{\chi}^{\prime}(G)=c+m-n+p=1+5-4+1=3,$	$\bar{\chi}'(G) = c + m - n + p = 1 + 5 - 4 + 1 = 3,$	$\overline{\chi}(G) = c + m - n + p = 1 + 5 - 4 + 1 = 3,$
voloshin-hypergraph_ EQ0186	215	■ 0.990 ● C -3	$M(\mathrm{\mathcal{H}})=m\left(y,\mathrm{\mathcal{H}}_{Y}\right)\leq m\left(x_{k},\mathrm{\mathcal{H}}_{k}\right)\leq t.$	$M(\mathcal{H}) = m(y, \mathcal{H}_Y) \leq m(x_k, \mathcal{H}_k) \leq t.$	$M(\mathcal{H}) = m(y, \mathcal{H}_Y) \leq m(x_k, \mathcal{H}_k) \leq t.$
voloshin-hypergraph_ EQ0254	266	■ 0.990 ● N +1	$R\left(\mathrm{\mathcal{H}}^{\wedge\prime}\right)=\left(0,r_{2}\left(\mathrm{\mathcal{H}}^{\prime}\right),0,\ldots,0,r_{\chi},\ldots,r_{\overline{\chi}},0,\ldots,0\right).$	$R(\mathcal{H}') = (0, r_2(\mathcal{H}'), 0 \dots, 0, r_{\chi}, \dots, r_{\overline{\chi}}, 0, \dots, 0).$	$R(\mathcal{H}') = (0, r_2(\mathcal{H}'), 0 \dots, 0, r_{\chi}, \dots, r_{\overline{\chi}}, 0, \dots, 0).$
voloshin-hypergraph_ EQ0054	108	■ 0.991 ● N +0	$\begin{gathered} P(G,\lambda)=\frac{P\left(G_{X_1\cup S},\lambda\right)}{\lambda^{(k)}}\frac{P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)}=\\ \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}. \end{gathered}$	$P(G,\lambda)=\frac{P\left(G_{X_1\cup S},\lambda\right)}{\lambda^{(k)}}\frac{P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)}=\\ \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}.$	$P(G,\lambda)=\frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)}=\\ \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}.$
voloshin-hypergraph_ EQ0132	170	■ 0.993 ●	$L(\mathrm{\mathcal{H}})=(\mathrm{\mathcal{D}},\mathrm{\mathcal{E}}),\text{ where }\left\{\left(D_i,D_j\right)\in\mathcal{E}\Leftrightarrow D_i\cap D_j\neq\emptyset\right\}$	$L(\mathcal{H}) = (\mathcal{D}, \mathcal{E}), \text{ where } (D_i, D_j) \in \mathcal{E} \Leftrightarrow D_i \cap D_j \neq \emptyset.$	$L(\mathcal{H}) = (\mathcal{D}, \mathcal{E}), \text{ where } (D_i, D_j) \in \mathcal{E} \Leftrightarrow D_i \cap D_j \neq \emptyset.$
voloshin-hypergraph_ EQ0150	192	■ 0.994 ● K +0	$\wedge(\mathrm{\mathcal{H}},T)=(2-1)+(1-1)+(1-1)+1=2.$	$\wedge(\mathcal{H},T) = (2 - 1) + (1 - 1) + (1 - 1) + 1 = 2.$	$\wedge(\mathcal{H},T) = (2 - 1) + (1 - 1) + (1 - 1) + 1 = 2.$
voloshin-hypergraph_ EQ0065 (5.11)	112	■ 0.994 ● K +0	$P(G,\lambda)=\lambda^{S_0}(\lambda-1)^{S_1}(\lambda-2)^{S_2}\cdots(\lambda-\chi(G)+1)^{S_{\chi-1}}$	$P(G,\lambda) = \lambda^{S_0}(\lambda - 1)^{S_1}(\lambda - 2)^{S_2} \cdots (\lambda - \chi(G) + 1)^{S_{\chi-1}}$	$P(G,\lambda) = \lambda^{S_0}(\lambda - 1)^{S_1}(\lambda - 2)^{S_2} \cdots (\lambda - \chi(G) + 1)^{S_{\chi-1}}$
voloshin-hypergraph_ EQ0223	243	■ 0.994 ● K +0	$P(\mathrm{\mathcal{H}},\lambda)=\lambda(\lambda-1)(\lambda-2)\cdots(\lambda-\chi+1)=\lambda^{(\chi)}.$	$P(\mathcal{H},\lambda) = \lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - \chi + 1) = \lambda^{(\chi)}.$	$P(\mathcal{H},\lambda) = \lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - \chi + 1) = \lambda^{(\chi)}.$
voloshin-hypergraph_ EQ0182	212	■ 0.995 ● K +0	$\chi(\mathrm{\mathcal{H}})\leq\wedge(\mathrm{\mathcal{H}})+2.$	$\chi(\mathcal{H}) \leq \wedge(\mathcal{H}) + 2.$	$\chi(\mathcal{H}) \leq \wedge(\mathcal{H}) + 2.$
voloshin-hypergraph_ EQ0257	268	■ 0.996 ● C -4	$\begin{array}{lll} 123 & \& 341 & \& 561 \\ 124 & \& 342 & \& 562 \\ 125 & \& 345 & \& 563 \\ 126 & \& 346 & \& 564 \end{array}.$	$\begin{array}{lll} 123 & 341 & 561 \\ 124 & 342 & 562 \\ 125 & 345 & 563 \\ 126 & 346 & 564. \end{array}.$	$\begin{array}{lll} 123 & 341 & 561 \\ 124 & 342 & 562 \\ 125 & 345 & 563 \\ 126 & 346 & 564. \end{array}.$
voloshin-hypergraph_ EQ0176	211	■ 0.996 ● K +0	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$
voloshin-hypergraph_ EQ0085	116	■ 0.996 ● N -1	$W(G,\lambda)=\frac{\lambda P(G,\lambda-1)}{P(G,\lambda)}.$	$W(G,\lambda) = \frac{\lambda P(G,\lambda - 1)}{P(G,\lambda)}.$	$W(G,\lambda) = \frac{\lambda P(G,\lambda - 1)}{P(G,\lambda)}.$
voloshin-hypergraph_ EQ0156	193	■ 0.996 ● N -1	$\wedge(\mathrm{\mathcal{H}}-x,T-x)+ \mathcal{D}(x) - \mathcal{D}_T(x) -m_2=2+3-1-1=3=\wedge(\mathrm{\mathcal{H}},T).$	$\wedge(\mathcal{H} - x, T - x) + \mathcal{D}(x) - \mathcal{D}_T(x) - m_2 = 2 + 3 - 1 - 1 = 3 = \wedge(\mathcal{H}, T).$	$\wedge(\mathcal{H} - x, T - x) + \mathcal{D}(x) - \mathcal{D}_T(x) - m_2 = 2 + 3 - 1 - 1 = 3 = \wedge(\mathcal{H}, T).$
voloshin-hypergraph_ EQ0258	273	■ 0.996 ● N +0	$r_k(G)=\sum_{i\geq 1}S(i,k-1)f_i(G^*),1\leq k\leq n(G),$	$r_k(G) = \sum_{i \geq 1} S(i, k - 1) f_i(G^*), 1 \leq k \leq n(G),$	$r_k(G) = \sum_{i \geq 1} S(i, k - 1) f_i(G^*), 1 \leq k \leq n(G),$
voloshin-hypergraph_ EQ0208	230	■ 0.996 ● C -3	$\mathcal{C}_1=\mathcal{C}-\mathcal{C}_t\text{ (contraction).}$	$\mathcal{C}_1 = \mathcal{C} - \mathcal{C}_t \text{ (contraction).}$	$\mathcal{C}_1 = \mathcal{C} - \mathcal{C}_t \text{ (contraction).}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0246	259	■ 0.997 ● C -3	$\begin{aligned} & \& V_{\{1\}}=(\{1,2,3,4\},\{\{1,2,3\}, \\ & \{1,3,4\},\{1,2,4\}\}) \text{ } \& \& \\ & V_{\{2\}}=(\{1,2,3,4,5\},\{\{1,2,3\},\{1,4,5\}\}) \text{ } \& \& \\ & (\text{monostar}); \& \& V_{\{3\}}=(\{1,2,3,4,5\},\{\{1,2,3\}, \\ & \{1,3,4\},\{1,4,5\}\}) \text{ } \& \& V_{\{4\}}=(\\ & \{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\},\{1,2,5\} \\ & \}) \text{ } \& \& \left.V_{\{5\}}=(\{1,2,3,4,5 \\ & \},\{\{1,2,3\},\{2,3,4\},\{3,4,5\},\{4,5,1\},\{5,1,2\} \\ & \}) \text{ } \& \& (\text{cycloid } C_5^3)\right) \& \& K_{\{1\}}=(\\ & \{1,2,3,4,5,6\},\{\{1,2,4\},\{2,3,5\},\{3,4,6\}, \quad \\ & \{4,5,1\}, \quad \quad \quad 2\}, \quad \quad \quad \{6,1,3\}, \quad \\ & \& \& 3,5\},\{2,4,6\}\}) \text{ } \& \& (\text{Král's example}). \end{aligned}$	$\begin{aligned} V_1 &= (\{1,2,3,4\},\{\{1,2,3\},\{1,3,4\},\{1,2,4\}\}) \text{ (monostar); } \\ V_2 &= (\{1,2,3,4,5\},\{\{1,2,3\},\{1,4,5\}\}) \text{ (monostar); } \\ V_3 &= (\{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\}\}) \text{ (monostar); } \\ V_4 &= (\{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\},\{1,2,5\}\}) \text{ (monostar); } \\ V_5 &= (\{1,2,3,4,5\},\{\{1,2,3\},\{2,3,4\},\{3,4,5\},\{4,5,1\},\{5,1,2\}\}) \text{ (cycloid } C_5^3); \\ K_1 &= (\{1,2,3,4,5,6\},\{\{1,2,4\},\{2,3,5\},\{3,4,6\}, \quad \{4,5,1\}, \quad \{5,6, \quad 2\}, \quad \{6,1,3\}, \\ & 3,5\},\{2,4,6\}\}) \text{ (Král's example). } \end{aligned}$	$V_1 = (\{1,2,3,4\},\{\{1,2,3\},\{1,3,4\},\{1,2,4\}\}) \text{ (monostar);}$ $V_2 = (\{1,2,3,4,5\},\{\{1,2,3\},\{1,4,5\}\}) \text{ (monostar);}$ $V_3 = (\{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\}\}) \text{ (monostar);}$ $V_4 = (\{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\},\{1,2,5\}\}) \text{ (monostar);}$ $V_5 = (\{1,2,3,4,5\},\{\{1,2,3\},\{2,3,4\},\{3,4,5\},\{4,5,1\},\{5,1,2\}\}) \text{ (cycloid } C_5^3);$ $K_1 = (\{1,2,3,4,5,6\},\{\{1,2,4\},\{2,3,5\},\{3,4,6\}, \quad \{4,5,1\}, \quad \{5,6, \quad 2\}, \quad \{6,1,3\}, \\ 3,5\},\{2,4,6\}\}) \text{ (Král's example).}$
voloshin-hypergraph_ EQ0195 (10.2)	220	■ 0.997 ●	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$
voloshin-hypergraph_ EQ0261	274	■ 0.997 ● N +0	$X=\{1,2,3,4\}, \mathcal{C}=\mathcal{D}=\{\{1,2,3\},\{2,3,4\},\{3,4,1\},\{4,1,2\}\};$	$X = \{1,2,3,4\}, \mathcal{C} = \mathcal{D} = \{\{1,2,3\},\{2,3,4\},\{3,4,1\},\{4,1,2\}\};$	$X = \{1,2,3,4\}, \mathcal{C} = \mathcal{D} = \{\{1,2,3\},\{2,3,4\},\{3,4,1\},\{4,1,2\}\};$
voloshin-hypergraph_ EQ0026	073	■ 0.997 ● W +0	$M(G)=\max_{X'\subseteq X} \min_{x\in G'} d(x).$	$M(G) = \max_{X' \subseteq X} \min_{x \in G'} d(x).$	$M(G) = \max_{X' \subseteq X} \min_{x \in G'} d(x).$
voloshin-hypergraph_ EQ0018	063	■ 0.998 ● K +0	$ \left N_G(S)\right \geq S \text{ .}$	$ N_G(S) \geq S .$	$ N_G(S) \geq S .$
voloshin-hypergraph_ EQ0071	113	■ 0.998 ● W +1	$\begin{aligned} & \& \& \lambda^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]= \\ & \& \& \lambda^{n-1}+(-1)^{n-1}(\lambda-1)\end{aligned}$	$\lambda(\lambda-1)^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]=$ $(\lambda-1)^n+(-1)^n(\lambda-1)$	$\lambda(\lambda-1)^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]=$ $(\lambda-1)^n+(-1)^n(\lambda-1).$
voloshin-hypergraph_ EQ0154	193	■ 0.999 ● K +0	$\wedge(\mathcal{H},T)=2+2+1+0+0-3+1=3 \text{ .}$	$\wedge(\mathcal{H},T) = 2 + 2 + 1 + 0 + 0 - 3 + 1 = 3.$	$\wedge(\mathcal{H},T) = 2 + 2 + 1 + 0 + 0 - 3 + 1 = 3.$
voloshin-hypergraph_ EQ0155	193	■ 0.999 ● K +0	$\wedge(\mathcal{H}-x,T-x)=2+1+0+0-2+1=2 \text{ .}$	$\wedge(\mathcal{H}-x,T-x) = 2 + 1 + 0 + 0 - 2 + 1 = 2.$	$\wedge(\mathcal{H}-x,T-x) = 2 + 1 + 0 + 0 - 2 + 1 = 2.$
voloshin-hypergraph_ EQ0185	215	■ 0.999 ● C -3	$m\left(y,\mathcal{H}_Y\right)=\min_{z} m\left(z,\mathcal{H}_Y\right)=M(\mathcal{H}) \text{ .}$	$m(y,\mathcal{H}_Y) = \min_z m(z,\mathcal{H}_Y) = M(\mathcal{H}).$	$m(y,\mathcal{H}_Y) = \min_z m(z,\mathcal{H}_Y) = M(\mathcal{H}).$
voloshin-hypergraph_ EQ0203	227	■ 0.999 ● N -2	$o\left(y,\mathcal{H}^{\prime}\right)=\min_{z} o\left(z,\mathcal{H}^{\prime}\right)=O(\mathcal{H}) \geq t+1 \text{ .}$	$o(y,\mathcal{H}') = \min_z o(z,\mathcal{H}') = O(\mathcal{H}) \geq t + 1.$	$o(y,\mathcal{H}') = \min_z o(z,\mathcal{H}') = O(\mathcal{H}) \geq t + 1.$
voloshin-hypergraph_ EQ0025	064	■ 0.999 ● N -1	$\tau(G)=\mathcal{V}(G) \text{ .}$	$\tau(G) = \mathcal{V}(G).$	$\mathfrak{T}(G) = \mathfrak{v}(G).$
voloshin-hypergraph_ EQ0013	040	■ 0.999 ● K +0	$\sigma=\left(\begin{array}{llllll} 1&2&3&4&5&6 \\ a&b&c&d&e&f \end{array}\right) \text{ .}$	$\sigma = \left(\begin{array}{llllll} 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & d & e & f \end{array}\right).$	$\sigma = \left(\begin{array}{llllll} 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & d & e & f \end{array}\right).$
voloshin-hypergraph_ EQ0224	244	■ 0.999 ● K +0	$x_{\{j\}} \not= x_{\{j+1\}} \not= x_{\{j+2\}} \not= x_{\{j\}} \rightarrow \left\{x_{\{j\}}, x_{\{j+1\}}, x_{\{j+2\}}\right\} \in \mathcal{C}, j=0,1, \ldots, k-2 \text{ .}$	$x_j \neq x_{j+1} \neq x_{j+2} \neq x_j \Rightarrow \{x_j, x_{j+1}, x_{j+2}\} \in \mathcal{C}, j = 0, 1, \dots, k - 2.$	$x_j \neq x_{j+1} \neq x_{j+2} \neq x_j \Rightarrow \{x_j, x_{j+1}, x_{j+2}\} \in \mathcal{C}, j = 0, 1, \dots, k - 2.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0248	260	■ 0.999 ● K +0	1122,1112,1121,1222,2122,1212,1221.	1122, 1112, 1121, 1222, 2122, 1212, 1221.	1122, 1112, 1121, 1222, 2122, 1212, 1221.
voloshin-hypergraph_ EQ0179	211	■ 0.999 ● N +1	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq\left S_i\right \leq\left\lceil\frac{n}{\lambda}\right\rceil$.	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq\left S_i\right \leq\left\lceil\frac{n}{\lambda}\right\rceil$.	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq\left S_i\right \leq\left\lceil\frac{n}{\lambda}\right\rceil$.
voloshin-hypergraph_ EQ0004	029	■ 1.000 ● S +0	$\begin{aligned} L(G)= & \left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ & \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\} . \end{aligned}$	$L(G)=\left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\} .$	$L(G)=\left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\} .$
voloshin-hypergraph_ EQ0006	030	■ 1.000 ● K +0	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right) .$	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right) .$	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right) .$
voloshin-hypergraph_ EQ0019 (2.1)	063	■ 1.000 ●	$ S <\left N_G(S)\right $.	$ S <\left N_G(S)\right $.	$ S <\left N_G(S)\right $.
voloshin-hypergraph_ EQ0039 (5.3)	102	■ 1.000 ● N +1	$P(G,\lambda)=\sum_{i=\chi}^n r_i(G)\lambda^{\left(i\right)}$	$P(G,\lambda)=\sum_{i=\chi}^n r_i(G)\lambda^{\left(i\right)}$	$P(G,\lambda)=\sum_{i=\chi}^n r_i(G)\lambda^{\left(i\right)}$.
voloshin-hypergraph_ EQ0050	106	■ 1.000 ● N +0	$P(G-e,\lambda)=\sum_{i=0}^n\left(-1\right)^ia_i\lambda^{n-i}$	$P(G-e,\lambda)=\sum_{i=0}^n\left(-1\right)^ia_i\lambda^{n-i}$	$P(G-e,\lambda)=\sum_{i=0}^n\left(-1\right)^ia_i\lambda^{n-i}$
voloshin-hypergraph_ EQ0066 (5.12)	112	■ 1.000 ● W +0	$\begin{gathered} P\left(G_1,\lambda\right)=\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) . \end{gathered}$	$P\left(G_1,\lambda\right)=\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) .$	$P\left(G_1,\lambda\right)=\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) .$
voloshin-hypergraph_ EQ0067	113	■ 1.000 ● K +0	$P\left(G_1,4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.	$P\left(G_1,4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.	$P\left(G_1,4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.
voloshin-hypergraph_ EQ0079	115	■ 1.000 ● K +0	$\left\{x_0\right\}\cup N\left(x_0\right)\subseteq X_1\cup A,\left\{x_0\right\}\cap A\subseteq X_i\cup A,1\leq i\leq k$.	$\left\{x_0\right\}\cup N\left(x_0\right)\subseteq X_1\cup A,\left\{x_0\right\}\cap A\subseteq X_i\cup A,1\leq i\leq k$.	$\left\{x_0\right\}\cup N\left(x_0\right)\subseteq X_1\cup A,\left\{x_0\right\}\cap A\subseteq X_i\cup A,1\leq i\leq k$.
voloshin-hypergraph_ EQ0084	116	■ 1.000 ● K +0	$P\left(C_k,\lambda\right)=\lambda\left(\lambda-1\right)^2\lambda\left(\lambda-1\right)^{k-2}\lambda^{-2}=\left(\lambda-1\right)^k\neq P\left(C_k,\lambda\right),$	$P\left(C_k,\lambda\right)=\lambda\left(\lambda-1\right)^2\lambda\left(\lambda-1\right)^{k-2}\lambda^{-2}=\left(\lambda-1\right)^k\neq P\left(C_k,\lambda\right),$	$P\left(C_k,\lambda\right)=\lambda\left(\lambda-1\right)^2\lambda\left(\lambda-1\right)^{k-2}\lambda^{-2}=\left(\lambda-1\right)^k\neq P\left(C_k,\lambda\right),$
voloshin-hypergraph_ EQ0090	117	■ 1.000 ● K +0	$P\left(C,\lambda\right)=P\left(C-x,\lambda\right)W\left(C_{N\left(x\right)},\lambda\right)=\left(\lambda-1\right)^k\neq P\left(C,\lambda\right),$	$P\left(C,\lambda\right)=P\left(C-x,\lambda\right)W\left(C_{N\left(x\right)},\lambda\right)=\left(\lambda-1\right)^k\neq P\left(C,\lambda\right),$	$P\left(C,\lambda\right)=P\left(C-x,\lambda\right)W\left(C_{N\left(x\right)},\lambda\right)=\left(\lambda-1\right)^k\neq P\left(C,\lambda\right),$
voloshin-hypergraph_ EQ0106	135	■ 1.000 ● N -1	$\overline{\chi}^{\prime}\left(A_1\right)\geq\overline{\chi}^{\prime}\left(A_0\right)+1\geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$	$\overline{\chi}^{\prime}\left(A_1\right)\geq\overline{\chi}^{\prime}\left(A_0\right)+1\geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$	$\overline{\chi}^{\prime}\left(A_1\right)\geq\overline{\chi}^{\prime}\left(A_0\right)+1\geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0146 (8.4)	192	■ 1.000 ● N +0	$\wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\mathcal{H}, T).$	$\wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\mathcal{H}, T).$	$\Lambda(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (c(T, D) - 1) + l(\mathcal{H}, T).$
voloshin-hypergraph_ EQ0180	211	■ 1.000 ● N +0	$\theta \leq a_j \leq b_j \leq D_j .$	$0 \leq a_j \leq b_j \leq D_j .$	$0 \leq a_j \leq b_j \leq D_j .$
voloshin-hypergraph_ EQ0194	220	■ 1.000 ● N +0	$R(\mathcal{H}) = (0, \dots, 0, r_\chi, \dots, r_{\bar{\chi}}, 0, \dots, 0).$	$R(\mathcal{H}) = (0, \dots, 0, r_\chi, \dots, r_{\bar{\chi}}, 0, \dots, 0).$	$R(\mathcal{H}) = (0, \dots, 0, r_\chi, \dots, r_{\bar{\chi}}, 0, \dots, 0).$
voloshin-hypergraph_ EQ0200	223	■ 1.000 ● N -1	$b(x, \mathcal{H}) = \max_y \{ \mathcal{C}(x) \cap \mathcal{C}(y) : y \in X, y \neq x \}$	$b(x, \mathcal{H}) = \max_y \{ \mathcal{C}(x) \cap \mathcal{C}(y) : y \in X, y \neq x \}$	$b(x, \mathcal{H}) = \max_y \{ \mathcal{C}(x) \cap \mathcal{C}(y) : y \in X, y \neq x \}$
voloshin-hypergraph_ EQ0204	227	■ 1.000 ● N -2	$o(x_i, \mathcal{H}_i) \leq O(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}/Y)$	$o(x_i, \mathcal{H}_i) \leq O(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}/Y)$	$o(x_i, \mathcal{H}_i) \leq O(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} o(x, \mathcal{H}/Y)$
voloshin-hypergraph_ EQ0229	248	■ 1.000 ● N +0	$M = \sum_{\substack{x \in T \\ d(x) \geq 2}} \binom{d(x)}{2},$	$M = \sum_{\substack{x \in T \\ d(x) \geq 2}} \binom{d(x)}{2},$	$M = \sum_{\substack{x \in T \\ d(x) \geq 2}} \binom{d(x)}{2},$
voloshin-hypergraph_ EQ0237	255	■ 1.000 ● N +0	$\begin{aligned} & n_{12} + 1 + n_{23} < r, \\ & n_{23} + 1 + n_{31} < r, \end{aligned}$	$\begin{aligned} & n_{12} + 1 + n_{23} < r, \\ & n_{23} + 1 + n_{31} < r, \end{aligned}$	$\begin{aligned} & n_{12} + 1 + n_{23} < r, \\ & n_{23} + 1 + n_{31} < r, \end{aligned}$
voloshin-hypergraph_ EQ0001	022	■ 1.000 ● K +0	$G = (X, E).$	$G = (X, E).$	$G = (X, E).$
voloshin-hypergraph_ EQ0002 (1.1)	023	■ 1.000 ● N +0	$\sum_{i=1}^n d(x_i) = 2m.$	$\sum_{i=1}^n d(x_i) = 2m.$	$\sum_{i=1}^n d(x_i) = 2m.$
voloshin-hypergraph_ EQ0003	023	■ 1.000 ●	$2 + 4 + 2 + 3 + 3 = 14 = 2 \times 7.$	$2 + 4 + 2 + 3 + 3 = 14 = 2 \times 7.$	$2 + 4 + 2 + 3 + 3 = 14 = 2 \times 7.$
voloshin-hypergraph_ EQ0005	029	■ 1.000 ● K +0	$A(G) = \left(\begin{array}{lllll} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{array} \right).$	$A(G) = \left(\begin{array}{lllll} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{array} \right).$	$A(G) = \left(\begin{array}{lllll} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{array} \right).$
voloshin-hypergraph_ EQ0008	032	■ 1.000 ● K +0	$L(G) = \{ \emptyset, \{2\}, \{2, 4\}, \{1, 1, 3\} \};$	$L(G) = \{ \emptyset, \{2\}, \{2, 4\}, \{1, 1, 3\} \};$	$L(G) = \{ \emptyset, \{2\}, \{2, 4\}, \{1, 1, 3\} \};$
voloshin-hypergraph_ EQ0010	033	■ 1.000 ● N +0	$I(G) = \left(\begin{array}{rrrrrr} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & l & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{array} \right);$	$I(G) = \left(\begin{array}{rrrrrr} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & l & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{array} \right);$	$I(G) = \left(\begin{array}{rrrrrr} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & l & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{array} \right);$
voloshin-hypergraph_ EQ0011	033	■ 1.000 ● K +0	$J(G) = \{ (4, 1), (4, 1), (3, 2), (2, 2), (4, 3), (3, 4) \}.$	$J(G) = \{ (4, 1), (4, 1), (3, 2), (2, 2), (4, 3), (3, 4) \}.$	$J(G) = \{ (4, 1), (4, 1), (3, 2), (2, 2), (4, 3), (3, 4) \}.$
voloshin-hypergraph_ EQ0012	036	■ 1.000 ● K +0	$m = \binom{n}{2} = \frac{n(n-1)}{2}.$	$m = \binom{n}{2} = \frac{n(n-1)}{2}.$	$m = \binom{n}{2} = \frac{n(n-1)}{2}.$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

LaTeX source MathPix’s reading, unchanged;
 Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ E00014	047	1.000 ● N -1	$\alpha(G)+\tau(G)= X $.	$\alpha(G)+\tau(G)= X .$	$\alpha(G)+\tau(G)= X .$
voloshin-hypergraph_ E00016	061	1.000 ● K +0	$X=N_{\{0\}}\cup N_{\{1\}}\cup N_{\{2\}}\cup N_{\{3\}}\cup\ldots\cup N_{\{k\}}$.	$X=N_0\cup N_1\cup N_2\cup N_3\cup\ldots\cup N_k.$	$X=N_0\cup N_1\cup N_2\cup N_3\cup\ldots\cup N_k.$
voloshin-hypergraph_ E00017	062	1.000 ● K +0	$N(S)=\cup_{x\in S}N(x)\backslash S$.	$N(S)=\cup_{x\in S}N(x)\backslash S.$	$N(S)=\cup_{x\in S}N(x)\backslash S.$
voloshin-hypergraph_ E00020	063	1.000 ● K +0	$\left S^{\prime}\right \leq\left N_{G}\left(S^{\prime}\right)\right $.	$ S' \leq N_G(S') .$	$ S' \leq N_G(S') .$
voloshin-hypergraph_ E00021 (2.2)	063	1.000 ● K +0	$\left S_{\{0\}}\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right $.	$ S_0 = N_G(S_0) .$	$ S_0 = N_G(S_0) .$
voloshin-hypergraph_ E00022	063	1.000 ● N -1	$A=S_{\{0\}}\cup N_{\{G\}}\left(S_{\{0\}}\right)\text { and } B=\left(X\cup Y\right) \backslash A$	$A=S_0\cup N_G\left(S_0\right) \text { and } B=\left(X\cup Y\right) \backslash A$	$A=S_0\cup N_G\left(S_0\right) \text { and } B=\left(X\cup Y\right) \backslash A$
voloshin-hypergraph_ E00023	063	1.000 ● K +0	$N_{G_B}(S)=N_{\{G\}}(S) \backslash N_{\{G\}}\left(S_{\{0\}}\right),$	$N_{G_B}(S)=N_G(S) \backslash N_G\left(S_0\right),$	$N_{G_B}(S)=N_G(S) \backslash N_G\left(S_0\right),$
voloshin-hypergraph_ E00024	063	1.000 ● K +0	$\left S_{\{0\}}\right + S =\left S_{\{0\}}\cup S\right \leq\left N_{\{G\}}\left(S_{\{0\}}\cup S\right)\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right +\left N_{G_B}(S)\right $.	$ S_0 + S = S_0\cup S \leq N_G(S_0\cup S) = N_G(S_0) + N_{G_B}(S) .$	$ S_0 + S = S_0\cup S \leq N_G(S_0\cup S) = N_G(S_0) + N_{G_B}(S) .$
voloshin-hypergraph_ E00027	074	1.000 ● N -1	$M(G)-(\omega(G)-1)$.	$M(G)-(\omega(G)-1).$	$M(G)-(\omega(G)-1).$
voloshin-hypergraph_ E00028	075	1.000 ● K +0	$X=N_{\{0\}}\cup N_{\{1\}}\cup N_{\{2\}}\cup\cdots\cup N_{\{k-1\}}\cup N_{\{k\}}$	$X=N_0\cup N_1\cup N_2\cup\cdots\cup N_{k-1}\cup N_k$	$X=N_0\cup N_1\cup N_2\cup\cdots\cup N_{k-1}\cup N_k$
voloshin-hypergraph_ E00031	097	1.000 ● N +0	$R(G)=\left(r_1, r_2, \ldots, r_n\right)$	$R(G)=\left(r_1, r_2, \ldots, r_n\right)$	$R(G)=\left(r_1, r_2, \ldots, r_n\right)$
voloshin-hypergraph_ E00032	097	1.000 ● N +0	$R(G)=\left(0,0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n\right)$.	$R(G)=\left(0,0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n\right).$	$R(G)=\left(0,0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n\right).$
voloshin-hypergraph_ E00033	098	1.000 ● K +0	$R(G)=(0,1,1)$.	$R(G)=(0,1,1).$	$R(G)=(0,1,1).$
voloshin-hypergraph_ E00034	099	1.000 ● N +0	$P\left(K_n, \lambda\right)=\lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1)$.	$P\left(K_n, \lambda\right)=\lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1).$	$P\left(K_n, \lambda\right)=\lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1).$
voloshin-hypergraph_ E00035 (5.1)	100	1.000 ●	$P(G, \lambda)=P\left(G_{\{1\}}, \lambda\right)+P\left(G_{\{2\}}, \lambda\right)$.	$P(G, \lambda)=P\left(G_1, \lambda\right)+P\left(G_2, \lambda\right).$	$P(G, \lambda)=P\left(G_1, \lambda\right)+P\left(G_2, \lambda\right).$
voloshin-hypergraph_ E00036	101	1.000 ● N +2	$P(G, \lambda)=P\left(K_{\{1\}}, \lambda\right)+P\left(K_{\{2\}}, \lambda\right)+\cdots+P\left(K_{\{s\}}, \lambda\right)$	$P(G, \lambda)=P\left(K_{i_1}, \lambda\right)+P\left(K_{i_2}, \lambda\right)+\cdots+P\left(K_{i_s}, \lambda\right)$	$P(G, \lambda)=P\left(K_{i_1}, \lambda\right)+P\left(K_{i_2}, \lambda\right)+\cdots+P\left(K_{i_s}, \lambda\right)$
voloshin-hypergraph_ E00037	101	1.000 ● W +0	$P(G, \lambda)=P\left(K_{\{1\}}, \lambda\right)+P\left(K_{\{2\}}, \lambda\right)+\cdots+P\left(K_{\{s\}}, \lambda\right)=\sum_{i=1}^n r_i(G) P\left(K_{\{i\}}, \lambda\right)$	$P(G, \lambda)=P\left(K_{i_1}, \lambda\right)+P\left(K_{i_2}, \lambda\right)+\cdots+P\left(K_{i_s}, \lambda\right)=\sum_{i=1}^n r_i(G) P\left(K_{i_1}, \lambda\right)+\sum_{i=1}^n r_i(G) P\left(K_{i_2}, \lambda\right)+\cdots+\sum_{i=1}^n r_i(G) P\left(K_{i_s}, \lambda\right)=\sum_{i=1}^n r_i(G) P\left(K_n, \lambda\right)$	$P(G, \lambda)=P\left(K_{i_1}, \lambda\right)+P\left(K_{i_2}, \lambda\right)+\cdots+P\left(K_{i_s}, \lambda\right)=\sum_{i=1}^n r_i(G) P\left(K_n, \lambda\right)$
voloshin-hypergraph_ E00038 (5.2)	102	1.000 ● K +0	$\sum_{i=1}^n r_i(G) P\left(K_{\{i\}}, \lambda\right)$.	$\sum_{i=1}^n r_i(G) P\left(K_{i_1}, \lambda\right)$.	$\sum_{i=1}^n r_i(G) P\left(K_{i_1}, \lambda\right).$
voloshin-hypergraph_ E00042	103	1.000 ● K +0	$R\left(C_4\right)=(0,1,2,1)$.	$R\left(C_4\right)=(0,1,2,1).$	$R\left(C_4\right)=(0,1,2,1).$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0043 (5.5)	105	■ 1.000 ● K +0	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda) \cdot P(G, \lambda) \cdot \dots P(G_k, \lambda) \cdot P(G, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda)$
voloshin-hypergraph_ EQ0044	105	■ 1.000 ● K +0	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$
voloshin-hypergraph_ EQ0045 (5.6)	105	■ 1.000 ● N +0	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$
voloshin-hypergraph_ EQ0046	106	■ 1.000 ● K +0	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n))$	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n))$	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n))$
voloshin-hypergraph_ EQ0047	106	■ 1.000 ● N +0	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$
voloshin-hypergraph_ EQ0048	106	■ 1.000 ● K +0	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$
voloshin-hypergraph_ EQ0049 (5.7)	106	■ 1.000 ● K +0	$P(G, \lambda) = P(G, \lambda) - P(G_2, \lambda)$	$P(G, \lambda) = P(G, \lambda) - P(G_2, \lambda)$	$P(G, \lambda) = P(G, \lambda) - P(G_2, \lambda)$
voloshin-hypergraph_ EQ0053 (5.8)	108	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}$
voloshin-hypergraph_ EQ0055 (5.9)	109	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}$
voloshin-hypergraph_ EQ0057 (5.10)	109	■ 1.000 ● K +0	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda)$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda)$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda)$
voloshin-hypergraph_ EQ0058	109	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$
voloshin-hypergraph_ EQ0059	110	■ 1.000 ● N +0	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda)$	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda)$	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda)$
voloshin-hypergraph_ EQ0060	110	■ 1.000 ● K +0	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2) (\lambda^2 - 3\lambda + 3)$	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2) (\lambda^2 - 3\lambda + 3)$	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2) (\lambda^2 - 3\lambda + 3)$
voloshin-hypergraph_ EQ0061	110	■ 1.000 ●	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x)$	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x)$	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x)$
voloshin-hypergraph_ EQ0062	110	■ 1.000 ● K +0	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1)$	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1)$	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1)$
voloshin-hypergraph_ EQ0063	110	■ 1.000 ● N +0	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25$	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25$	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25$
voloshin-hypergraph_ EQ0068 (5.13)	113	■ 1.000 ● N +1	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}$	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}$	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}$
voloshin-hypergraph_ EQ0069 (5.14)	113	■ 1.000 ● N +0	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1)$	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1)$	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1)$
voloshin-hypergraph_ EQ0070	113	■ 1.000 ● K +0	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$
voloshin-hypergraph_ EQ0073	114	■ 1.000 ● K +0	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0074 (5.15)	114	■ 1.000 ● N +0	$P(\text{G}^{\backslash\prime}, \lambda) = \frac{P(\text{G}_1^{\backslash\prime}, \lambda) P(\text{G}_2^{\backslash\prime}, \lambda) \dots P(\text{G}_k^{\backslash\prime}, \lambda)}{P(\text{G}_0^{\backslash\prime}, \lambda)^{k-1}}.$		
voloshin-hypergraph_ EQ0075	114	■ 1.000 ● N +0	$P(\text{G}_1, \lambda) = (\lambda - p) P(\text{G}_1 - x_0, \lambda).$		$P(\text{G}_1, \lambda) = (\lambda - p) P(\text{G}_1 - x_0, \lambda).$
voloshin-hypergraph_ EQ0076	114	■ 1.000 ● N +0	$P(\text{G}, \lambda) = (\lambda - p) P(\text{G} - x_0, \lambda).$		$P(\text{G}, \lambda) = (\lambda - p) P(\text{G} - x_0, \lambda).$
voloshin-hypergraph_ EQ0077 (5.16)	114	■ 1.000 ● N +0	$P(\text{G} - x_0, \lambda) = \frac{P(\text{G}_1 - x_0, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$		$P(\text{G} - x_0, \lambda) = \frac{P(\text{G}_1 - x_0, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ EQ0078	115	■ 1.000 ● N +0	$P(\text{G}, \lambda) = \frac{P(\text{G}_1, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$		$P(\text{G}, \lambda) = \frac{P(\text{G}_1, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ EQ0080	115	■ 1.000 ● K +0	$P(\text{G}, \lambda) = (\lambda - p) P(\text{G} - x_0, \lambda).$		$P(\text{G}, \lambda) = (\lambda - p) P(\text{G} - x_0, \lambda).$
voloshin-hypergraph_ EQ0081 (5.17)	115	■ 1.000 ● N +0	$P(\text{G} - x_0, \lambda) = \frac{P(\text{G}_1^{\backslash\prime}, \lambda) P(\text{G}_2^{\backslash\prime}, \lambda) \dots P(\text{G}_k^{\backslash\prime}, \lambda)}{P(\text{G}_0^{\backslash\prime}, \lambda)^{k-1}},$		$P(\text{G} - x_0, \lambda) = \frac{P(\text{G}_1^{\backslash\prime}, \lambda) P(\text{G}_2^{\backslash\prime}, \lambda) \dots P(\text{G}_k^{\backslash\prime}, \lambda)}{P(\text{G}_0^{\backslash\prime}, \lambda)^{k-1}},$
voloshin-hypergraph_ EQ0082	115	■ 1.000 ● N -1	$P(\text{G}_1, \lambda) = (\lambda - p) P(\text{G}_1^{\backslash\prime}, \lambda), \text{ and } P(\text{G}_i, \lambda) = (\lambda - p_1) P(\text{G}_i^{\backslash\prime}, \lambda)$		$P(\text{G}_1, \lambda) = (\lambda - p) P(\text{G}_1^{\backslash\prime}, \lambda), \text{ and } P(\text{G}_i, \lambda) = (\lambda - p_1) P(\text{G}_i^{\backslash\prime}, \lambda)$
voloshin-hypergraph_ EQ0083	116	■ 1.000 ● N +0	$P(\text{G}, \lambda) = \frac{P(\text{G}_1, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$		$P(\text{G}, \lambda) = \frac{P(\text{G}_1, \lambda) P(\text{G}_2, \lambda) \dots P(\text{G}_k, \lambda)}{P(\text{G}_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ EQ0086 (5.18)	116	■ 1.000 ● K +0	$P(\text{G}^{\backslash\prime}, \lambda) = P(\text{G}^{\backslash\prime} - x, \lambda) W(\text{G}_{N(x)}^{\backslash\prime}, \lambda).$		$P(\text{G}^{\backslash\prime}, \lambda) = P(\text{G}^{\backslash\prime} - x, \lambda) W(\text{G}_{N(x)}^{\backslash\prime}, \lambda).$
voloshin-hypergraph_ EQ0088	116	■ 1.000 ●	$P(\text{G}, \lambda) = \frac{P(\text{G} - x, \lambda) P(\text{G}_{\{x\} \cup N(x)}, \lambda)}{P(\text{G}_{N(x)}, \lambda)} =$		$P(\text{G}, \lambda) = \frac{P(\text{G} - x, \lambda) P(\text{G}_{\{x\} \cup N(x)}, \lambda)}{P(\text{G}_{N(x)}, \lambda)} =$
voloshin-hypergraph_ EQ0089	117	■ 1.000 ● N -1	$P(\text{G} - x, \lambda) \frac{\lambda P(\text{G}_{N(x)}, \lambda - 1)}{P(\text{G}_{N(x)}, \lambda)} = P(\text{G} - x, \lambda) W(\text{G}_{N(x)}, \lambda).$		$P(\text{G} - x, \lambda) \frac{\lambda P(\text{G}_{N(x)}, \lambda - 1)}{P(\text{G}_{N(x)}, \lambda)} = P(\text{G} - x, \lambda) W(\text{G}_{N(x)}, \lambda).$
voloshin-hypergraph_ EQ0091	117	■ 1.000 ● K +0	$P(\text{G}, \lambda) = (\lambda - k) P(\text{G} - x, \lambda).$		$P(\text{G}, \lambda) = (\lambda - k) P(\text{G} - x, \lambda).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0092	117	■ 1.000 ● N +0	$W\left(G_{N(x)}, \lambda\right)=W\left(K_k, \lambda\right)=\frac{\lambda(\lambda-1)^{(k)}}{\lambda^{(k)}}=(\lambda-k).$	$W\left(G_{N(x)}, \lambda\right)=W\left(K_k, \lambda\right)=\frac{\lambda(\lambda-1)^{(k)}}{\lambda^{(k)}}=(\lambda-k).$	$W\left(G_{N(x)}, \lambda\right)=W\left(K_k, \lambda\right)=\frac{\lambda(\lambda-1)^{(k)}}{\lambda^{(k)}}=(\lambda-k).$
voloshin-hypergraph_ EQ0093	117	■ 1.000 ● K +0	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda).$	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda).$	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda).$
voloshin-hypergraph_ EQ0094	117	■ 1.000 ● N -1	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1}.$	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1}.$	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1}.$
voloshin-hypergraph_ EQ0095	117	■ 1.000 ● K +0	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3).$	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3).$	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3).$
voloshin-hypergraph_ EQ0097	118	■ 1.000 ● W +0	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$
voloshin-hypergraph_ EQ0098 (5.19)	124	■ 1.000 ● N -1	$\chi(G) \geq \omega(G).$	$\chi(G) \geq \omega(G).$	$\chi(G) \geq \omega(G).$
voloshin-hypergraph_ EQ0100	133	■ 1.000 ● K +0	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G).$	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G).$	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G).$
voloshin-hypergraph_ EQ0101	134	■ 1.000 ● K +0	$\bar{\chi}^{\prime}(G)=c+m-n+p.$	$\bar{\chi}^{\prime}(G)=c+m-n+p.$	$\bar{\chi}^{\prime}(G)=c+m-n+p.$
voloshin-hypergraph_ EQ0102	135	■ 1.000 ● N +0	$n-v^{\prime}-p+r.$	$n-v^{\prime}-p+r.$	$n-v^{\prime}-p+r.$
voloshin-hypergraph_ EQ0103	135	■ 1.000 ● N -2	$m^{\prime}=v^{\prime}+(n-v^{\prime}-p+r)=n-p+r.$	$m^{\prime}=v^{\prime}+(n-v^{\prime}-p+r)=n-p+r.$	$m^{\prime}=v^{\prime}+(n-v^{\prime}-p+r)=n-p+r.$
voloshin-hypergraph_ EQ0104	135	■ 1.000 ● N -1	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+(m-m^{\prime})=c^{\prime}+m-n+p,$	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+(m-m^{\prime})=c^{\prime}+m-n+p,$	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+(m-m^{\prime})=c^{\prime}+m-n+p,$
voloshin-hypergraph_ EQ0105	135	■ 1.000 ● N +0	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p.$	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p.$	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p.$
voloshin-hypergraph_ EQ0108	136	■ 1.000 ● K +0	$\bar{\chi}^{\prime}(G)=p-1.$	$\bar{\chi}^{\prime}(G)=p-1.$	$\bar{\chi}^{\prime}(G)=p-1.$
voloshin-hypergraph_ EQ0109	141	■ 1.000 ● N -1	$d(x)+d(y) \geq n,$	$d(x)+d(y) \geq n,$	$d(x)+d(y) \geq n,$
voloshin-hypergraph_ EQ0110	141	■ 1.000 ● N +0	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$
voloshin-hypergraph_ EQ0111	142	■ 1.000 ● K +0	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$
voloshin-hypergraph_ EQ0112	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$
voloshin-hypergraph_ EQ0113	144	■ 1.000 ●	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v.$	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v.$	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v.$
voloshin-hypergraph_ EQ0114	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v.$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v.$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v.$
voloshin-hypergraph_ EQ0115	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v.$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v.$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0116	152	■ 1.000 ● N +0	$\Delta(\mathcal{H})=\max_{x\in X} \mathcal{D}(x) $.	$\Delta(\mathcal{H})=\max_{x\in X} \mathcal{D}(x) .$	$\Delta(\mathcal{H})=\max_{x\in X} \mathcal{D}(x) .$
voloshin-hypergraph_ EQ0117	152	■ 1.000 ● N +0	$r(\mathcal{H})=\max_{D\in\mathcal{D}} D $.	$r(\mathcal{H})=\max_{D\in\mathcal{D}} D .$	$r(\mathcal{H})=\max_{D\in\mathcal{D}} D .$
voloshin-hypergraph_ EQ0121 (7.1)	158	■ 1.000 ● S +0	$\sum_{i=1}^n \mathcal{D}(x_i) =\sum_{j=1}^m D_j $.	$\sum_{i=1}^n \mathcal{D}(x_i) =\sum_{j=1}^m D_j .$	$\sum_{i=1}^n \mathcal{D}(x_i) =\sum_{j=1}^m D_j .$
voloshin-hypergraph_ EQ0122	158	■ 1.000 ● K +0	$3+3+2+2+0=12=1+2+3+3$.	$3+3+2+2+2+0=12=1+2+3+3+3.$	$3+3+2+2+2+0=12=1+2+3+3+3.$
voloshin-hypergraph_ EQ0123	159	■ 1.000 ● N +0	$L=\{\{1\},\{1,2\},\{1,2,4\},\{2,3,5\},\{3,4,5\}\}$.	$L=\{\{1\},\{1,2\},\{1,2,4\},\{2,3,5\},\{3,4,5\}\}.$	$L=\{\{1\},\{1,2\},\{1,2,4\},\{2,3,5\},\{3,4,5\}\}.$
voloshin-hypergraph_ EQ0126	160	■ 1.000 ● K +0	$ \mathcal{D}(K_4^3) =\frac{4\cdot 3\cdot 2}{1\cdot 2\cdot 3}=4$.	$ \mathcal{D}(K_4^3) =\binom{4}{3}=\frac{4\cdot 3\cdot 2}{1\cdot 2\cdot 3}=4.$	$ \mathcal{D}(K_4^3) =\binom{4}{3}=\frac{4\cdot 3\cdot 2}{1\cdot 2\cdot 3}=4.$
voloshin-hypergraph_ EQ0127	160	■ 1.000 ● K +0	$\mu=x_0D_0x_1D_1x_2\ldots x_{t-1}D_{t-1}x_t$	$\mu=x_0D_0x_1D_1x_2\ldots x_{t-1}D_{t-1}x_t$	$\mu=x_0D_0x_1D_1x_2\ldots x_{t-1}D_{t-1}x_t$
voloshin-hypergraph_ EQ0128	168	■ 1.000 ● N -2	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X $.	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X .$	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X .$
voloshin-hypergraph_ EQ0129 (7.2)	168	■ 1.000 ● N -1	$\tau(\mathcal{H})\geq v(\mathcal{H})$.	$\tau(\mathcal{H})\geq v(\mathcal{H}).$	$\tau(\mathcal{H})\geq v(\mathcal{H}).$
voloshin-hypergraph_ EQ0131	169	■ 1.000 ● K +0	$\rho(\mathcal{H})=\tau(\mathcal{H}^*)$.	$\rho(\mathcal{H})=\tau(\mathcal{H}^*).$	$\rho(\mathcal{H})=\tau(\mathcal{H}^*).$
voloshin-hypergraph_ EQ0134	170	■ 1.000 ● K +0	$(\mathcal{H})_2=L(\mathcal{H}^*)$.	$(\mathcal{H})_2=L(\mathcal{H}^*).$	$(\mathcal{H})_2=L(\mathcal{H}^*).$
voloshin-hypergraph_ EQ0135	170	■ 1.000 ● N +0	$L(\mathcal{H})=L(\mathcal{H}^*)_2$.	$L(\mathcal{H})=(\mathcal{H}^*)_2.$	$L(\mathcal{H})=(\mathcal{H}^*)_2.$
voloshin-hypergraph_ EQ0139	173	■ 1.000 ● K +0	$Y=(Y_1\cap Y_2)\cup(Y_1\cap Y_3)\cup(Y_2\cap Y_3)$.	$Y=(Y_1\cap Y_2)\cup(Y_1\cap Y_3)\cup(Y_2\cap Y_3).$	$Y=(Y_1\cap Y_2)\cup(Y_1\cap Y_3)\cup(Y_2\cap Y_3).$
voloshin-hypergraph_ EQ0140	173	■ 1.000 ● K +0	$Y\subseteq(D_1\cap D_2)\cup(D_1\cap D_3)\cup(D_2\cap D_3)$.	$Y\subseteq(D_1\cap D_2)\cup(D_1\cap D_3)\cup(D_2\cap D_3).$	$Y\subseteq(D_1\cap D_2)\cup(D_1\cap D_3)\cup(D_2\cap D_3).$
voloshin-hypergraph_ EQ0142 (8.2)	190	■ 1.000 ● N +0	$w(T)\leq\sum_{D\in\mathcal{D}}(D -1)$.	$w(T)\leq\sum_{D\in\mathcal{D}}(D -1).$	$w(T)\leq\sum_{D\in\mathcal{D}}(D -1).$
voloshin-hypergraph_ EQ0147 (8.5)	192	■ 1.000 ● K +0	$c(T,D)= D - E_D $.	$c(T,D)= D - E_D .$	$c(T,D)= D - E_D .$
voloshin-hypergraph_ EQ0148	192	■ 1.000 ● K +0	$w(T)=\sum_{ D \geq 2} E_D $.	$w(T)=\sum_{ D \geq 2} E_D .$	$w(T)=\sum_{ D \geq 2} E_D .$
voloshin-hypergraph_ EQ0151	193	■ 1.000 ● N +0	$w(T)=\sum_{D\in\mathcal{D}}(D -1)$.	$w(T)=\sum_{D\in\mathcal{D}}(D -1).$	$w(T)=\sum_{D\in\mathcal{D}}(D -1).$
voloshin-hypergraph_ EQ0157	194	■ 1.000 ● N -1	$w(T_1)=w(T)- \mathcal{D}_T(x) $.	$w(T_1)=w(T)- \mathcal{D}_T(x) .$	$w(T_1)=w(T)- \mathcal{D}_T(x) .$
voloshin-hypergraph_ EQ0158	194	■ 1.000 ●	$w(T')=w(T_1)+ \mathcal{D}_T(x) \geq w(T_1)+1+ \mathcal{D}_T(x) =w(T)+1$	$w(T')=w(T_1)+ \mathcal{D}_T(x) \geq w(T_1)+1+ \mathcal{D}_T(x) =w(T)+1$	$w(T')=w(T_1)+ \mathcal{D}_T(x) \geq w(T_1)+1+ \mathcal{D}_T(x) =w(T)+1$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0159	194	■ 1.000 ● K +0	$w(\mathrm{mathcal{H}})=\max_{\mathcal{T}} w(\mathcal{T})$.	$w(\mathcal{H}) = \max_T w(T).$	$w(\mathcal{H}) = \max_T w(T).$
voloshin-hypergraph_ EQ0163	197	■ 1.000 ● K +0	$\mu=x_{\{0\}}D_{\{0\}}x_{\{1\}}D_{\{1\}}x_{\{2\}}\ldots x_{\{t-1\}}D_{\{t-1\}}x_{\{t\}}$	$\mu = x_0D_0x_1D_1x_2\ldots x_{t-1}D_{t-1}x_t$	$\mu = x_0D_0x_1D_1x_2\ldots x_{t-1}D_{t-1}x_t$
voloshin-hypergraph_ EQ0164	201	■ 1.000 ● N -1	$\chi^{\mathrm{\prime}}(\mathrm{mathcal{H}}) \geq \omega(L(\mathrm{mathcal{H}})) \geq \Delta(\mathrm{mathcal{H}}) \geq \Delta(\mathrm{mathcal{H}})$.	$\chi'(\mathcal{H}) \geq \omega(L(\mathcal{H})) \geq \Delta(\mathcal{H}).$	$\chi'(\mathcal{H}) \geq \omega(L(\mathcal{H})) \geq \Delta(\mathcal{H}).$
voloshin-hypergraph_ EQ0165	201	■ 1.000 ● K +0	$\chi^{\mathrm{\prime}}(\mathrm{mathcal{H}}) \geq \omega(L(\mathrm{mathcal{H}})) \geq \Delta(\mathrm{mathcal{H}})$.	$\chi'(\mathcal{H}) = \Delta(\mathcal{H}).$	$\chi'(\mathcal{H}') = \Delta(\mathcal{H}').$
voloshin-hypergraph_ EQ0166	201	■ 1.000 ● N -1	$\chi^{\mathrm{\prime}}(\mathrm{mathcal{H}}) \geq \omega(L(\mathrm{mathcal{H}})) \geq \Delta(\mathrm{mathcal{H}})$.	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H})) = \omega(L(\mathcal{H})) = \Delta(\mathcal{H}).$	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H})) = \omega(L(\mathcal{H})) = \Delta(\mathcal{H}).$
voloshin-hypergraph_ EQ0167	204	■ 1.000 ● N +0	$n-\sum_{i=1}^m(E_i -1)+f=m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2$.	$n-\sum_{i=1}^m(E_i -1)+f=m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2.$	$n-\sum_{i=1}^m(E_i -1)+f=m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2.$
voloshin-hypergraph_ EQ0168 (9.1)	204	■ 1.000 ● N +0	$m^{\mathrm{\prime}}=\sum_{j=1}^m D_j =\sum_{i=1}^n \mathcal{D}(x_i) $	$m' = \sum_{j=1}^m D_j = \sum_{i=1}^n \mathcal{D}(x_i) $	$m' = \sum_{j=1}^m D_j = \sum_{i=1}^n \mathcal{D}(x_i) $
voloshin-hypergraph_ EQ0169	205	■ 1.000 ● N +1	$n^{\mathrm{\prime}}-m^{\mathrm{\prime}}+f^{\mathrm{\prime}}=2$	$n' - m' + f' = 2$	$n' - m' + f' = 2.$
voloshin-hypergraph_ EQ0170 (9.2)	205	■ 1.000 ● N +1	$n-\sum_{i=1}^m(D_i -1)+f=2$	$n-\sum_{i=1}^m(D_i -1)+f=2$	$n-\sum_{i=1}^m(D_i -1)+f=2,$
voloshin-hypergraph_ EQ0171 (9.3)	205	■ 1.000 ● N +1	$m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2$	$m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2$	$m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2.$
voloshin-hypergraph_ EQ0172	205	■ 1.000 ● N +0	$n-\sum_{i=1}^m(E_i -1)+f=4-(2+2)+2=2$	$n-\sum_{i=1}^m(E_i -1)+f=4-(2+2)+2=2$	$n-\sum_{i=1}^m(E_i -1)+f=4-(2+2)+2=2$
voloshin-hypergraph_ EQ0173	205	■ 1.000 ● N -1	$\left(m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2-(0+1+0+1)+2\right)=2$.	$m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2-(0+1+0+1)+2\bigg)=2.$	$m-\sum_{j=1}^n(\mathcal{D}(x_j) -1)+f=2-(0+1+0+1)+2)=2.$
voloshin-hypergraph_ EQ0174 (10.1)	209	■ 1.000 ● K +0	$X=\cup_{i=1}^k S_{\{i\}}, \quad S_{\{i\}} \neq \emptyset, \quad S_{\{i\}} \cap S_{\{j\}}=\emptyset, \quad i \neq j$.	$X = \cup_{i=1}^k S_i, \quad S_i \neq \emptyset, \quad S_i \cap S_j = \emptyset, i \neq j.$	$X = \cup_{i=1}^k S_i, \quad S_i \neq \emptyset, \quad S_i \cap S_j = \emptyset, i \neq j.$
voloshin-hypergraph_ EQ0175	210	■ 1.000 ● W -2	$\begin{gathered} \alpha(\mathrm{mathcal{H}}) \chi(\mathrm{mathcal{H}}) \geq n; \\ \chi(\mathrm{mathcal{H}}) \leq \tau(\mathrm{mathcal{H}}) + 1 = n - \alpha(\mathrm{mathcal{H}}) + 1. \end{gathered}$	$\alpha(\mathcal{H})\chi(\mathcal{H}) \geq n;$ $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1 = n - \alpha(\mathcal{H}) + 1.$	$\alpha(\mathcal{H})\chi(\mathcal{H}) \geq n;$ $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1 = n - \alpha(\mathcal{H}) + 1.$
voloshin-hypergraph_ EQ0177	211	■ 1.000 ● K +0	$\min\{ D , \lambda\}$	$\min\{ D , \lambda\}$	$\min\{ D , \lambda\}$
voloshin-hypergraph_ EQ0178	211	■ 1.000 ● N +0	$\lambda \leq \min_{D \in \mathcal{D}} D ,$	$\lambda \leq \min_{D \in \mathcal{D}} D ,$	$\lambda \leq \min_{D \in \mathcal{D}} D ,$
voloshin-hypergraph_ EQ0181	211	■ 1.000 ● K +0	$a_j \leq D_j \cap S_i \leq b_j$.	$a_j \leq D_j \cap S_i \leq b_j.$	$a_j \leq D_j \cap S_i \leq b_j.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_EQ0183	213	1.000 •	$D_{\{i\}}, D_{\{j\}} \in \mathcal{D}_1(x) \Rightarrow D_i \cap D_j = \{x\}.$	$D_i, D_j \in \mathcal{D}_1(x) \Rightarrow D_i \cap D_j = \{x\}.$	$D_i, D_j \in \mathcal{D}_1(x) \Rightarrow D_i \cap D_j = \{x\}.$
voloshin-hypergraph_EQ0184	214	1.000 • N -2	$M(\mathcal{H})=\max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$	$M(\mathcal{H})=\max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$	$M(\mathcal{H})=\max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$
voloshin-hypergraph_EQ0187	215	1.000 • N +0	$\chi(\mathcal{H}) \leq M(\mathcal{H})+1.$	$\chi(\mathcal{H}) \leq M(\mathcal{H}) + 1.$	$\chi(\mathcal{H}) \leq M(\mathcal{H}) + 1.$
voloshin-hypergraph_EQ0188	215	1.000 • N +0	$m(1, \mathcal{H})=2, m(2, \mathcal{H})=2, m(3, \mathcal{H})=2, m(4, \mathcal{H})=1, m(5, \mathcal{H})=1.$	$m(1, \mathcal{H}) = 2, m(2, \mathcal{H}) = 2, m(3, \mathcal{H}) = 2, m(4, \mathcal{H}) = 1, m(5, \mathcal{H}) = 1.$	$m(1, \mathcal{H}) = 2, m(2, \mathcal{H}) = 2, m(3, \mathcal{H}) = 2, m(4, \mathcal{H}) = 1, m(5, \mathcal{H}) = 1.$
voloshin-hypergraph_EQ0191	216	1.000 • K +0	$c=(2,1,2,1,1).$	$c = (2, 1, 2, 1, 1).$	$c = (2, 1, 2, 1, 1).$
voloshin-hypergraph_EQ0192	218	1.000 • K +0	$c: X \rightarrow \{1,2,\ldots,\lambda\}.$	$c: X \rightarrow \{1,2,\ldots,\lambda\}.$	$c: X \rightarrow \{1,2,\ldots,\lambda\}.$
voloshin-hypergraph_EQ0193	219	1.000 • N +0	$R(\mathcal{H})=\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$R(\mathcal{H}) = (r_1, r_2, \ldots, r_n)$	$R(\mathcal{H}) = (r_1, r_2, \ldots, r_n)$
voloshin-hypergraph_EQ0196	220	1.000 • N +0	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$
voloshin-hypergraph_EQ0205	230	1.000 • N -1	$P(\mathcal{H}, \lambda)=P\left(\mathcal{H}_{\{1\}}, \lambda\right)+P\left(\mathcal{H}_{\{2\}}, \lambda\right), \quad R(\mathcal{H})=R\left(\mathcal{H}_{\{1\}}\right)+R\left(\mathcal{H}_{\{2\}}\right),$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) + P(\mathcal{H}_2, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1) + R(\mathcal{H}_2),$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) + P(\mathcal{H}_2, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1) + R(\mathcal{H}_2),$
voloshin-hypergraph_EQ0209	231	1.000 • W +0	$\begin{gathered} Z=\left\{K_3, K_3, K_2\right\}, \\ R(\mathcal{H})=(0,1,2,0), \quad P(\mathcal{H}, \lambda)=2 \lambda^3-5 \lambda^2+3 \lambda, \end{gathered}$	$Z=\left\{K_3, K_3, K_2\right\}, R(\mathcal{H})=(0,1,2,0), \\ P(\mathcal{H}, \lambda)=2 \lambda^3-5 \lambda^2+3 \lambda,$	$Z=\left\{K_3, K_3, K_2\right\}, R(\mathcal{H})=(0,1,2,0), \\ P(\mathcal{H}, \lambda)=2 \lambda^3-5 \lambda^2+3 \lambda,$
voloshin-hypergraph_EQ0210	231	1.000 • K +0	$L=\{(1,2,3,2),(1,2,1,3),(1,2,1,2)\}.$	$L = \{(1, 2, 3, 2), (1, 2, 1, 3), (1, 2, 1, 2)\}.$	$L = \{(1, 2, 3, 2), (1, 2, 1, 3), (1, 2, 1, 2)\}.$
voloshin-hypergraph_EQ0211	233	1.000 • K +0	$R(\mathcal{H})=R(\mathcal{H}-C)(R(\mathcal{H})=R(\mathcal{H}-D)).$	$R(\mathcal{H}) = R(\mathcal{H} - C)(R(\mathcal{H}) = R(\mathcal{H} - D)).$	$R(\mathcal{H}) = R(\mathcal{H} - C)(R(\mathcal{H}) = R(\mathcal{H} - D)).$
voloshin-hypergraph_EQ0212	233	1.000 • N +0	$R(\mathcal{H})=R\left(\mathcal{H}-C_j\right)=(1,7,0,0), j=1,2,3,4.$	$R(\mathcal{H}) = R(\mathcal{H} - C_j) = (1, 7, 0, 0), j = 1, 2, 3, 4.$	$R(\mathcal{H}) = R(\mathcal{H} - C_j) = (1, 7, 0, 0), j = 1, 2, 3, 4.$
voloshin-hypergraph_EQ0213	233	1.000 • K +0	$\mathcal{D}=\left\{\{1,2\},\{2,3\},\{3,4\},\{4,1\}\right\}$	$\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$
voloshin-hypergraph_EQ0215	237	1.000 • K +0	$v(k)=k+4.$	$v(k) = k + 4.$	$v(k) = k + 4.$
voloshin-hypergraph_EQ0217	240	1.000 • N +0	$n \geq (l-1)(m-1)+1.$	$n \geq (l-1)(m-1) + 1.$	$n \geq (l-1)(m-1) + 1.$
voloshin-hypergraph_EQ0219	240	1.000 • N +0	$N_n=\sum_{k=1}^n\left\lfloor\frac{n}{k}\right\rfloor \simeq n \log n$	$N_n=\sum_{k=1}^n\left\lfloor\frac{n}{k}\right\rfloor \simeq n \log n$	$N_n=\sum_{k=1}^n\left\lfloor\frac{n}{k}\right\rfloor \simeq n \log n$
voloshin-hypergraph_EQ0220	240	1.000 • W -2	$\lim _{n \rightarrow \infty} \frac{N_n}{n^2}=\lim _{n \rightarrow \infty} \frac{\log n}{n}=0,$	$\lim _{n \rightarrow \infty} \frac{N_n}{n^2}=\lim _{n \rightarrow \infty} \frac{\log n}{n}=0,$	$\lim _{n \rightarrow \infty} \frac{N_n}{n^2}=\lim _{n \rightarrow \infty} \frac{\log n}{n}=0,$
voloshin-hypergraph_EQ0221	243	1.000 • N +0	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=\chi, r_{\chi}(\mathcal{H})=1,$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = \chi, r_{\chi}(\mathcal{H}) = 1,$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = \chi, r_{\chi}(\mathcal{H}) = 1,$

Conf. MathPix confidence:

≥0.9

0.5-0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0222	243	■ 1.000 ● K +0	$R(\mathcal{H})=(0,\ldots,0,1,0,\ldots,0)$.	$R(\mathcal{H})=(0,\ldots,0,1,0,\ldots,0).$	$R(\mathcal{H})=(0,\ldots,0,1,0,\ldots,0).$
voloshin-hypergraph_ EQ0225	244	■ 1.000 ●	$R\left(\mathcal{H}_{\{1\}}\right)=R\left(\mathcal{H}_{\{2\}}\right)=(0,1,0,0),P\left(\mathcal{H}_{\{1\}},\lambda\right)=P\left(\mathcal{H}_{\{2\}},\lambda\right)=\lambda(\lambda-1).$ $P\left(\mathcal{H}_{\{1\}},\lambda\right)=\lambda(\lambda-1).$	$R(\mathcal{H}_1)=R(\mathcal{H}_2)=(0,1,0,0),P(\mathcal{H}_1,\lambda)=P(\mathcal{H}_2,\lambda)=\lambda(\lambda-1).$	$R(\mathcal{H}_1)=R(\mathcal{H}_2)=(0,1,0,0),P(\mathcal{H}_1,\lambda)=P(\mathcal{H}_2,\lambda)=\lambda(\lambda-1).$
voloshin-hypergraph_ EQ0227	248	■ 1.000 ● K +0	$L_{\{1\}}=\left(X_{\{1\}},E_{\{1\}}\right),L_{\{2\}}=\left(X_{\{2\}},E_{\{2\}}\right),\ldots,L_{\{\chi\}}=\left(X_{\{\chi\}},E_{\{\chi\}}\right).$	$L_1=(X_1,E_1),L_2=(X_2,E_2),\ldots,L_\chi=(X_\chi,E_\chi).$	$L_1=(X_1,E_1),L_2=(X_2,E_2),\ldots,L_\chi=(X_\chi,E_\chi).$
voloshin-hypergraph_ EQ0230	249	■ 1.000 ● K +0	$\left\{x_{\{1\}},x_{\{2\}},\ldots,x_{\{i\}}\right\},i=1,2,\ldots,n$	$\left\{x_1,x_2,\ldots,x_i\right\},i=1,2,\ldots,n$	$\left\{x_1,x_2,\ldots,x_i\right\},i=1,2,\ldots,n$
voloshin-hypergraph_ EQ0234	253	■ 1.000 ● K +0	$C_{\{i\}}\cap C_{\{j\}}=\{1\},\quad i,j\in I=\{1,2,\ldots k+2\},\quad i\neq j$.	$C_i\cap C_j=\{1\},\quad i,j\in I=\{1,2,\ldots k+2\},\quad i\neq j.$	$C_i\cap C_j=\{1\},\quad i,j\in I=\{1,2,\ldots k+2\},\quad i\neq j.$
voloshin-hypergraph_ EQ0236	255	■ 1.000 ● N +0	$\begin{aligned} & n_{12}+n_{23}+n_{34}+2<r, \quad \& \\ & n_{12}+n_{14}+n_{34}+2<r. \end{aligned}$	$\begin{aligned} & n_{12}+n_{23}+n_{34}+2<r, \\ & n_{12}+n_{14}+n_{34}+2<r. \end{aligned}$	$\begin{aligned} & n_{12}+n_{23}+n_{34}+2<r, \\ & n_{12}+n_{14}+n_{34}+2<r. \end{aligned}$
voloshin-hypergraph_ EQ0238	256	■ 1.000 ● N +0	$n_{\{31\}}+1+n_{12}<r$.	$n_{31}+1+n_{12}<r.$	$n_{31}+1+n_{12}<r.$
voloshin-hypergraph_ EQ0239	256	■ 1.000 ● N +0	$X=\{1,2,3,\ldots,2r\},$	$X=\{1,2,3,\ldots,2r\},$	$X=\{1,2,3,\ldots,2r\},$
voloshin-hypergraph_ EQ0240	256	■ 1.000 ● K +0	$\mathcal{C}=\left(\mathcal{C}_{\{1\}},\mathcal{C}_{\{2\}},\ldots,\mathcal{C}_{\{2r\}},\mathcal{C}_{\{o\}},\mathcal{C}_{\{e\}}\right)$	$\mathcal{C}=\{C_1,C_2,\ldots,C_{2r},C_o,C_e\}$	$\mathcal{C}=\{C_1,C_2,\ldots,C_{2r},C_o,C_e\}$
voloshin-hypergraph_ EQ0241	256	■ 1.000 ● N +0	$C_{\{1\}}=\{1,2,4,6,8,\ldots,2r-2\}$.	$C_1=\{1,2,4,6,8,\ldots,2r-2\}.$	$C_1=\{1,2,4,6,8,\ldots,2r-2\}.$
voloshin-hypergraph_ EQ0247	260	■ 1.000 ● K +0	$R(\mathcal{H})=(0,7,0,0)$	$R(\mathcal{H})=(0,7,0,0)$	$R(\mathcal{H})=(0,7,0,0)$
voloshin-hypergraph_ EQ0249	261	■ 1.000 ● N -1	$R(\mathcal{H})=\left(0,r_{\{2\}},0,1,0,0,0,0\right).$	$R(\mathcal{H})=(0,r_2,0,1,0,0,0,0).$	$R(\mathcal{H})=(0,r_2,0,1,0,0,0,0).$
voloshin-hypergraph_ EQ0250	263	■ 1.000 ● K +0	$\left\{x_{\{1\}},x_{\{2\}},a_{\{1\}},\ldots,a_{\{t-2\}},b_{\{1\}},\ldots,b_{\{t-2\}}\right\}.$	$\left\{x_1,x_2,a_1,\ldots,a_{t-2},b_1,\ldots,b_{t-2}\right\}.$	$\left\{x_1,x_2,a_1,\ldots,a_{t-2},b_1,\ldots,b_{t-2}\right\}.$
voloshin-hypergraph_ EQ0251	264	■ 1.000 ● K +0	$R\left(\mathcal{H}_{\{2,4\}}\right)=(0,4,0,1,0,0),$	$R(\mathcal{H}_{2,4})=(0,4,0,1,0,0),$	$R(\mathcal{H}_{2,4})=(0,4,0,1,0,0),$
voloshin-hypergraph_ EQ0252	264	■ 1.000 ● K +0	$P(\mathcal{H},\lambda)=\lambda(\lambda-1)\left(\lambda^2-5\lambda+10\right).$	$P(\mathcal{H},\lambda)=\lambda(\lambda-1)\left(\lambda^2-5\lambda+10\right).$	$P(\mathcal{H},\lambda)=\lambda(\lambda-1)(\lambda^2-5\lambda+10).$
voloshin-hypergraph_ EQ0253	266	■ 1.000 ● N +0	$R(\mathcal{H})=\left(0,\ldots,0,r_{\{\chi\}},\ldots,r_{\{\bar{\chi}\}},0,\ldots,0\right).$	$R(\mathcal{H})=(0,\ldots,0,r_\chi,\ldots,r_{\bar{\chi}},0,\ldots,0).$	$R(\mathcal{H})=(0,\ldots,0,r_\chi,\ldots,r_{\bar{\chi}},0,\ldots,0).$
voloshin-hypergraph_ EQ0255	268	■ 1.000 ● K +0	(A,A,B,B,C,C,D) .	$(A,A,B,B,C,C,D).$	$(A,A,B,B,C,C,D).$
voloshin-hypergraph_ EQ0256	268	■ 1.000 ● K +0	$\{1,2,7\},\{3,4,7\},\{5,6,7\}\in\mathcal{E}.$	$\{1,2,7\},\{3,4,7\},\{5,6,7\}\in\mathcal{E}.$	$\{1,2,7\},\{3,4,7\},\{5,6,7\}\in\mathcal{E}.$
voloshin-hypergraph_ EQ0260	274	■ 1.000 ● K +0	$\mathcal{H}=\left(X,\binom{X}{3},\binom{X}{3}\right)$	$\mathcal{H}=\left(X,\binom{X}{3},\binom{X}{3}\right)$	$\mathcal{H}=\left(X,\binom{X}{3},\binom{X}{3}\right)$
voloshin-hypergraph_ EQ0262	276	■ 1.000 ● K +0	$R\left(\mathcal{H}_{\{2,4\}}^{\prime}\right)=(0,1,0,1,0,0),$	$R(\mathcal{H}_{2,4}')=(0,1,0,1,0,0),$	$R(\mathcal{H}_{2,4}')=(0,1,0,1,0,0),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0263	276	■ 1.000 ● K +0	$P\left(\mathcal{H}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$	$P\left(\mathcal{H}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$	$P\left(\mathcal{H}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$
voloshin-hypergraph_ EQ0264	285	■ 1.000 ● N +0	$f(n)\leq Cg(n)$	$f(n)\leq Cg(n)$	$f(n)\leq Cg(n)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																																																
voloshin-hypergraph_TAB_011	110	—	$\begin{tabular}{t}{ l l l l l l }\hline&1&2&3&4&5\\\hline1&1&0&0&0&0\\\hline2&1&1&0&0&0\\\hline3&1&3&1&0&0\\\hline4&1&7&6&1&0\\\hline5&1&15&25&10&1\\\hline\end{tabular}$	<table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>2</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>3</td><td>1</td><td>3</td><td>1</td><td>0</td><td>0</td></tr><tr><td>4</td><td>1</td><td>7</td><td>6</td><td>1</td><td>0</td></tr><tr><td>5</td><td>1</td><td>15</td><td>25</td><td>10</td><td>1</td></tr></table>		1	2	3	4	5	1	1	0	0	0	0	2	1	1	0	0	0	3	1	3	1	0	0	4	1	7	6	1	0	5	1	15	25	10	1	<table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>2</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>3</td><td>1</td><td>3</td><td>1</td><td>0</td><td>0</td></tr><tr><td>4</td><td>1</td><td>7</td><td>6</td><td>1</td><td>0</td></tr><tr><td>5</td><td>1</td><td>15</td><td>25</td><td>10</td><td>1</td></tr></table>		1	2	3	4	5	1	1	0	0	0	0	2	1	1	0	0	0	3	1	3	1	0	0	4	1	7	6	1	0	5	1	15	25	10	1																																																								
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Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value

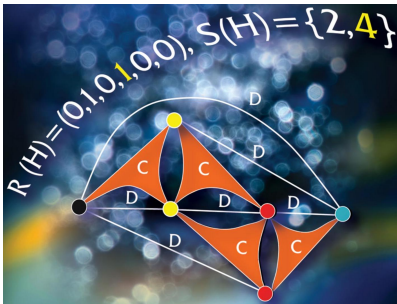
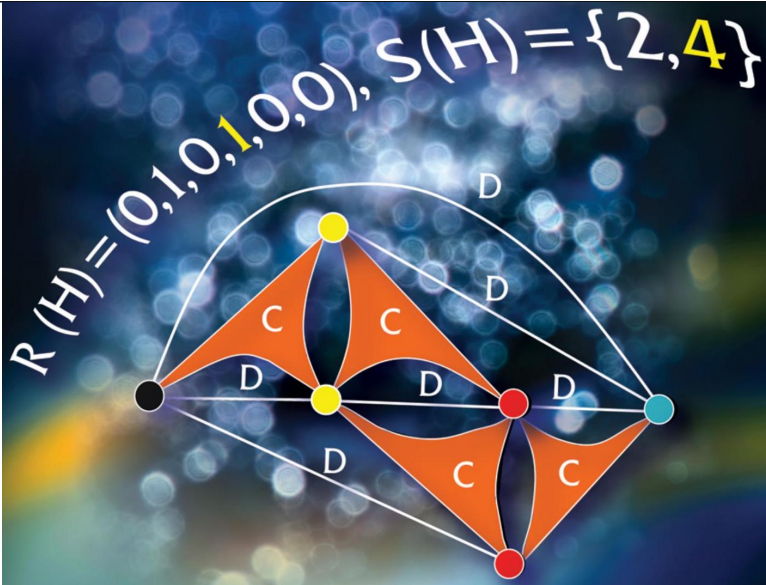
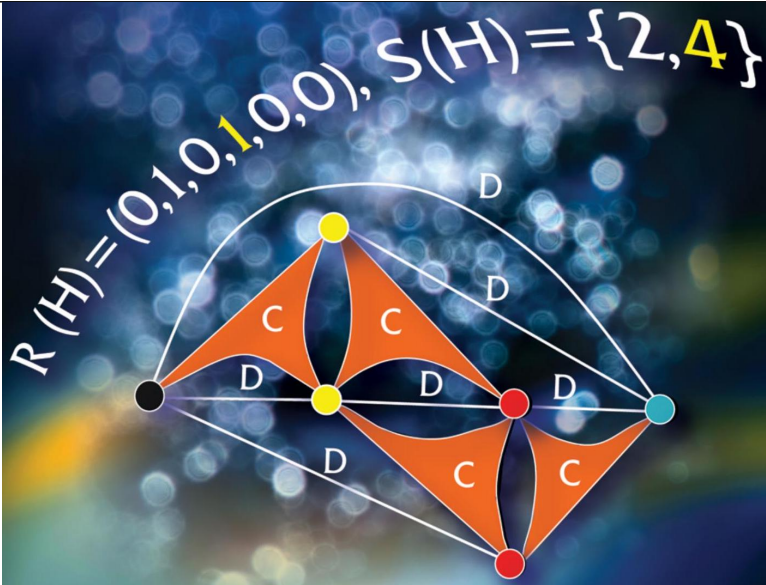
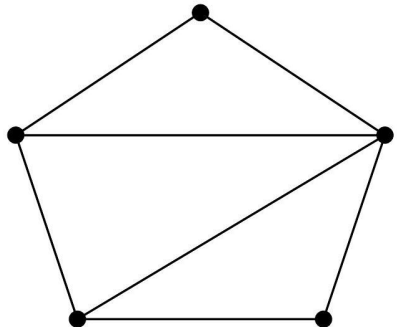
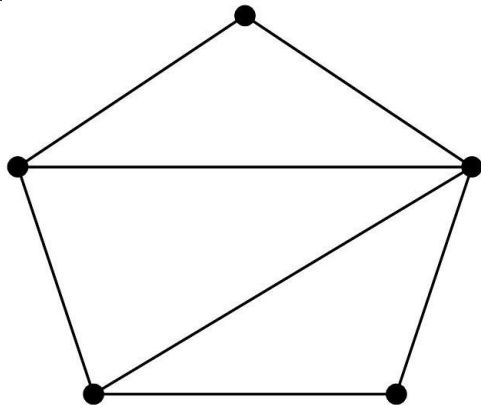
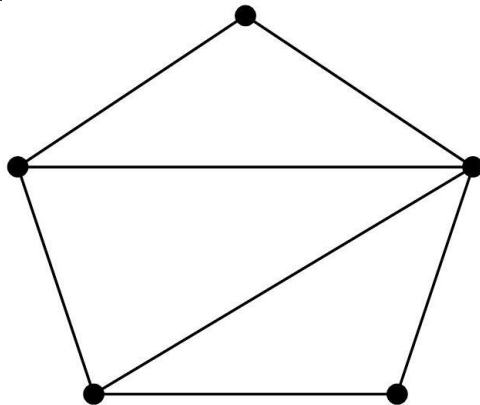
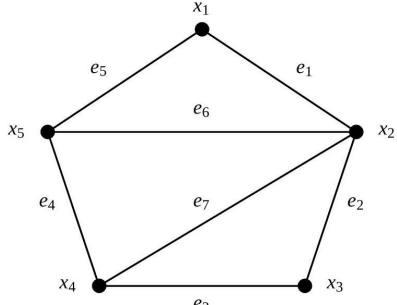
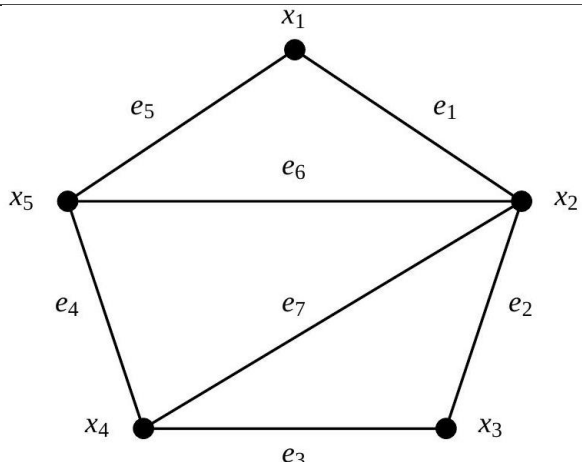
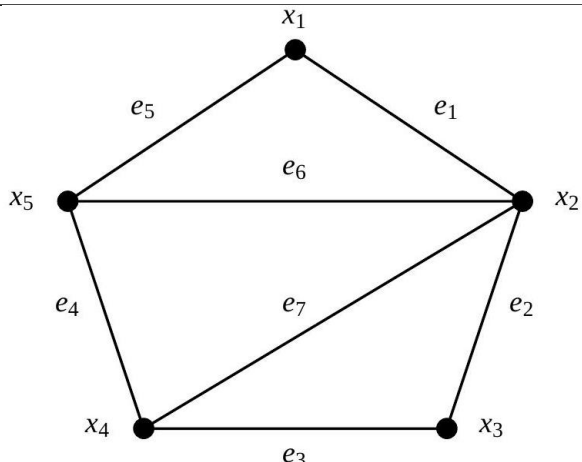
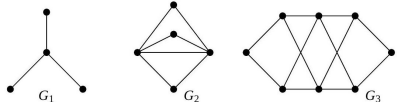
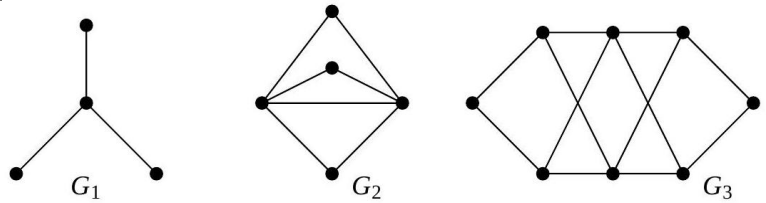
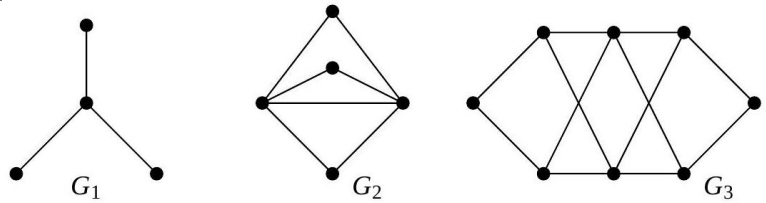
LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

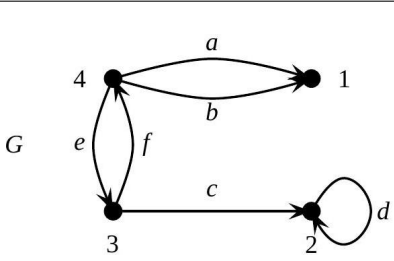
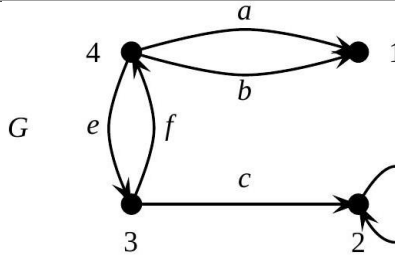
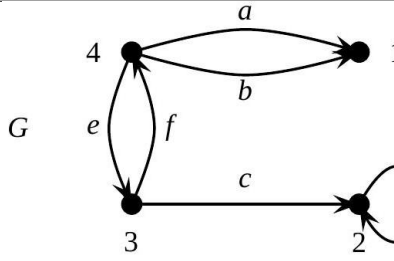
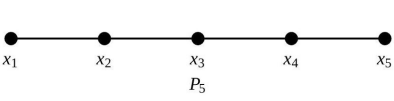
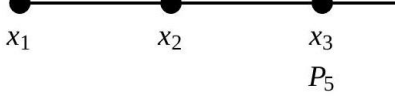
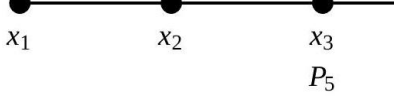
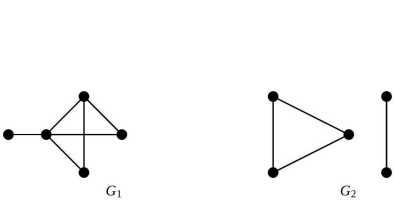
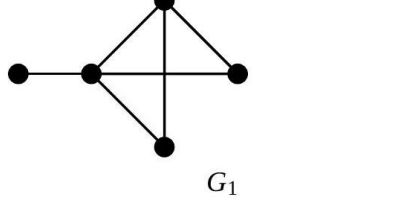
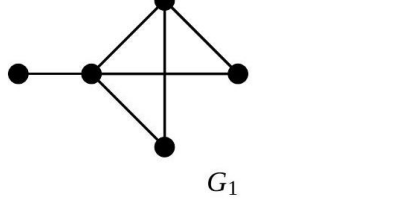
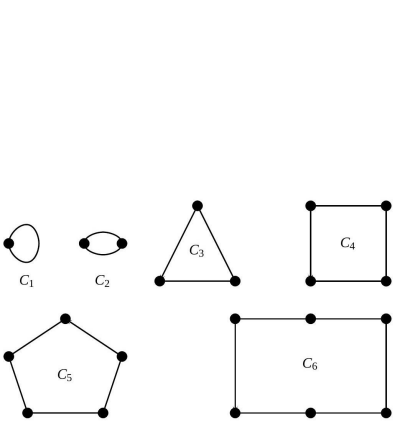
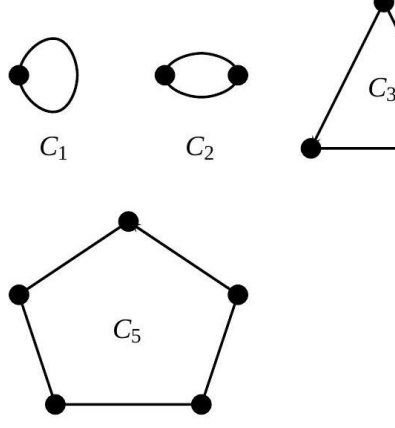
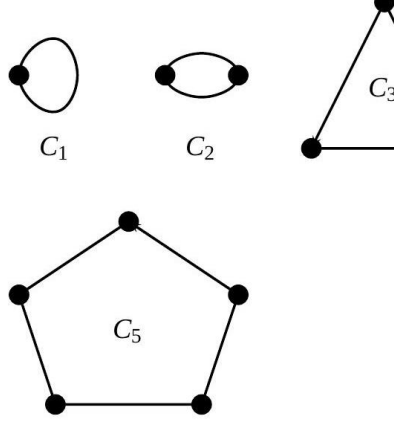
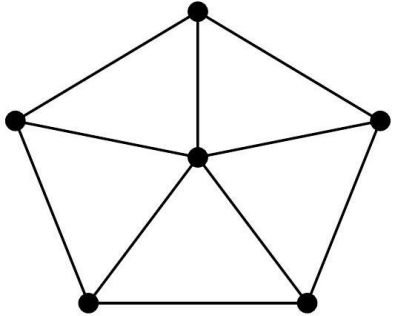
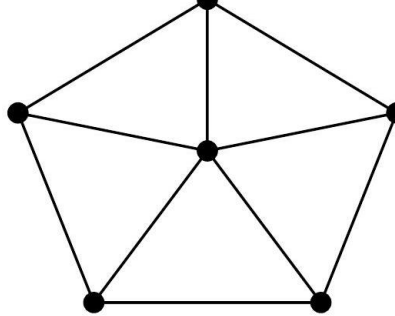
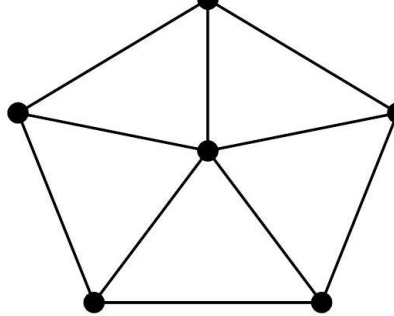
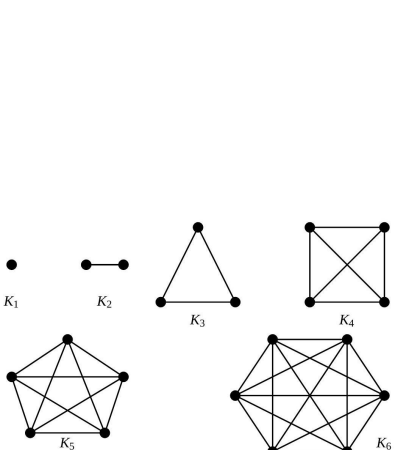
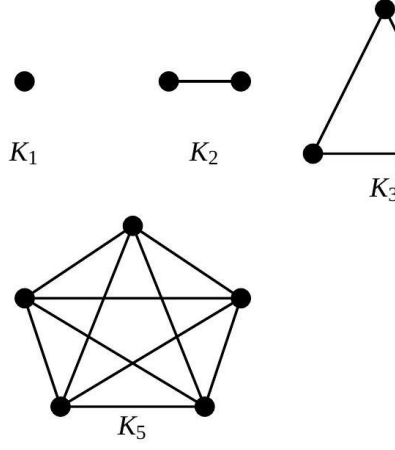
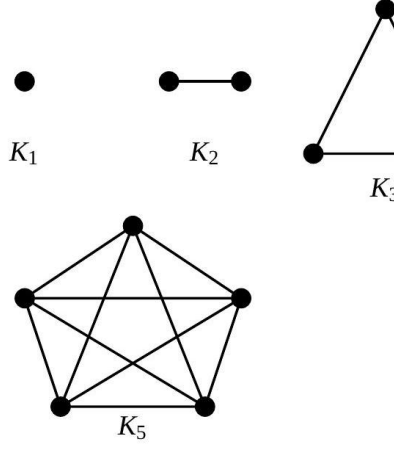
[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Image regions — rendered against scan

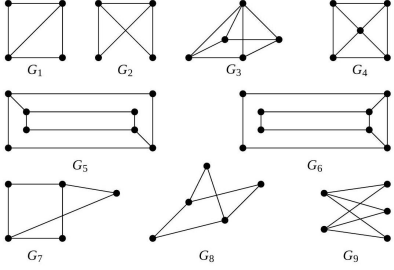
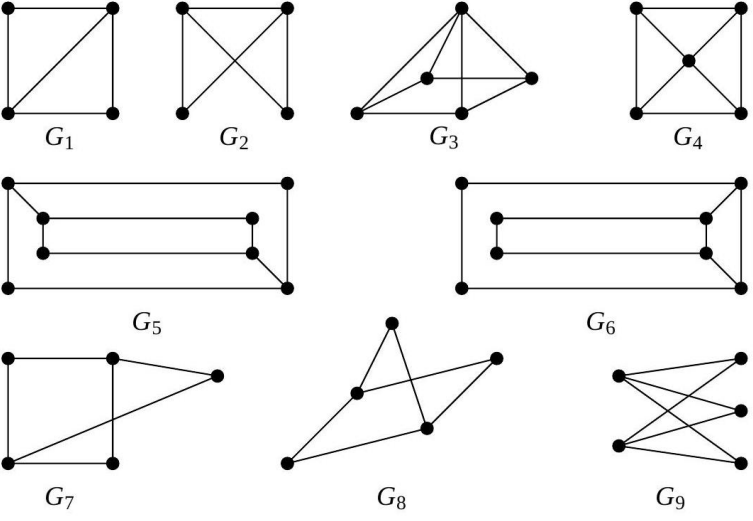
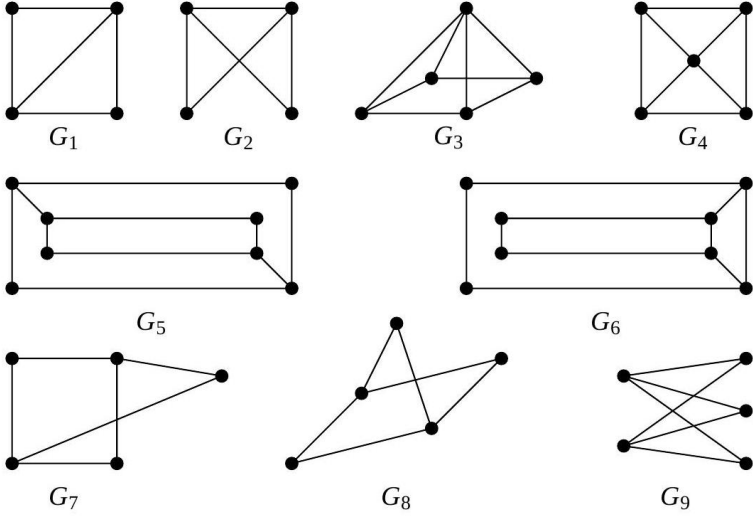
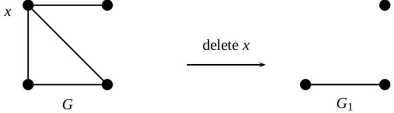
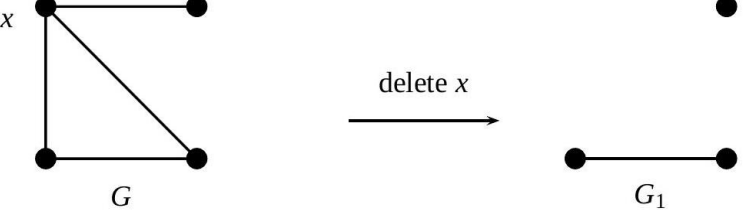
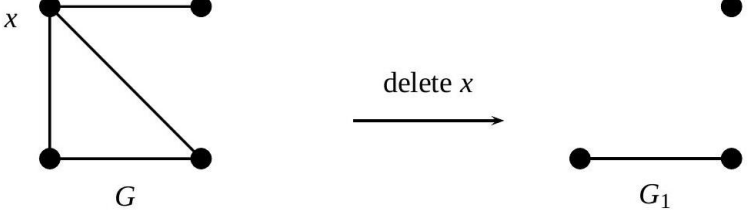
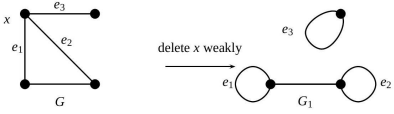
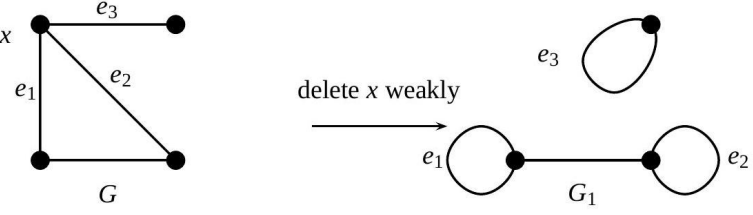
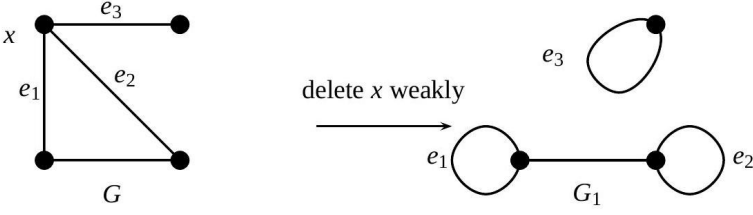
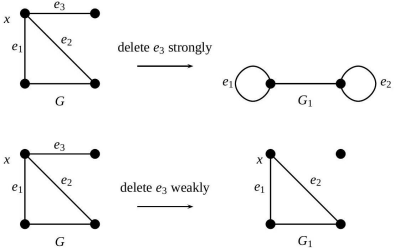
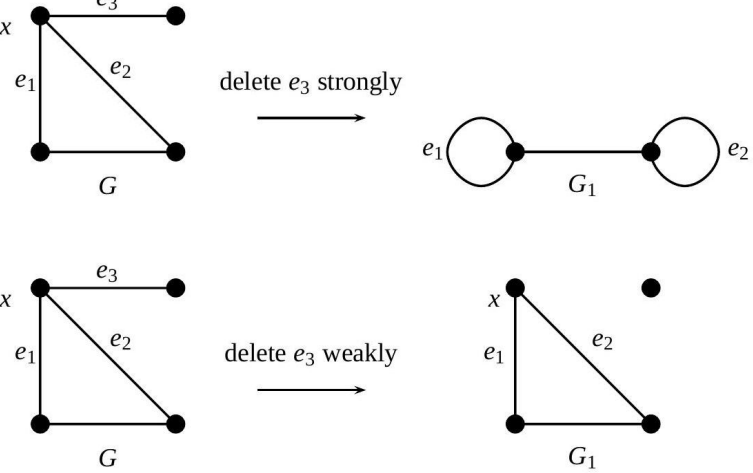
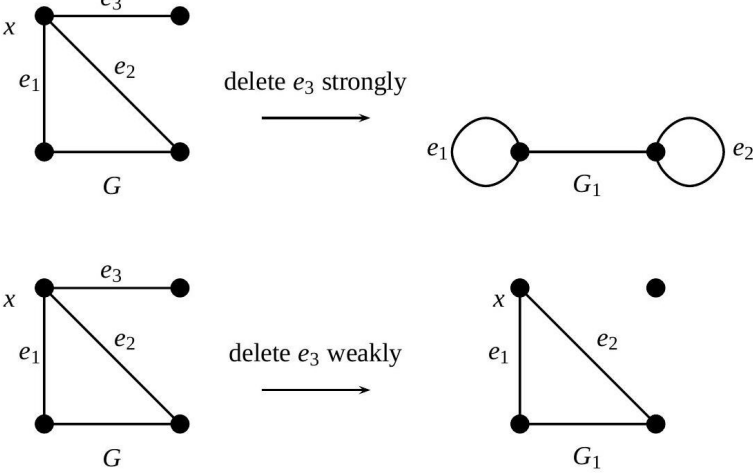
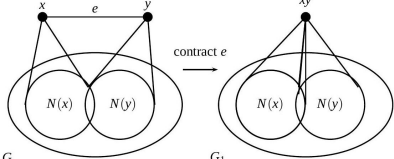
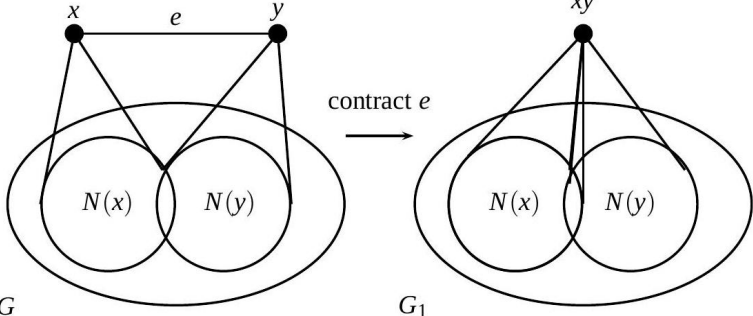
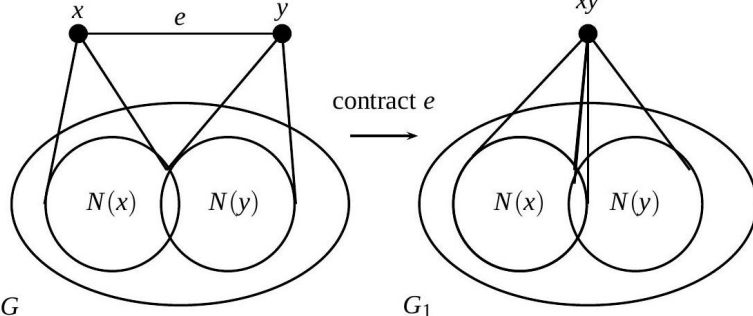
Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
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voloshin-hypergraph_DIA_002	022	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-022_608_756_249_502.jpg			
voloshin-hypergraph_DIA_003	024	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-024_290_1071_293_371.jpg			

voloshin-hypergraph_DIA_00025	025	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-025_709_999_285_388.jpg			
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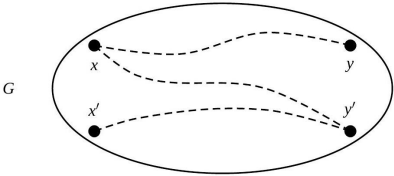
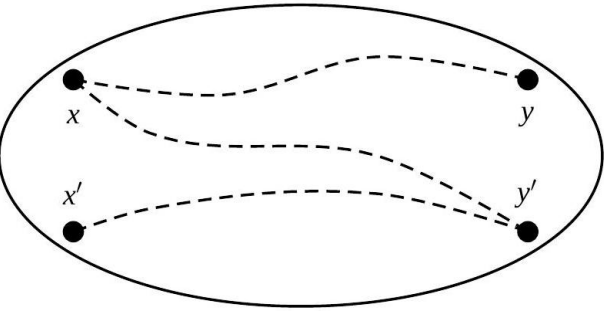
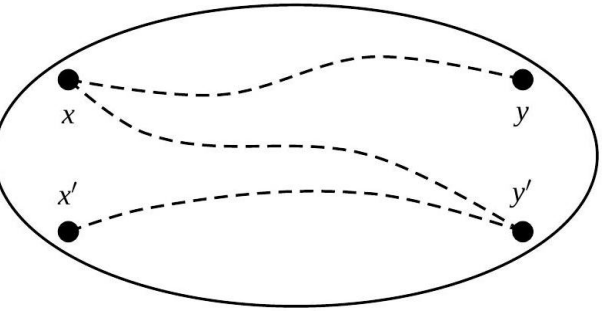
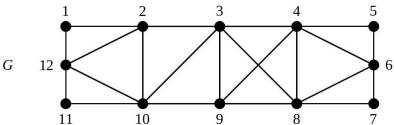
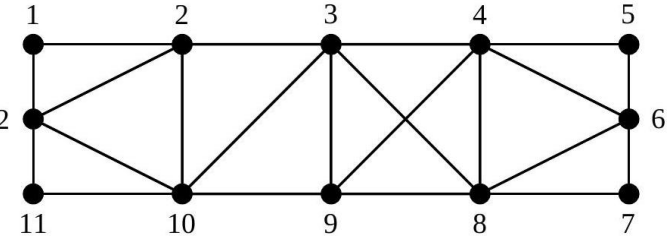
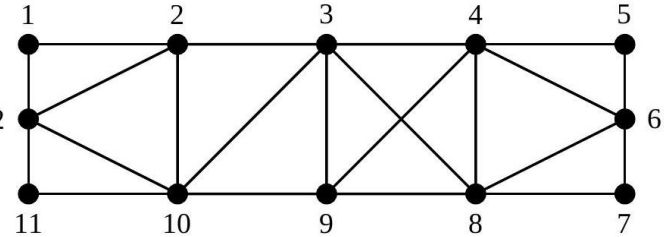
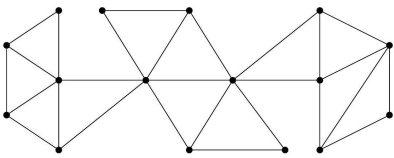
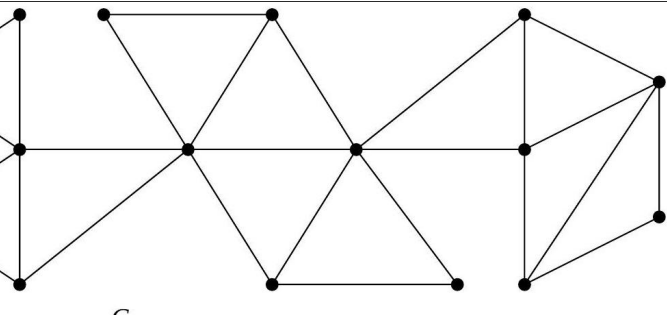
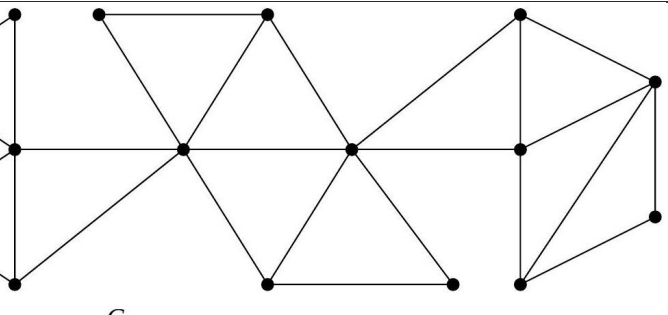
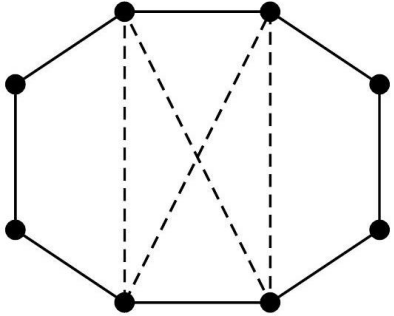
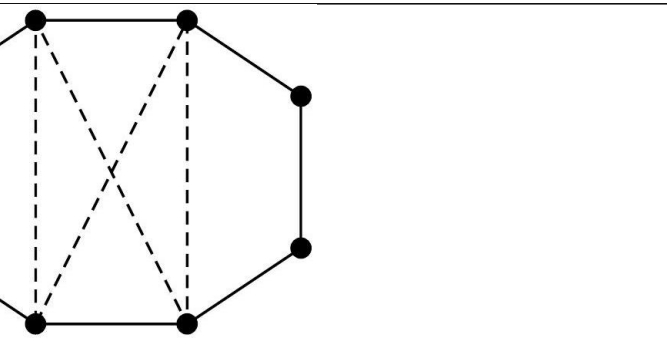
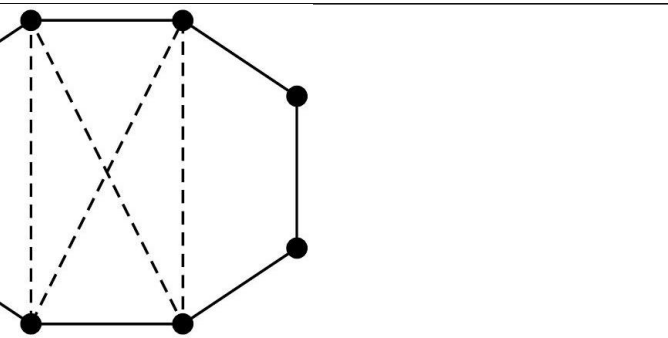
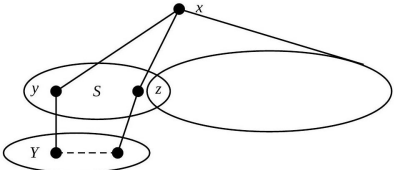
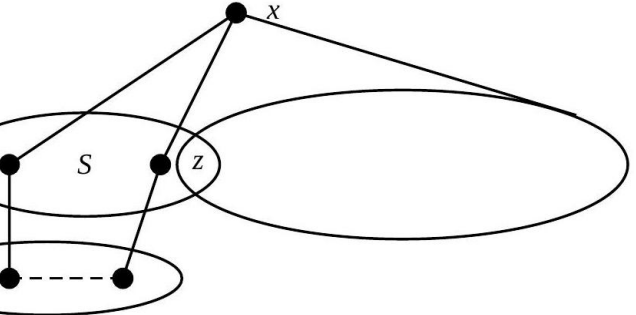
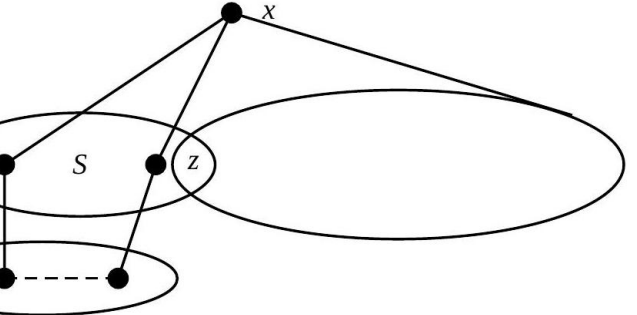
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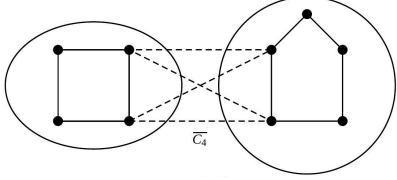
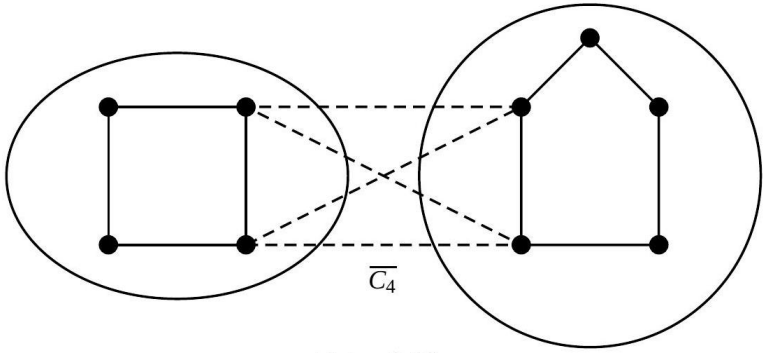
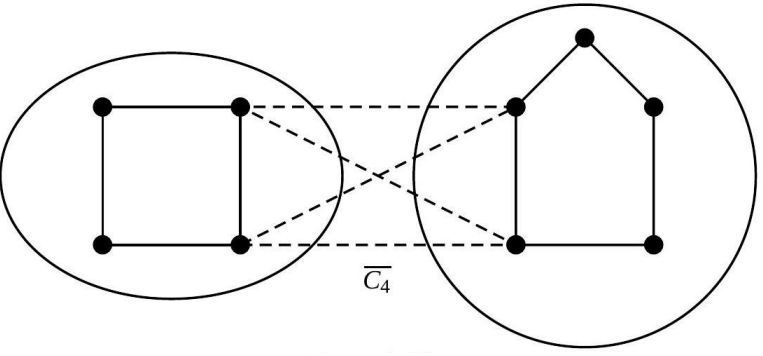
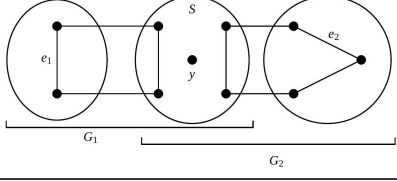
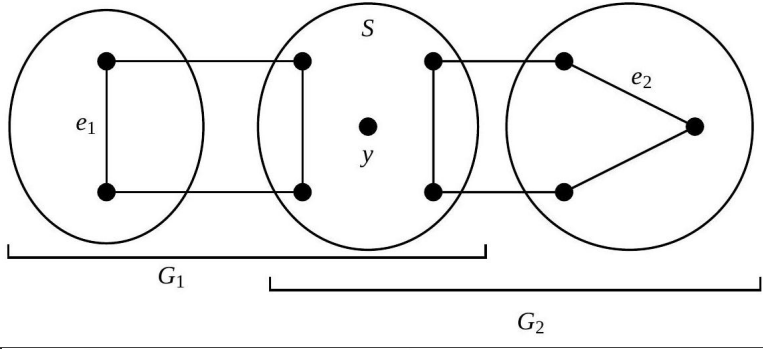
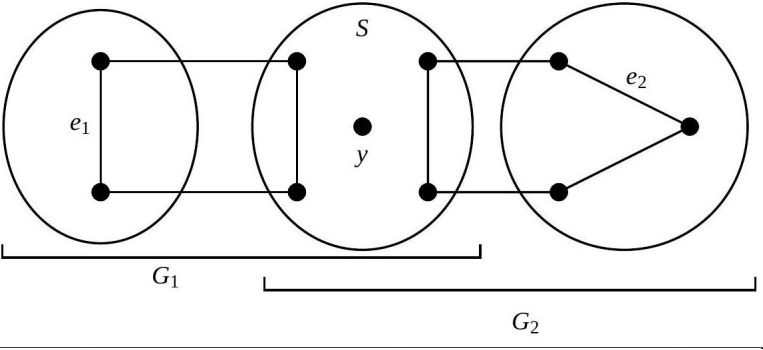
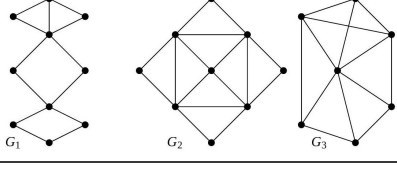
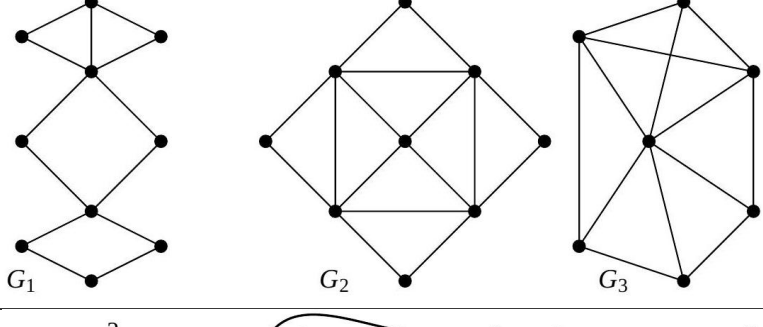
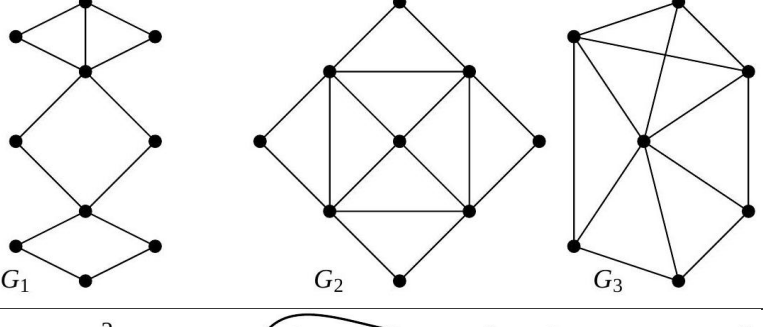
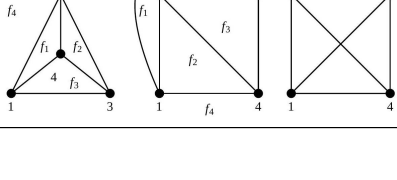
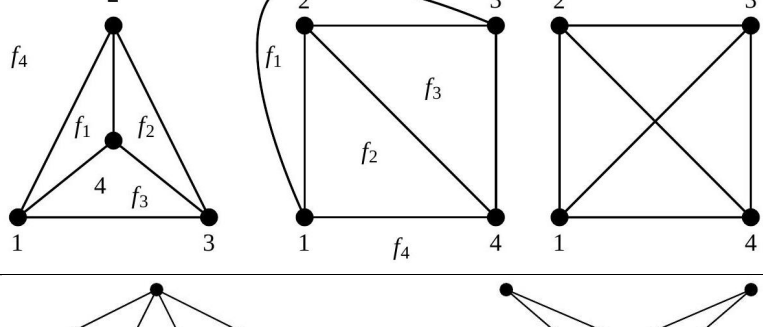
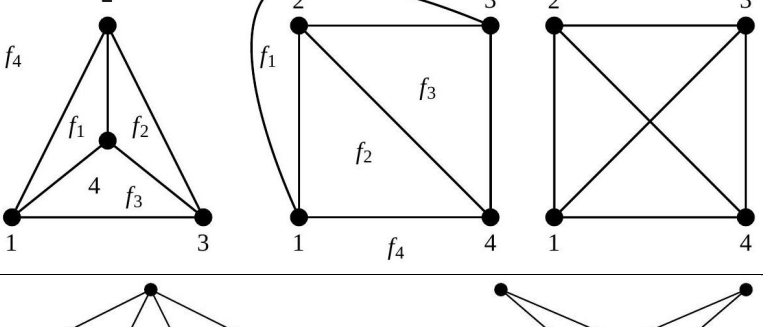
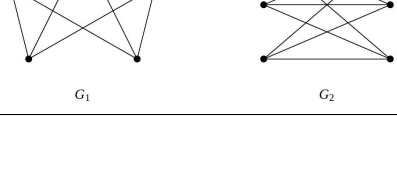
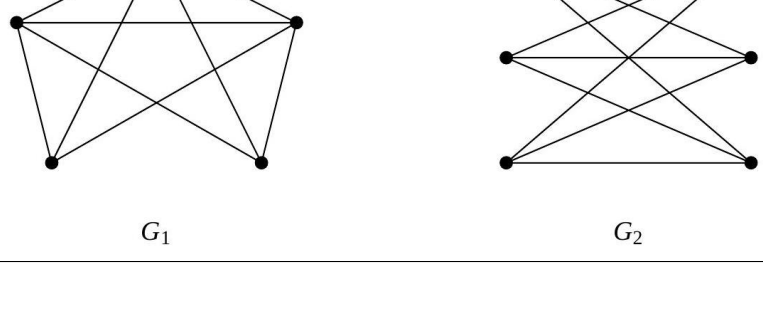
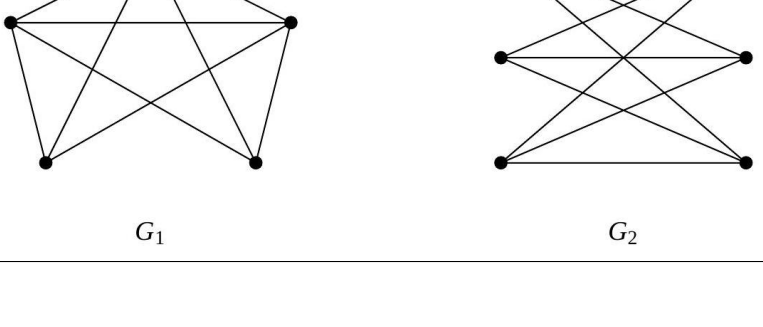
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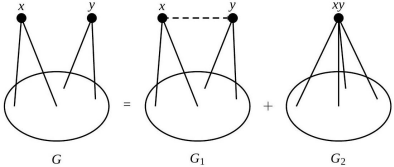
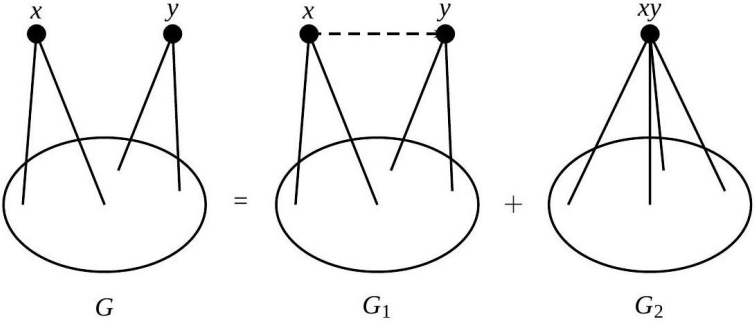
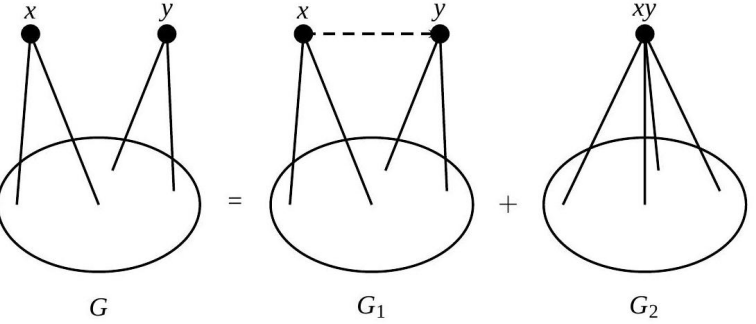
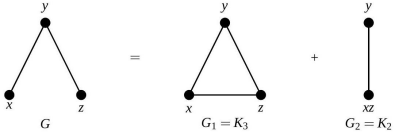
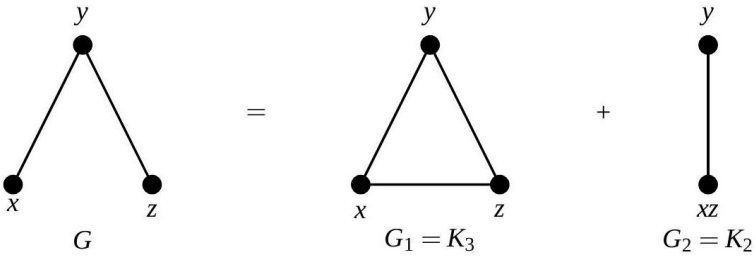
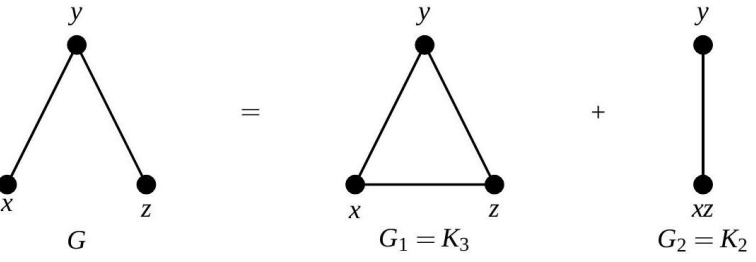
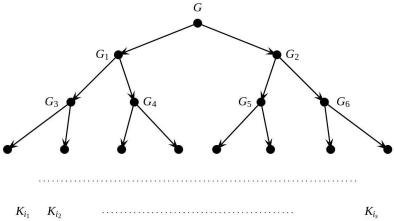
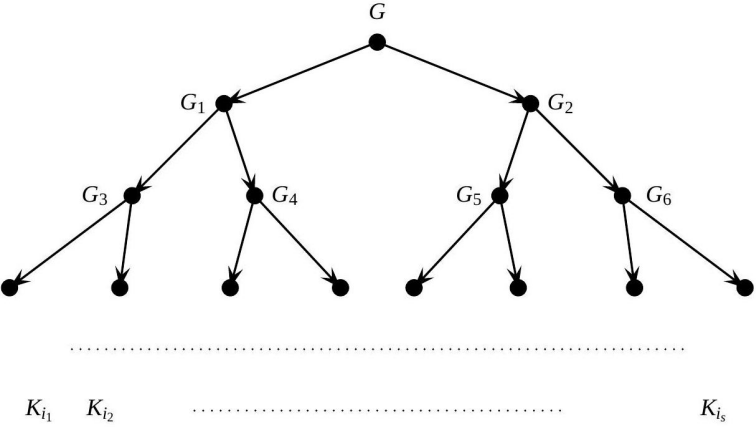
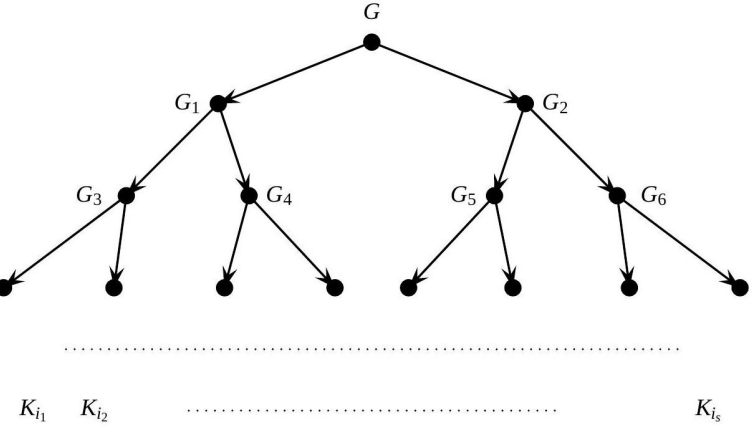
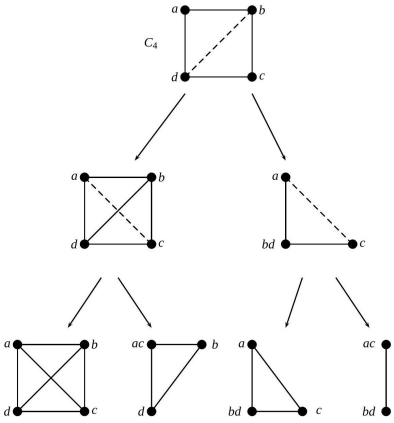
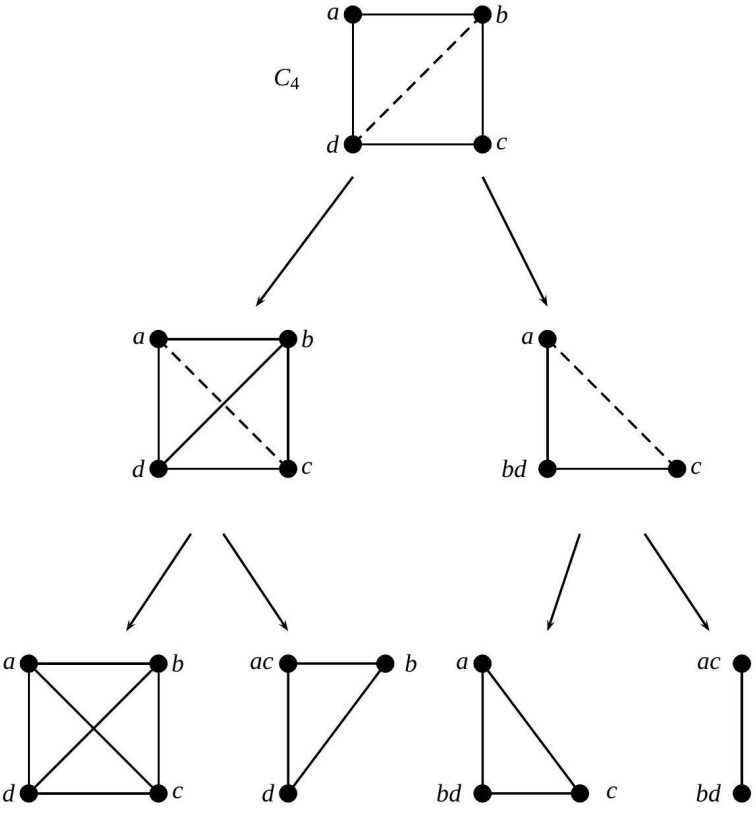
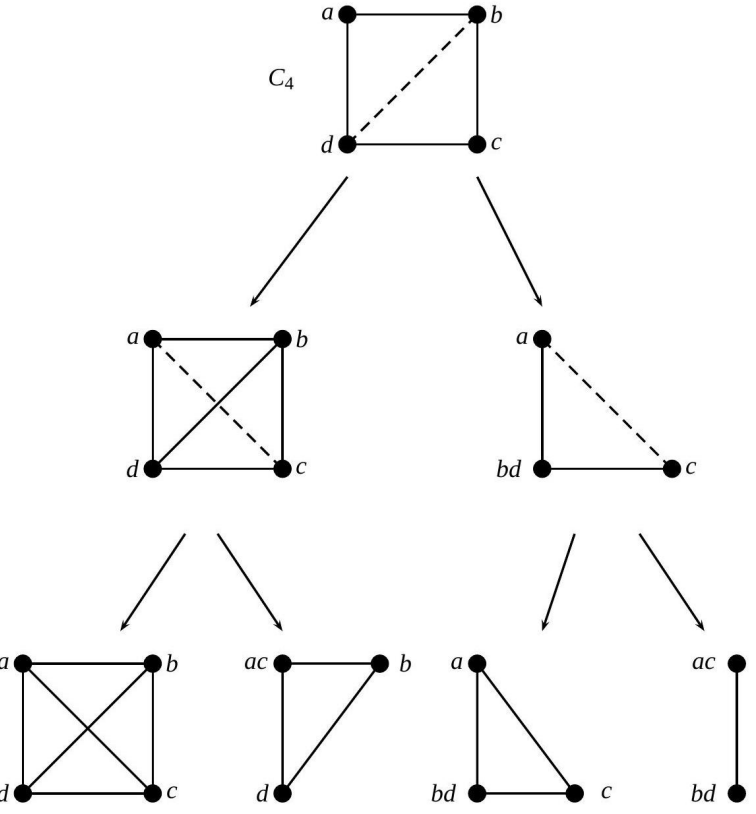
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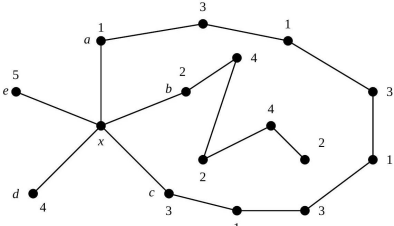
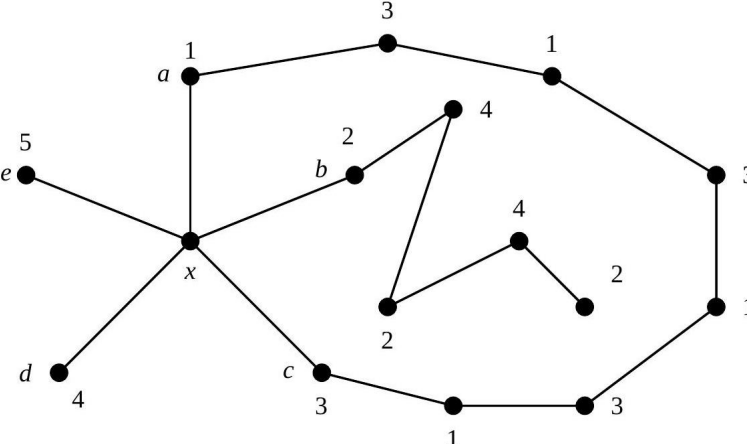
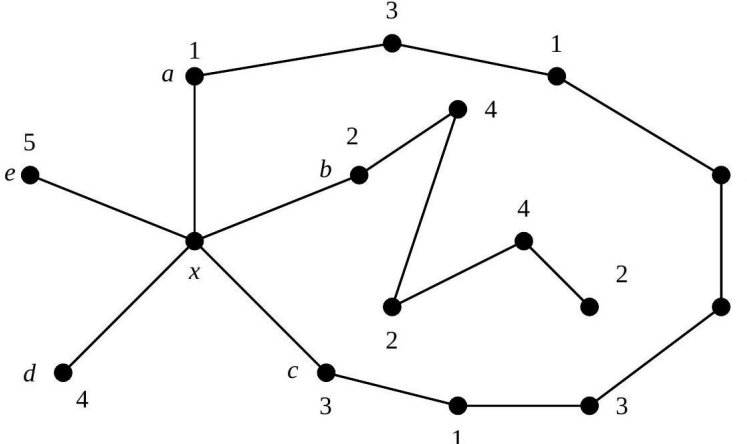
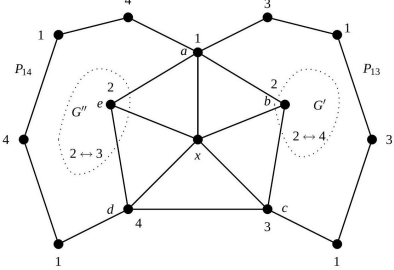
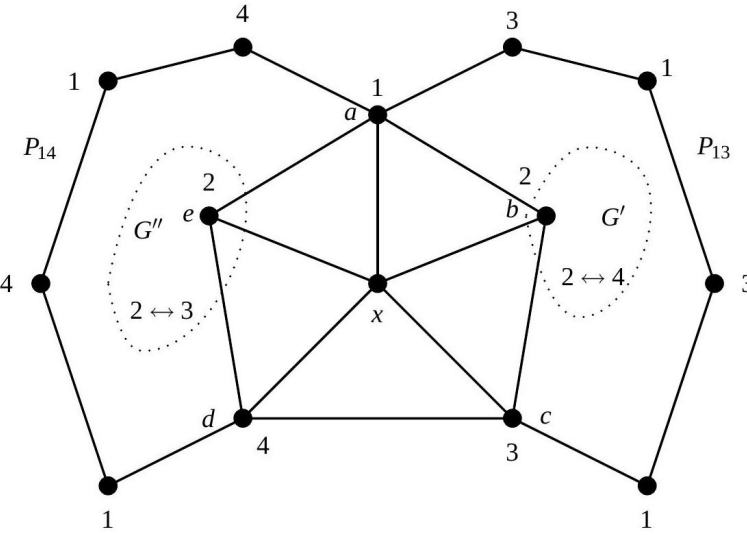
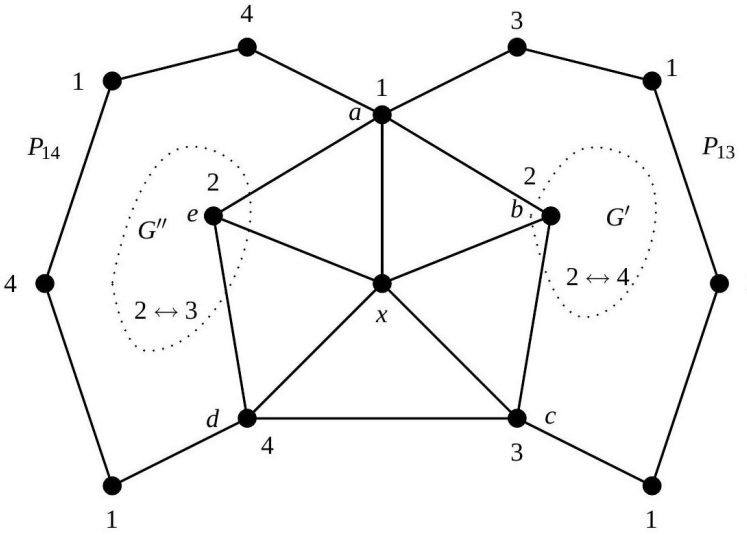
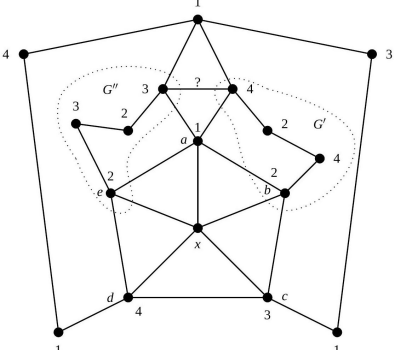
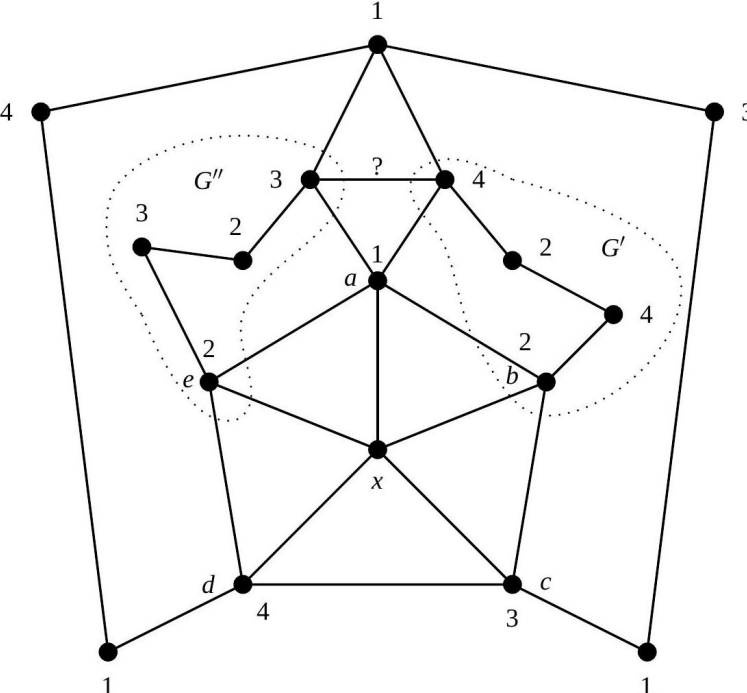
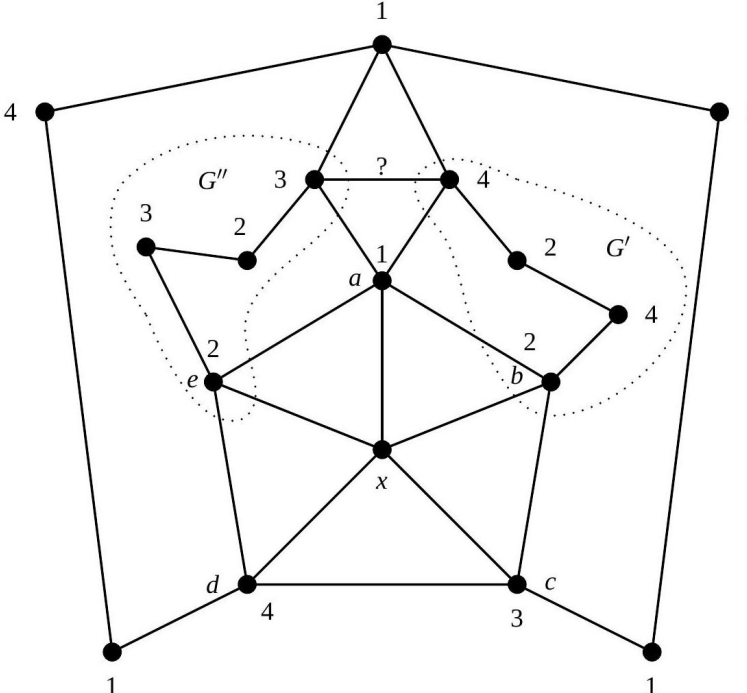
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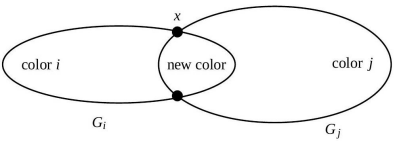
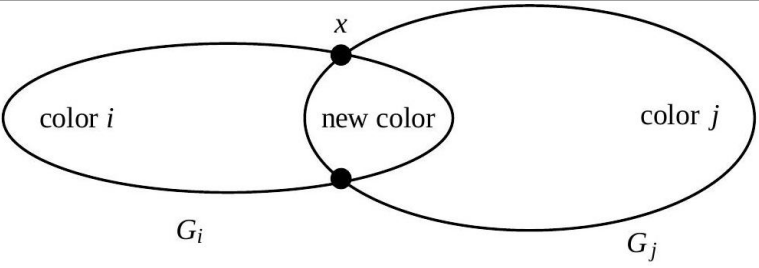
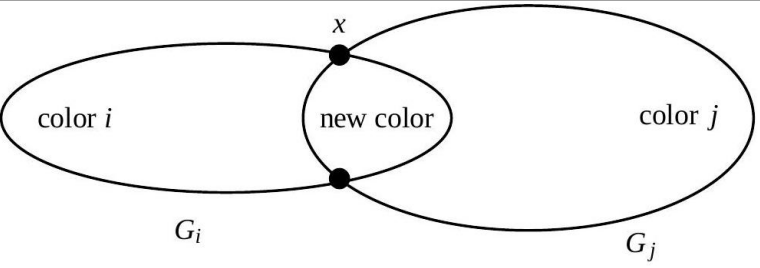
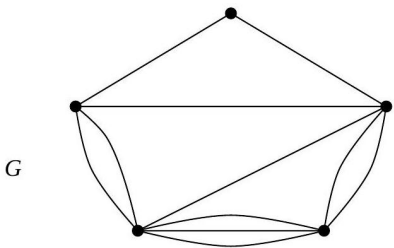
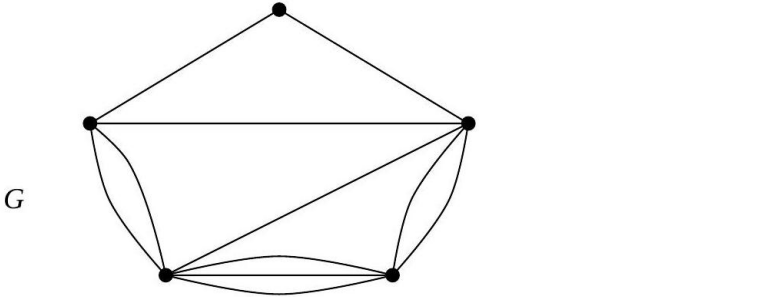
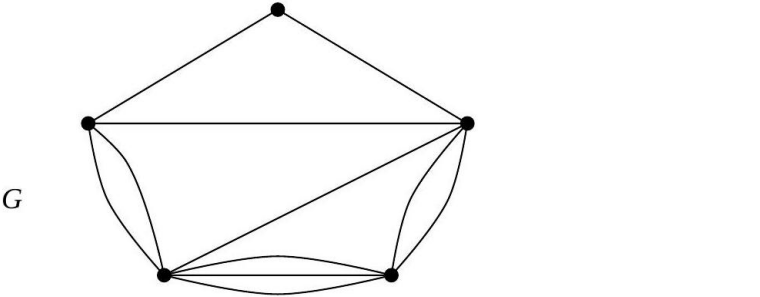
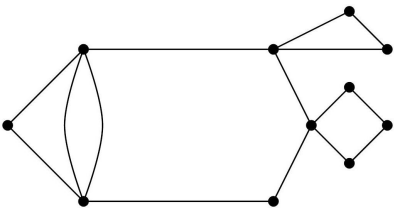
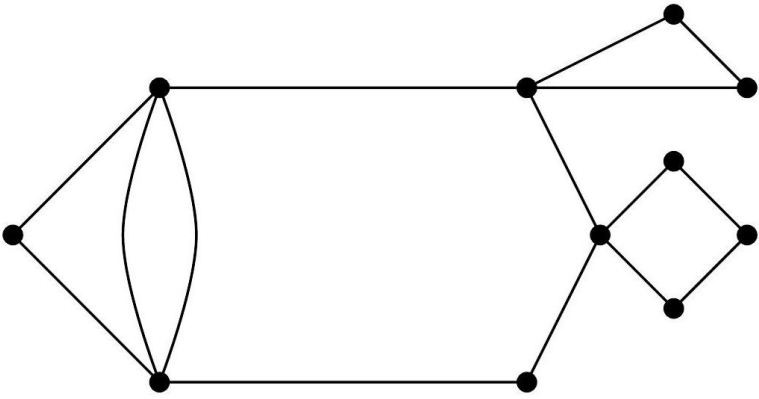
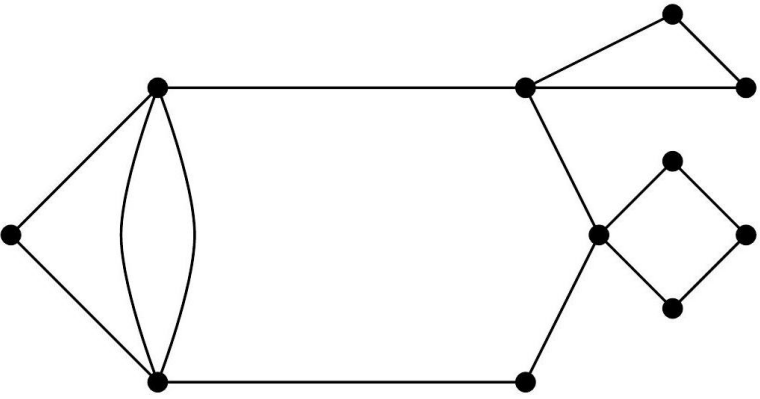
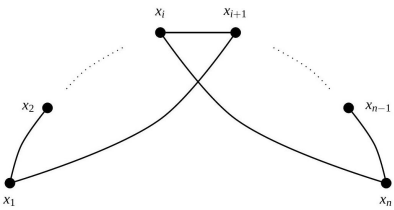
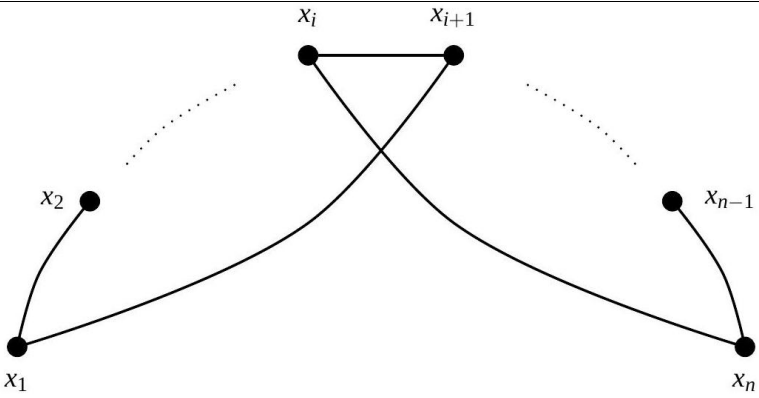
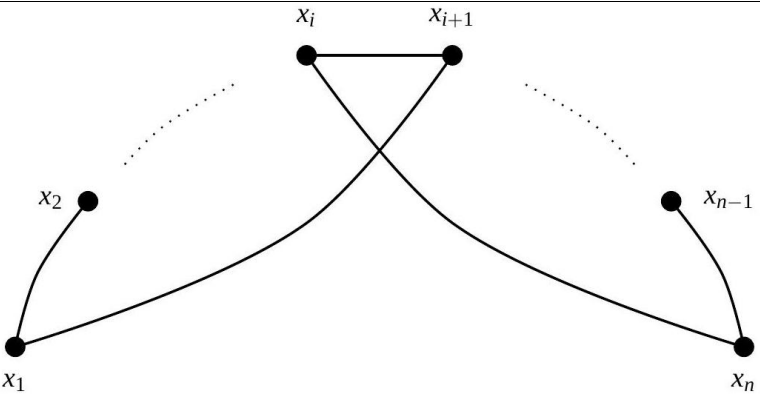
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voloshin-hypergraph_DIA_0110	105	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-105_233_1071_216_321.jpg			
voloshin-hypergraph_DIA_0111	107	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-107_790_1175_236_269.jpg	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$
voloshin-hypergraph_DIA_0112	108	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-108_378_1196_1315_285.jpg		 $G_1 = G_{X_1 \cup S}$ $G_2 = G_{X_2 \cup S}$	 $G_1 = G_{X_1 \cup S}$ $G_2 = G_{X_2 \cup S}$
voloshin-hypergraph_DIA_0113	109	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-109_557_1085_816_313.jpg			
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voloshin-hypergraph_DIA_0115	112	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-112_316_935_1473_362.jpg			

voloshin-hypergraph_DIA_0113	113	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-113_316_935_536_362.jpg			
voloshin-hypergraph_DIA_0115	115	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-115_782_999_917_383.jpg			
voloshin-hypergraph_DIA_0117	117	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-117_316_924_1372_370.jpg			
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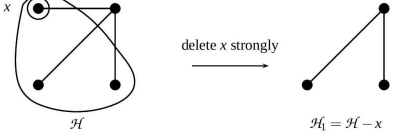
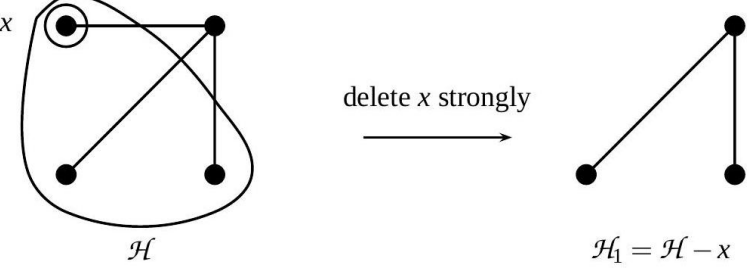
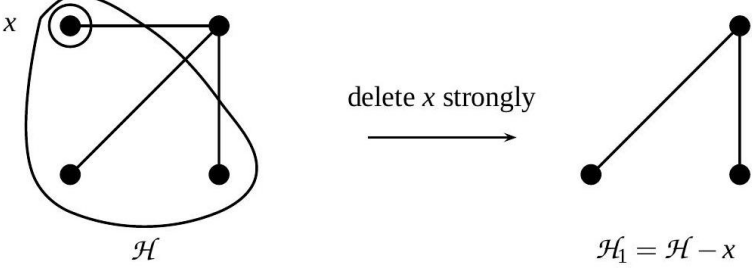
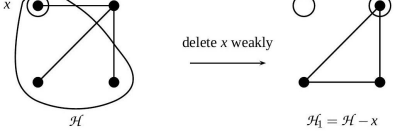
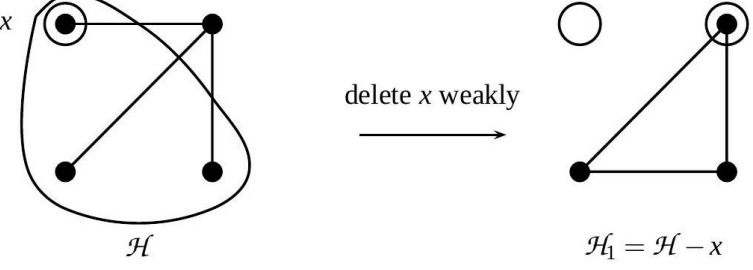
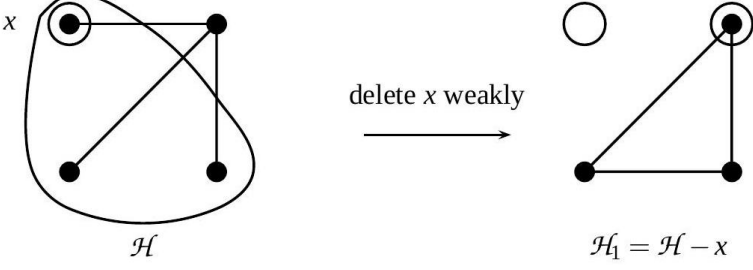
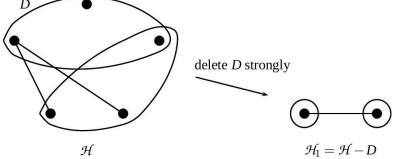
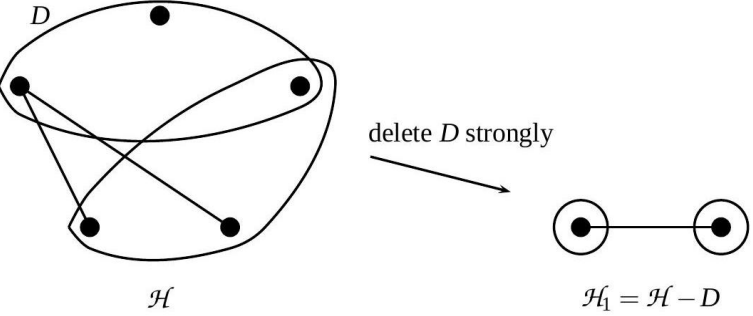
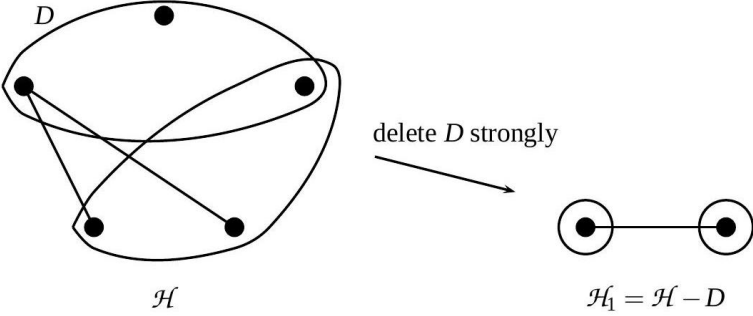
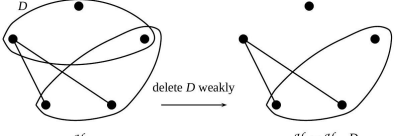
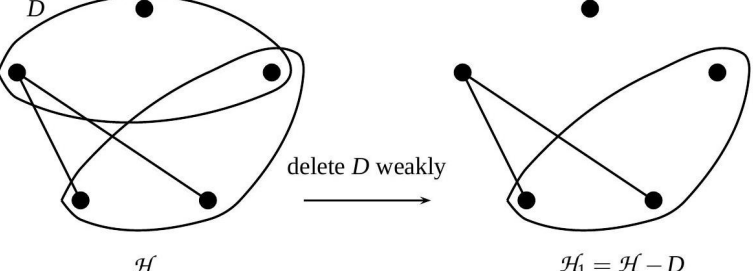
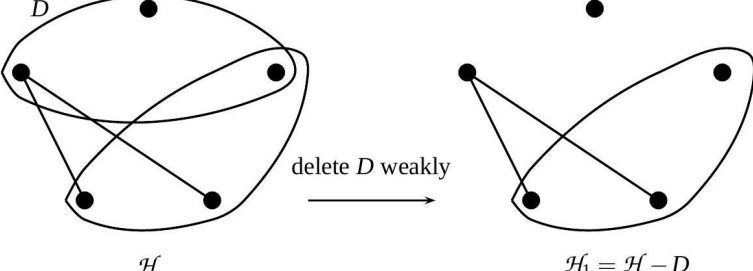
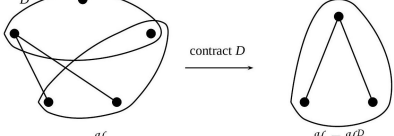
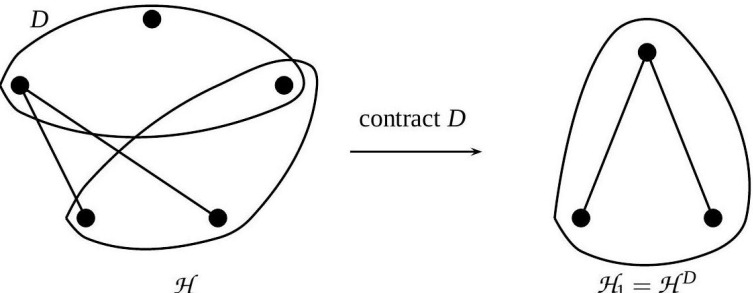
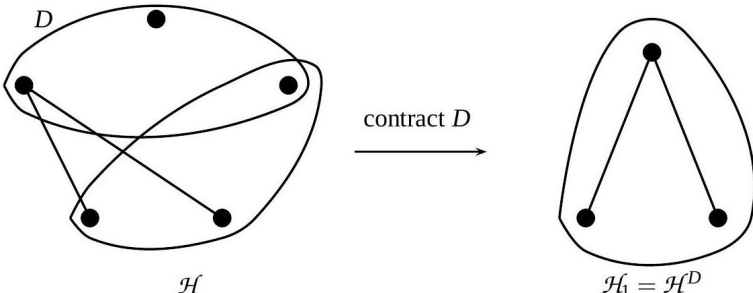
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voloshin-hypergraph_DIA_0182	122	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-122_1025_1116_676_324.jpg			

voloshin-hypergraph_DIA_0182	124	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-124_437_674_237_554.jpg			
voloshin-hypergraph_DIA_0192	125	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-125_311_930_621_417.jpg			
voloshin-hypergraph_DIA_0192	125	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-125_531_627_1362_570.jpg			
voloshin-hypergraph_DIA_0192	126	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-126_557_1116_909_318.jpg			

voloshin-hypergraph_DIA_0134	134	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-134_344_997_254_386.jpg			
voloshin-hypergraph_DIA_0136	136	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-136_393_626_1323_517.jpg			
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voloshin-hypergraph_DIA_0141	141	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-141_533_1033_1315_365.jpg			

voloshin-hypergraph_DIA_0146	146	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-146_1212_1015_249_316.jpg		
voloshin-hypergraph_DIA_0151	151	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-151_596_650_1043_554.jpg		
voloshin-hypergraph_DIA_0155	155	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-155_388_1074_243_409.jpg		

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voloshin-hypergraph_DIA_0158	158	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-158_381_1134_246_331.jpg		
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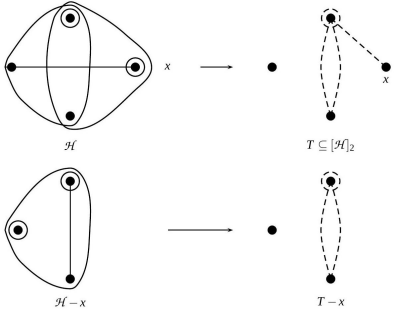
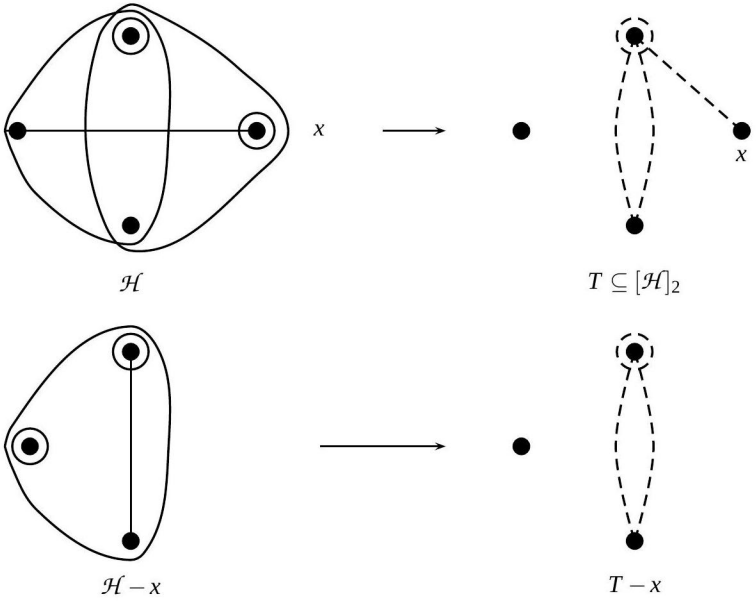
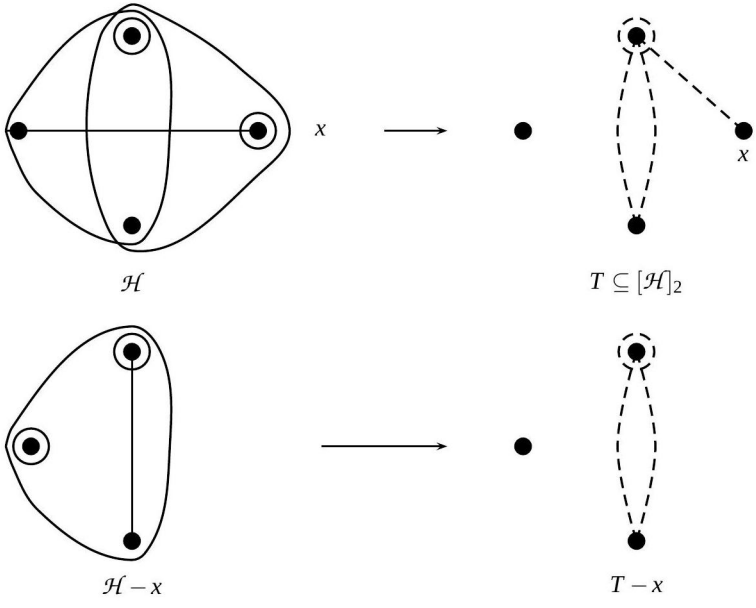
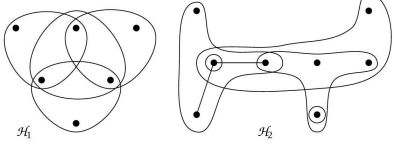
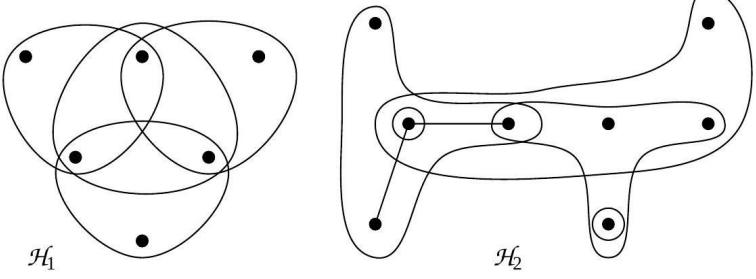
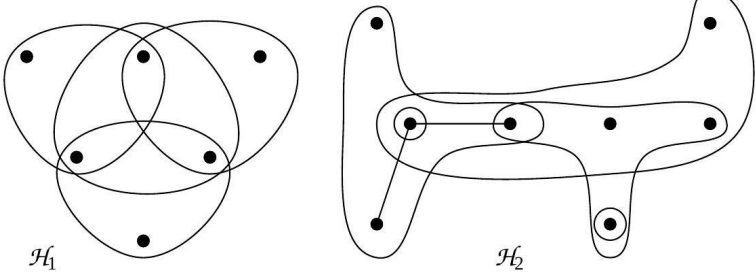
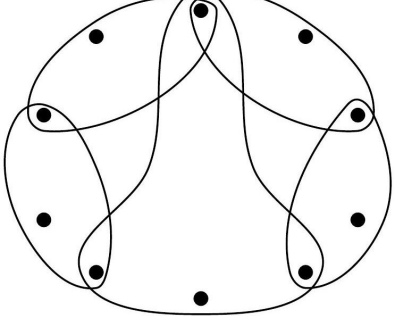
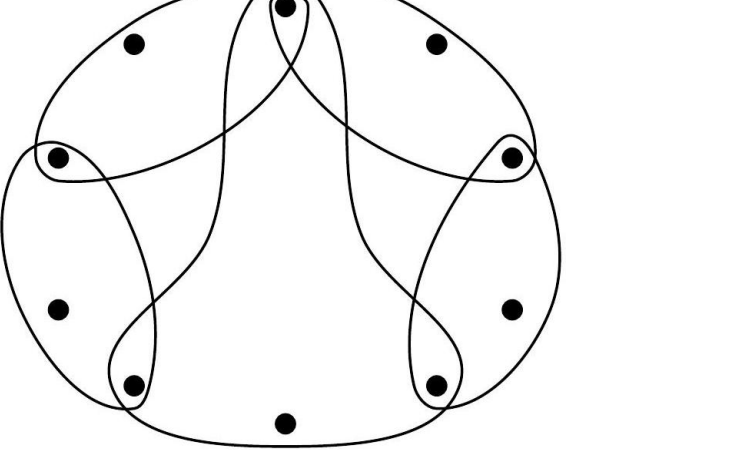
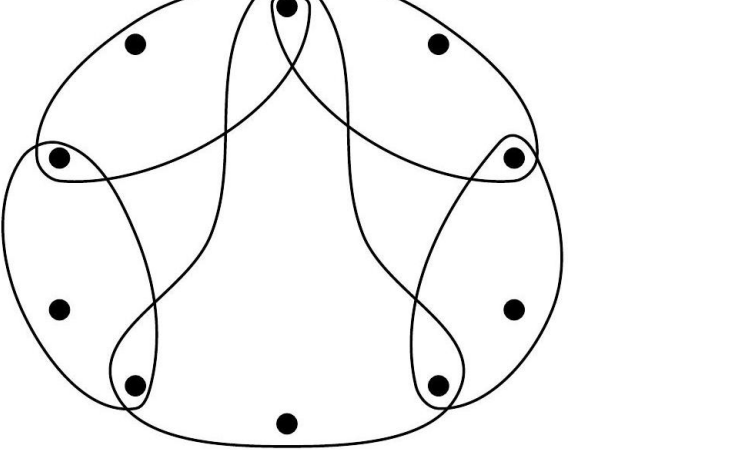
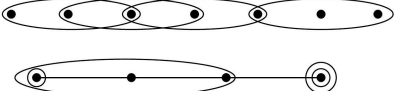
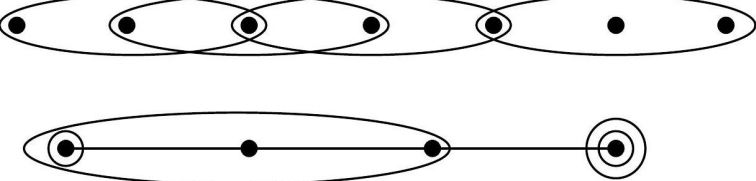
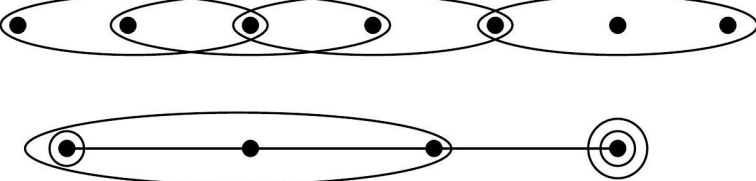
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voloshin-hypergraph_DIA_01164	164	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-164_451_1072_228_352.jpg			
voloshin-hypergraph_DIA_01164	164	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-164_451_1181_1271_303.jpg			
voloshin-hypergraph_DIA_01165	165	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-165_451_1139_1036_300.jpg			

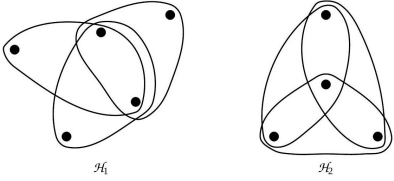
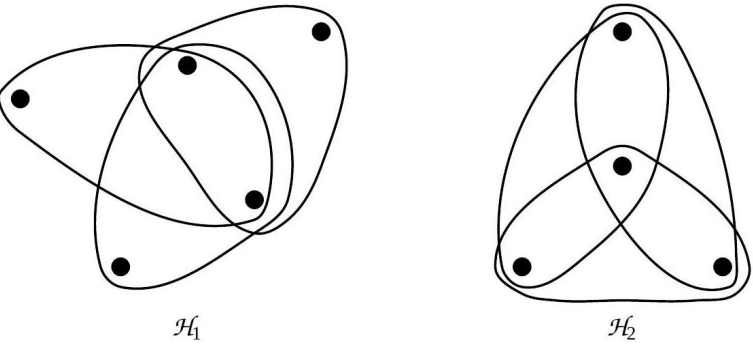
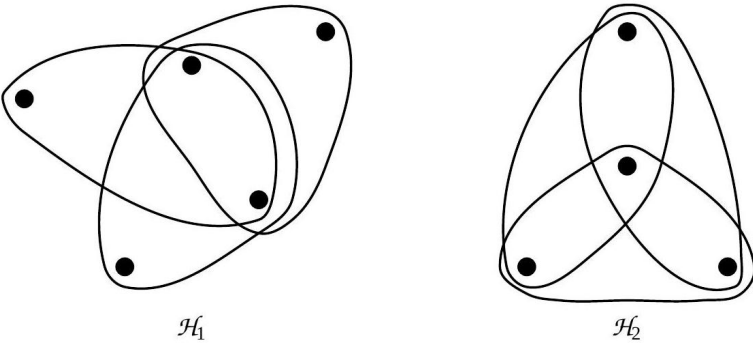
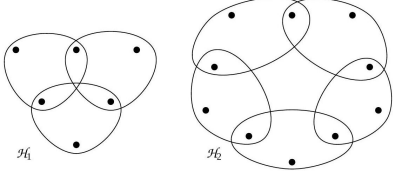
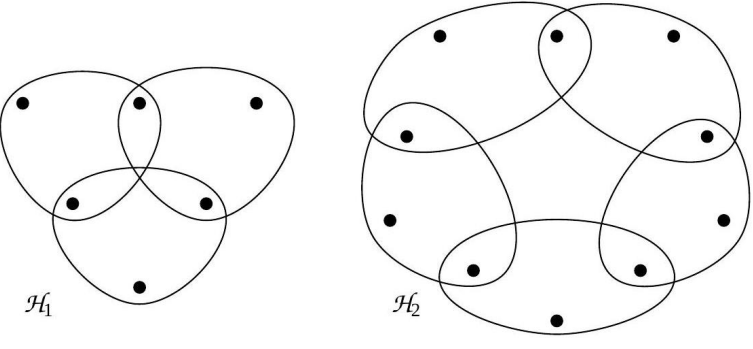
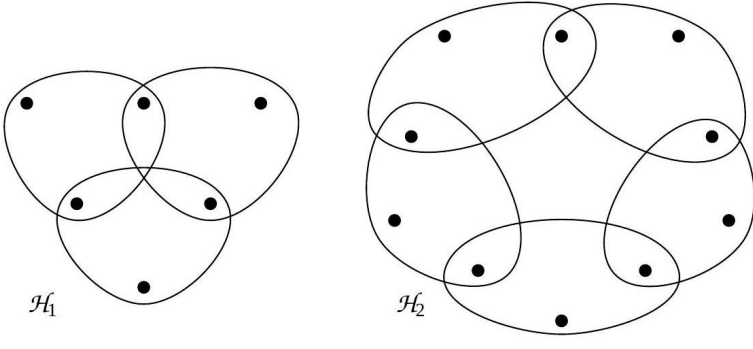
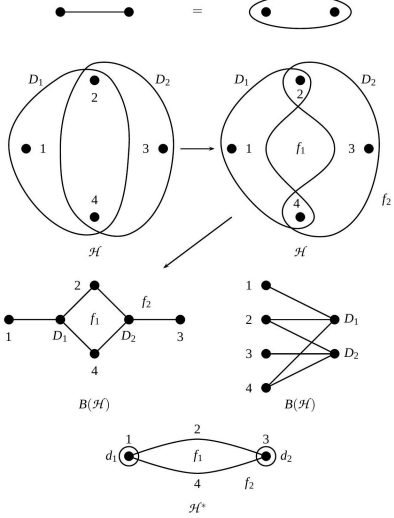
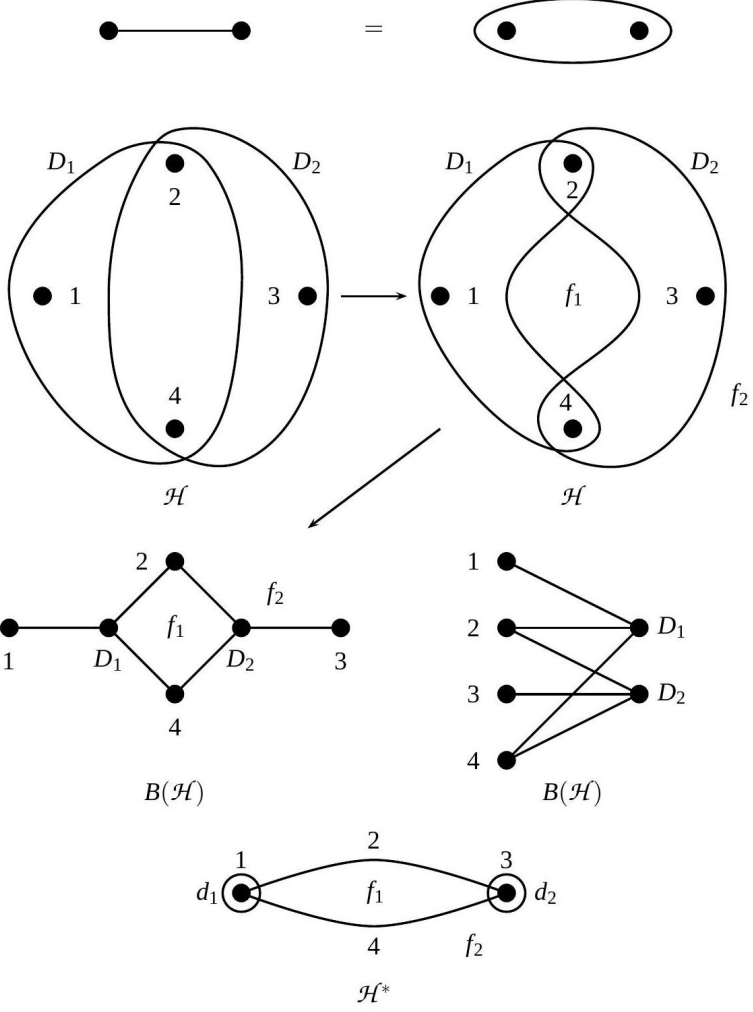
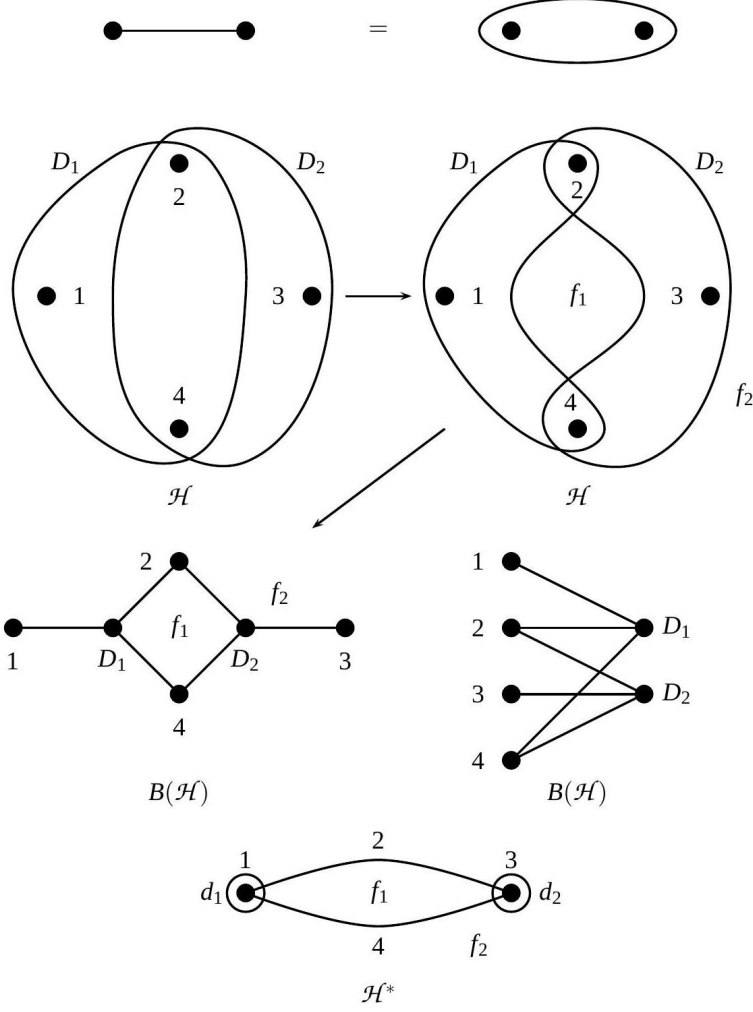
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voloshin-hypergraph_DIA_01168	168	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-168_414_1168_217_295.jpg		
voloshin-hypergraph_DIA_01171	171	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-171_466_1126_590_298.jpg		
voloshin-hypergraph_DIA_01172	172	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-172_469_1142_217_282.jpg		

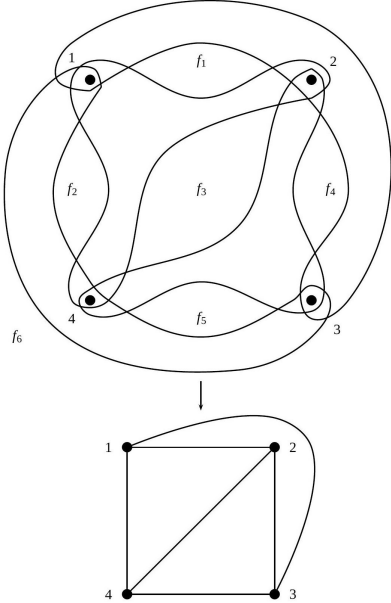
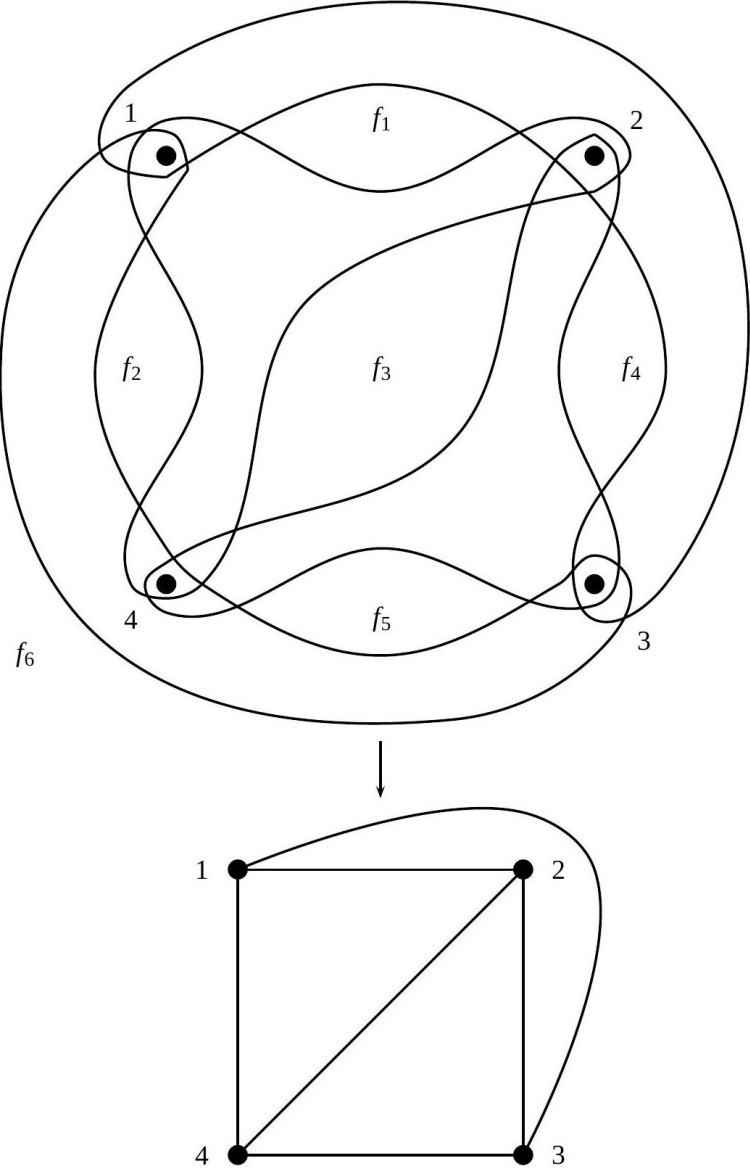
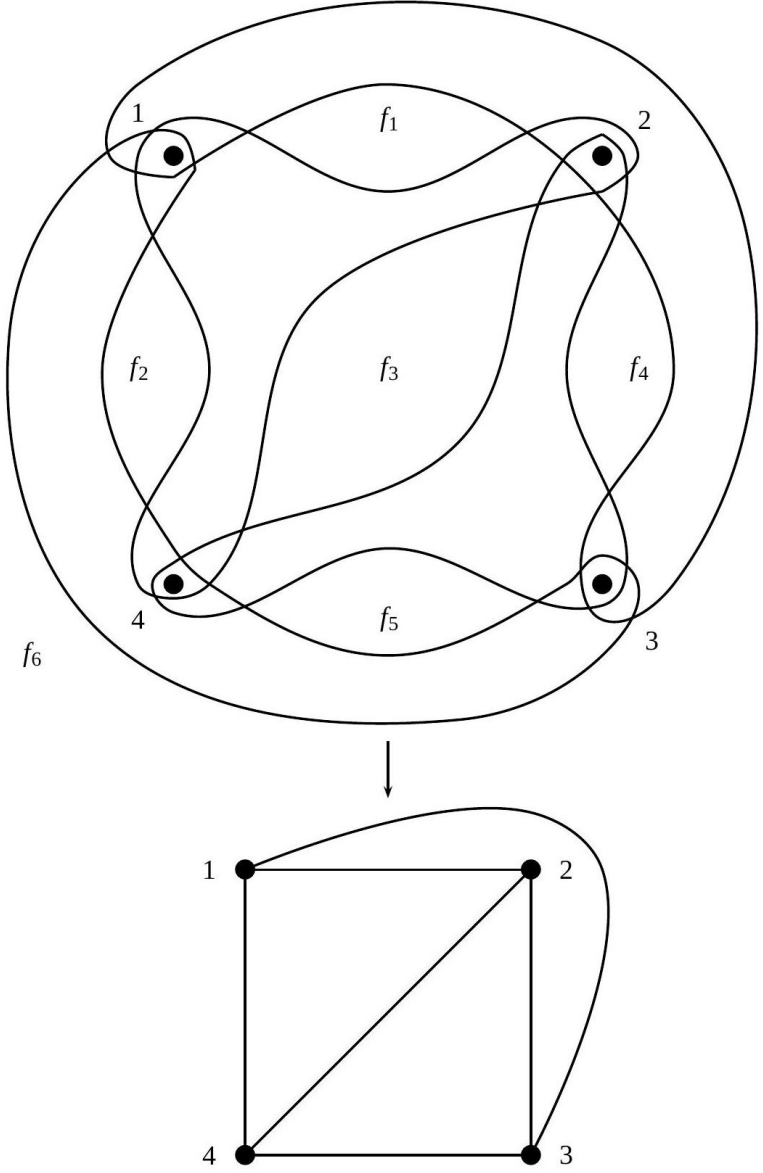
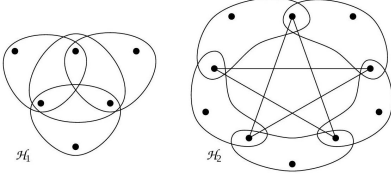
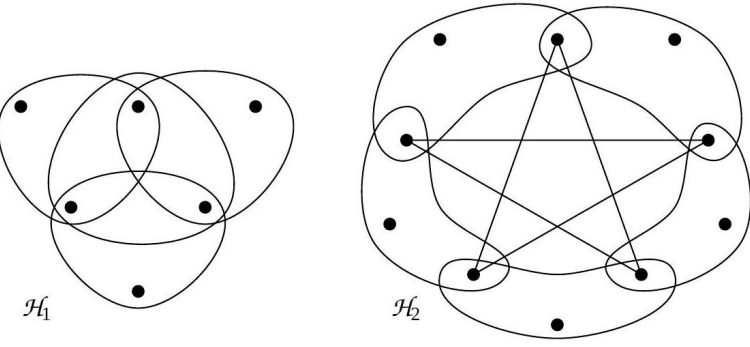
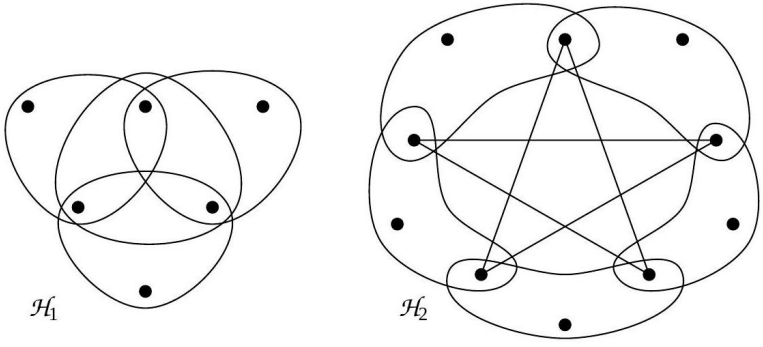
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voloshin-hypergraph_DIA_0178	178	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-178_746_1245_256_267.jpg	T C_4	T C_4
voloshin-hypergraph_DIA_0179	179	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-179_502_1126_246_347.jpg	$\mathcal{H}_1$ $\mathcal{H}_2$	$\mathcal{H}_1$ $\mathcal{H}_2$
voloshin-hypergraph_DIA_0180	180	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-180_406_1139_388_274.jpg	$\mathcal{H}$ $\mathcal{H}_0$	$\mathcal{H}$ $\mathcal{H}_0$

voloshin-hypergraph_DIA_0181	181	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-181_406_1033_207_381.jpg			
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voloshin-hypergraph_DIA_0182	182	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-182_404_1240_225_267.jpg			
voloshin-hypergraph_DIA_0183	183	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-183_269_1186_841_272.jpg			
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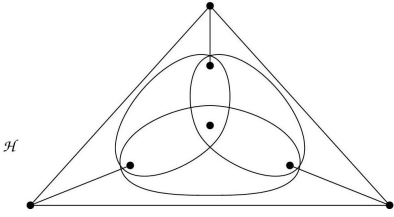
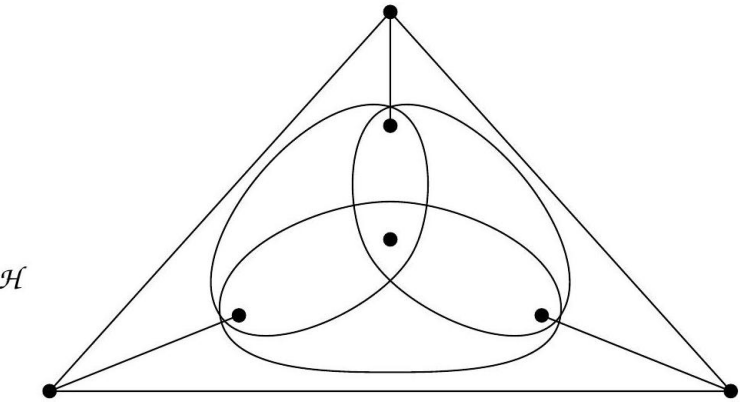
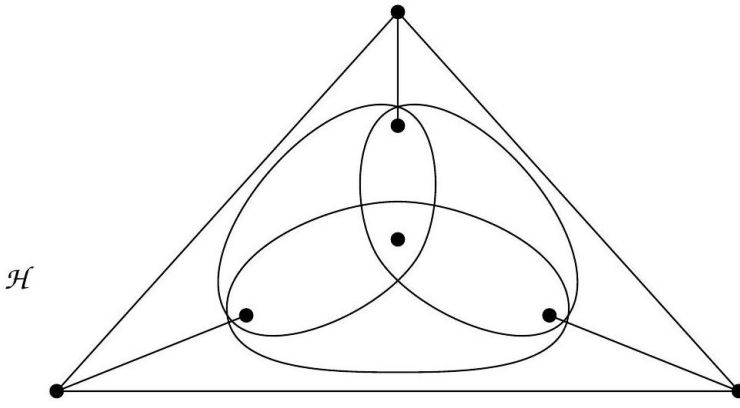
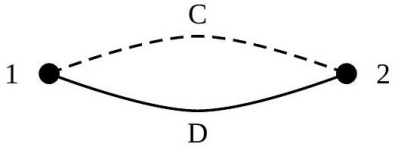
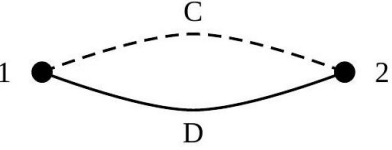
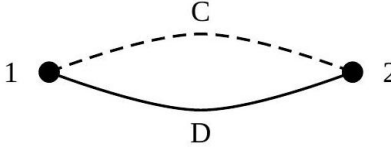
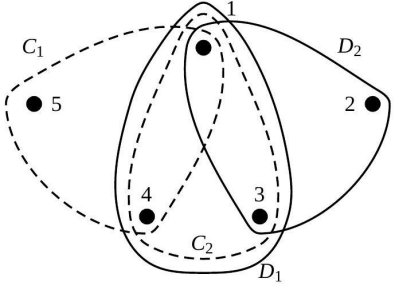
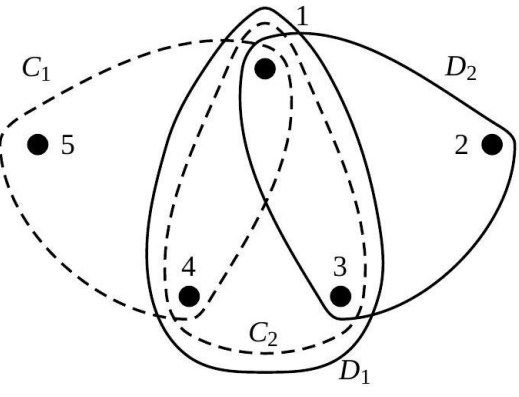
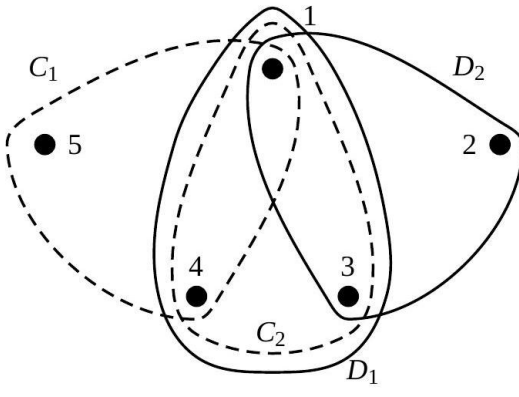
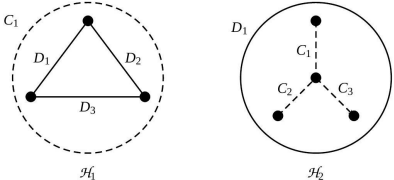
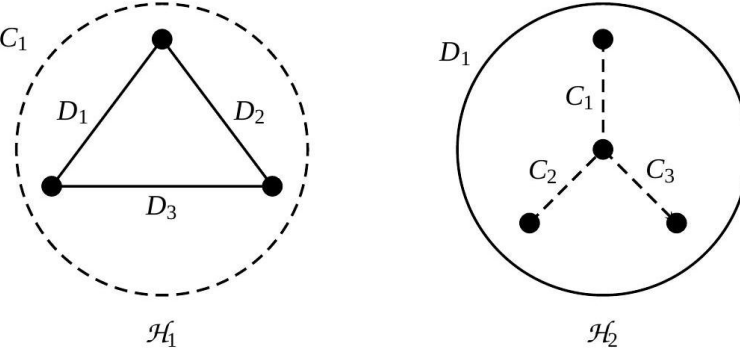
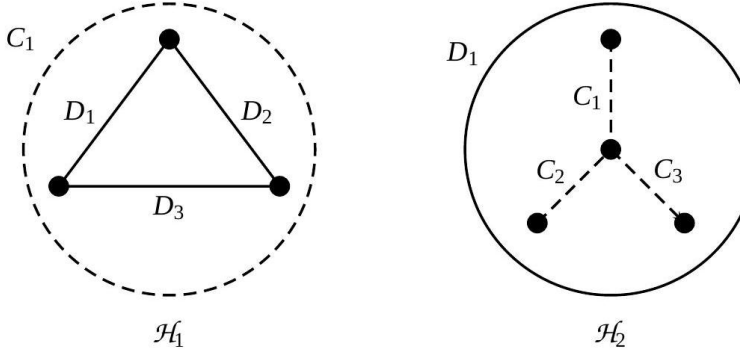
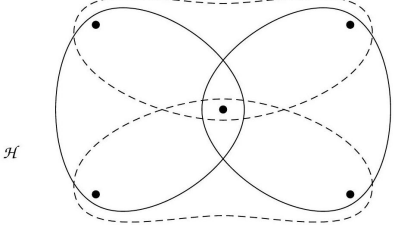
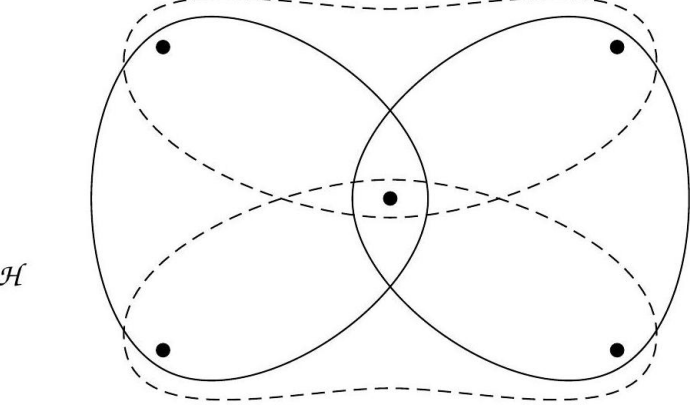
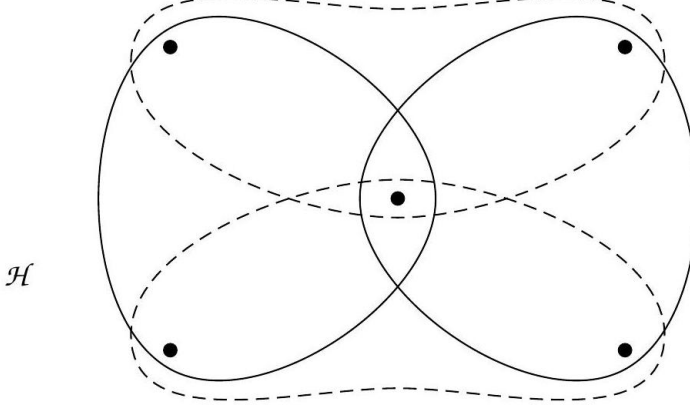
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voloshin-hypergraph_DIA_01189	189	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-189_479_1113_751_342.jpg			
voloshin-hypergraph_DIA_01191	191	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-191_927_1227_197_269.jpg			

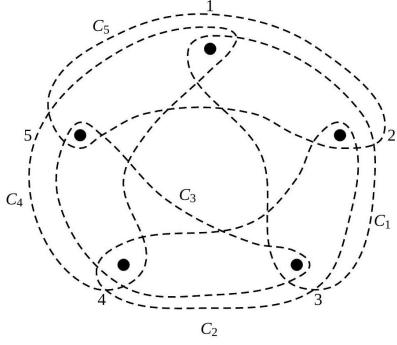
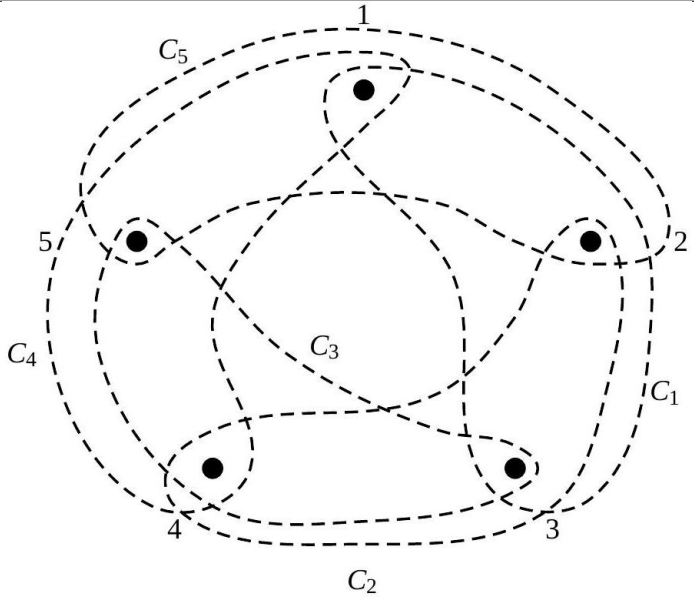
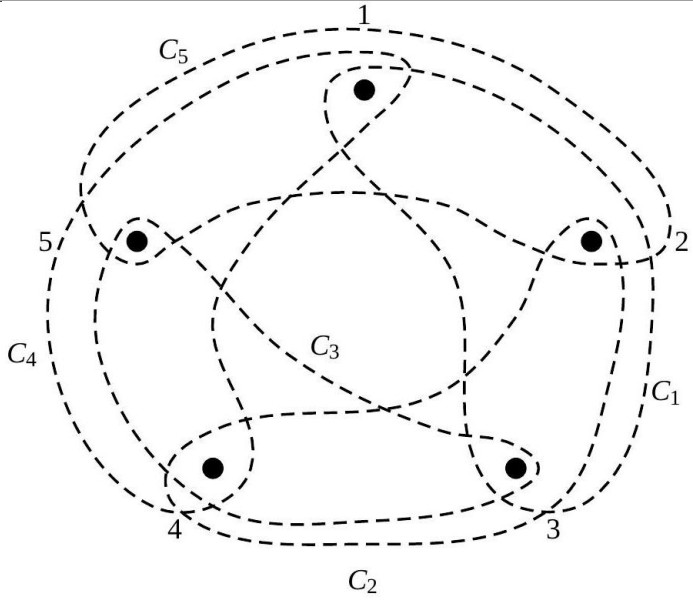
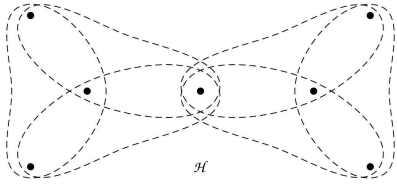
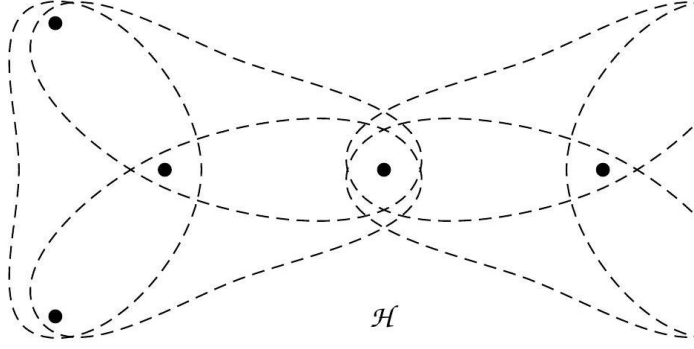
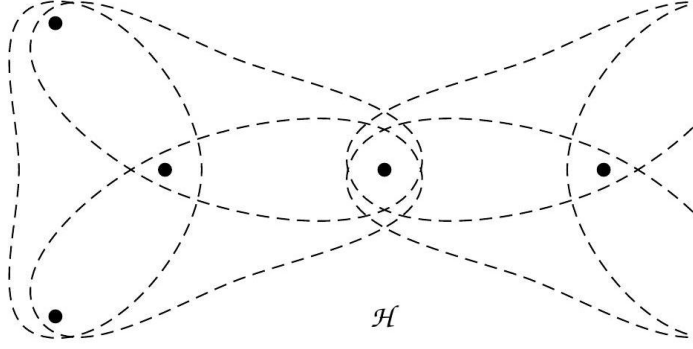
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voloshin-hypergraph_DIA_0139	196	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-196_432_1129_792_321.jpg			
voloshin-hypergraph_DIA_0139	198	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-198_624_743_212_505.jpg			
voloshin-hypergraph_DIA_0139	199	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-199_316_1230_1465_277.jpg			

voloshin-hypergraph_DIA_0200	200	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-200_507_1124_981_300.jpg			
voloshin-hypergraph_DIA_0202	202	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-202_507_1126_948_321.jpg			
voloshin-hypergraph_DIA_0203	203	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-203_1515_1134_562_316.jpg			

voloshin-hypergraph_DIA_0230	206	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-206_1623_1054_236_347.jpg			
voloshin-hypergraph_DIA_0230	207	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-207_507_1121_422_324.jpg			

voloshin-hypergraph_DIA_0233	210	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-210_598_709_259_531.jpg			
voloshin-hypergraph_DIA_0241	213	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-213_458_924_225_417.jpg			
voloshin-hypergraph_DIA_0244	216	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-216_1323_1230_194_267.jpg			

voloshin-hypergraph_DIA_0217	217	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-217_523_974_492_365.jpg			
voloshin-hypergraph_DIA_0218	218	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-218_194_523_1686_621.jpg			
voloshin-hypergraph_DIA_0221	221	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-221_507_689_217_528.jpg			
voloshin-hypergraph_DIA_0222	222	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-222_482_1023_199_360.jpg			
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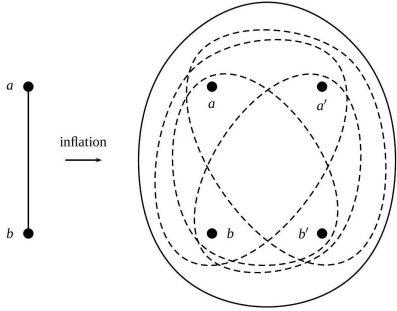
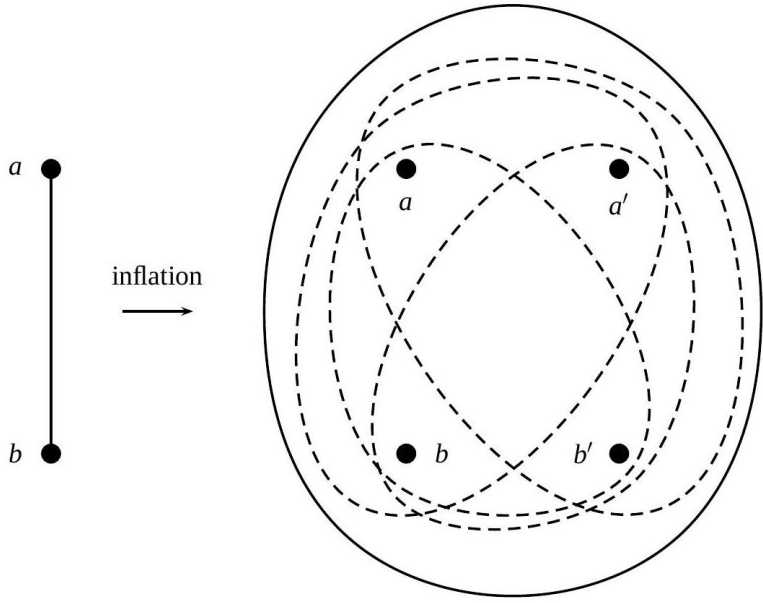
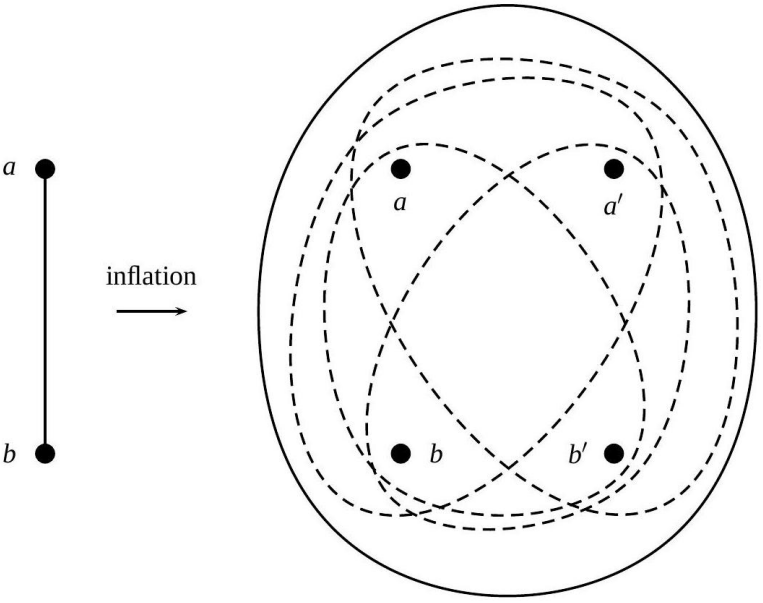
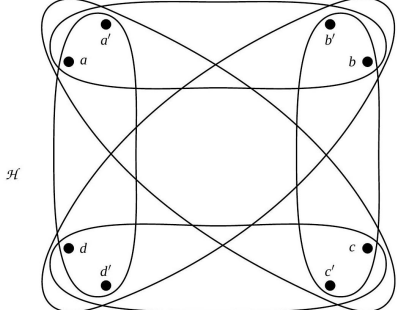
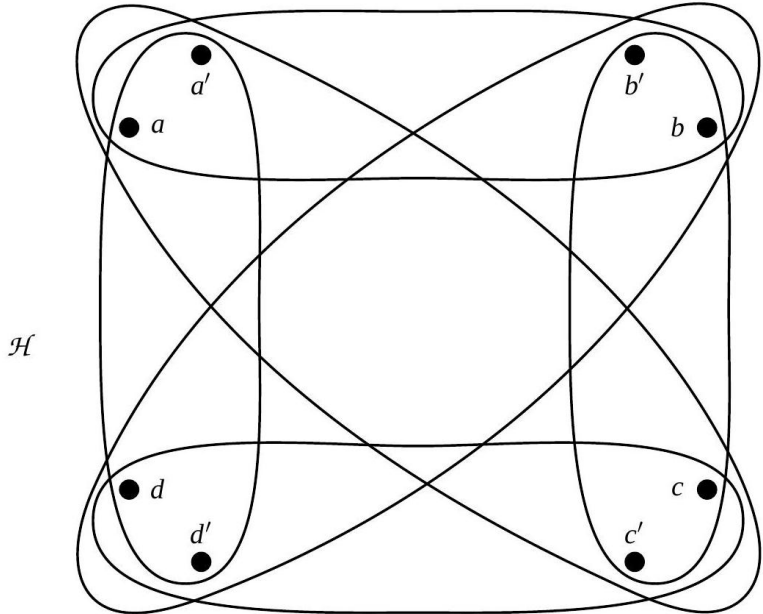
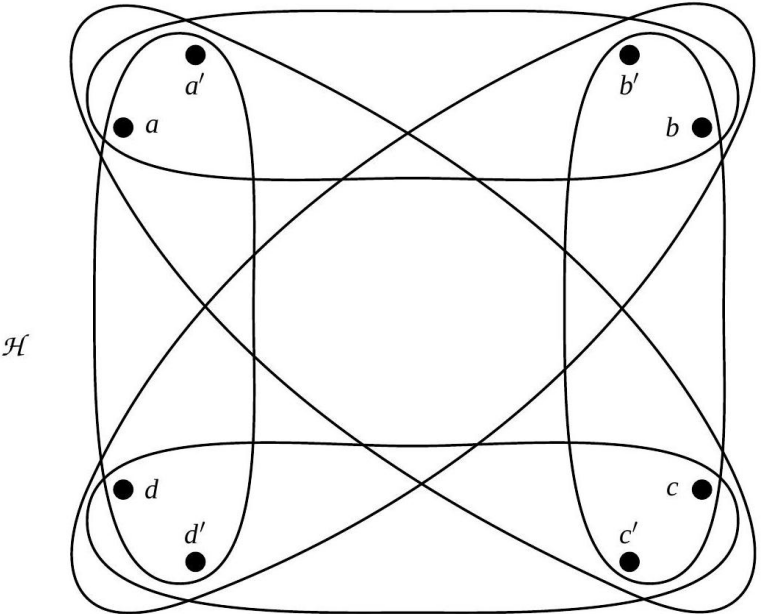
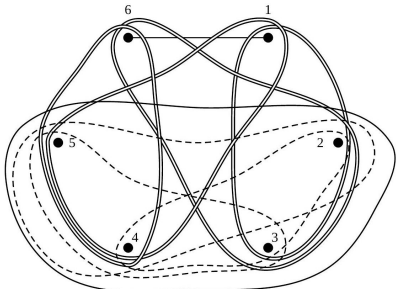
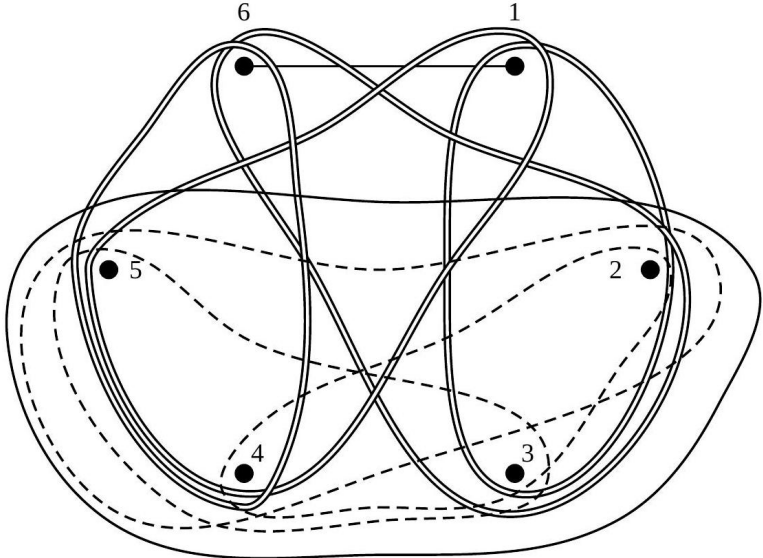
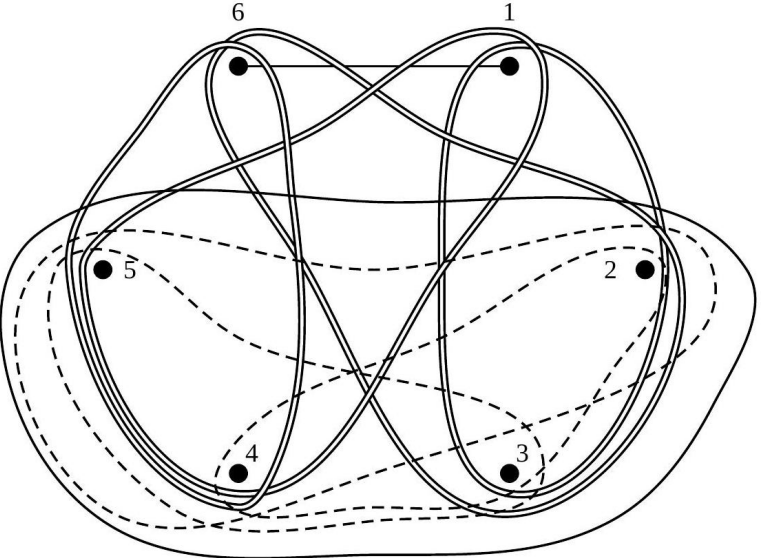
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	232 — (dup)				
voloshin-hypergraph_DIA_0234	234 — (dup)				

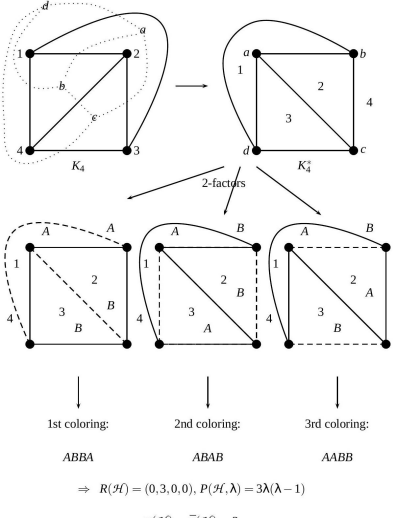
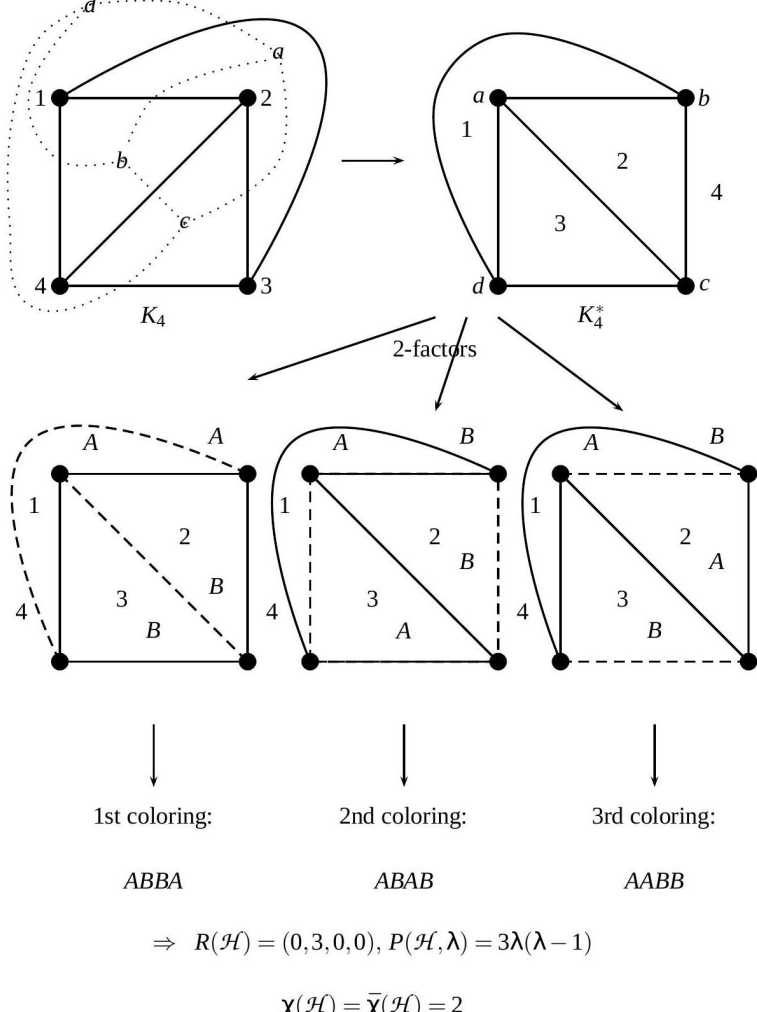
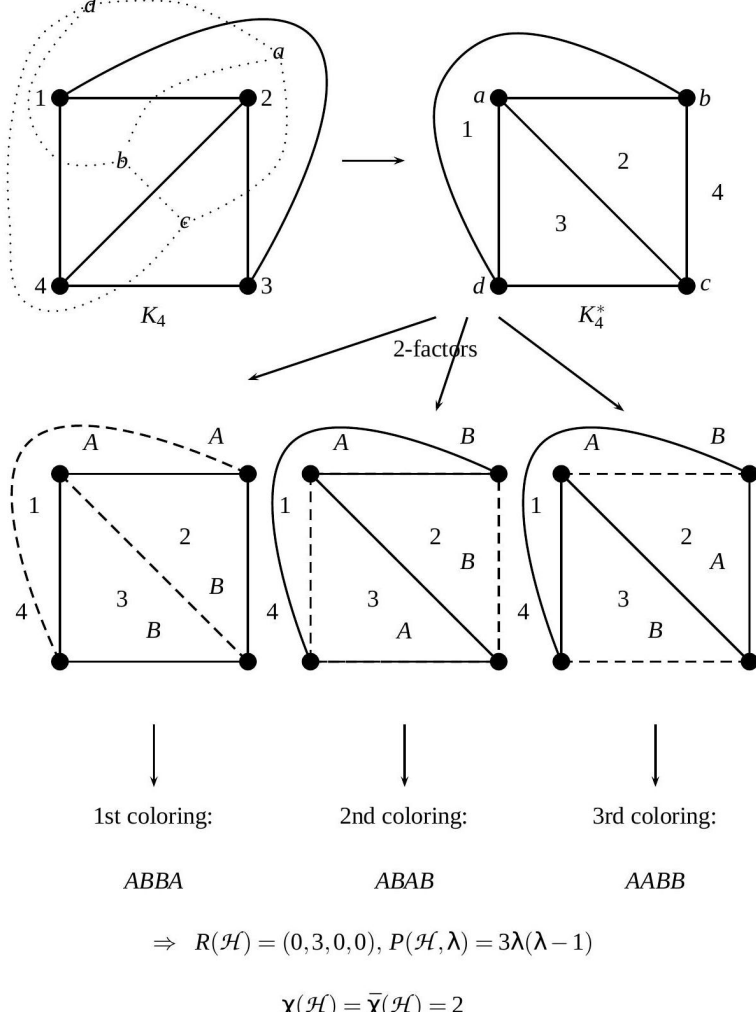
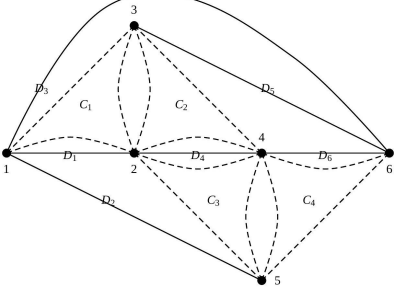
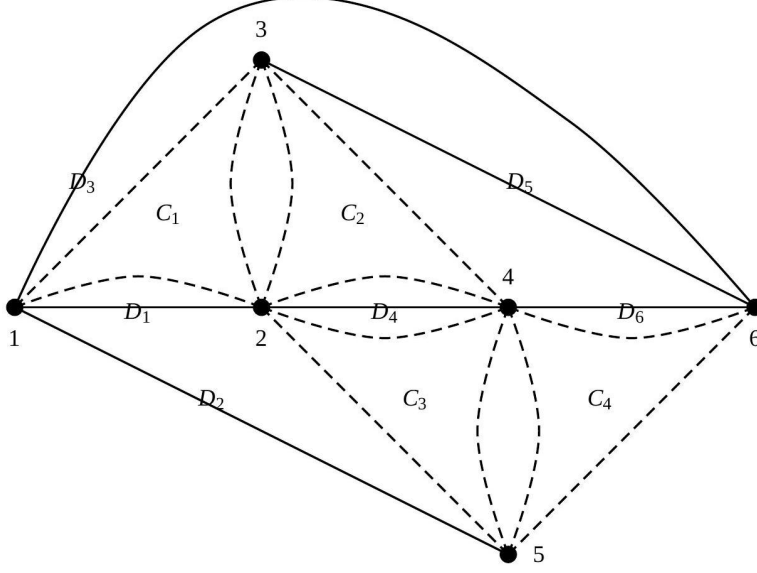
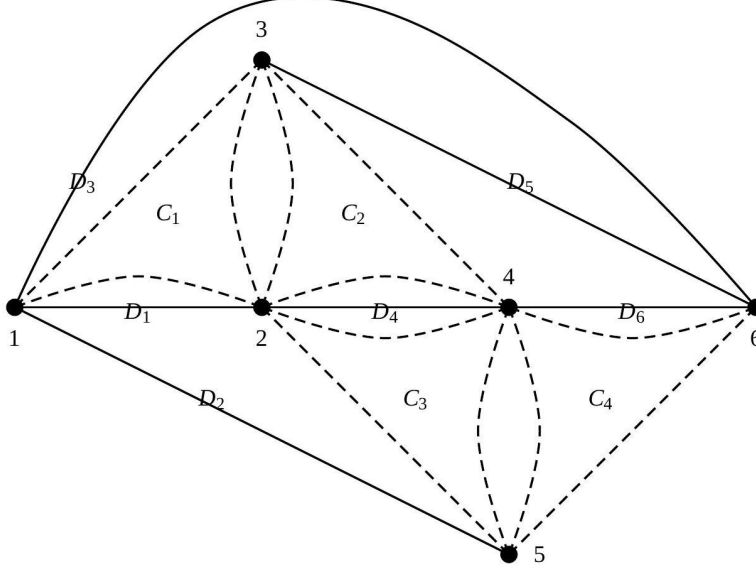
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voloshin-hypergraph_DIA_0253	237	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-237_466_466_269_647.jpg			
voloshin-hypergraph_DIA_0253	238	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-238_893_1121_241_321.jpg			
voloshin-hypergraph_DIA_0253	239	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-239_362_368_1828_699.jpg			

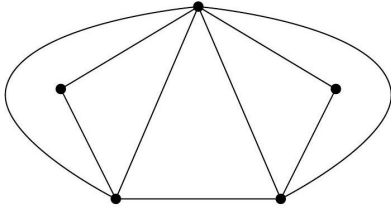
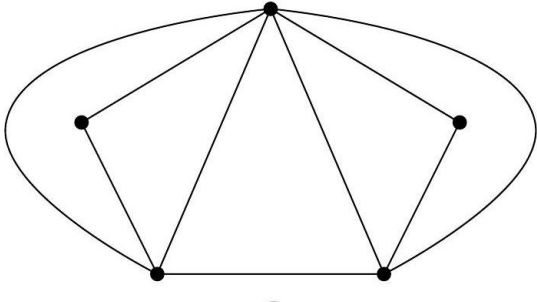
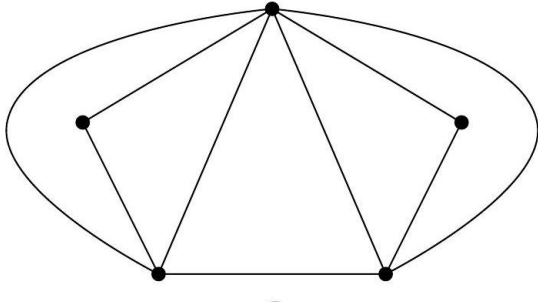
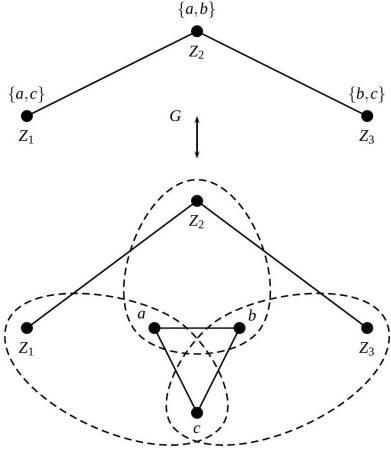
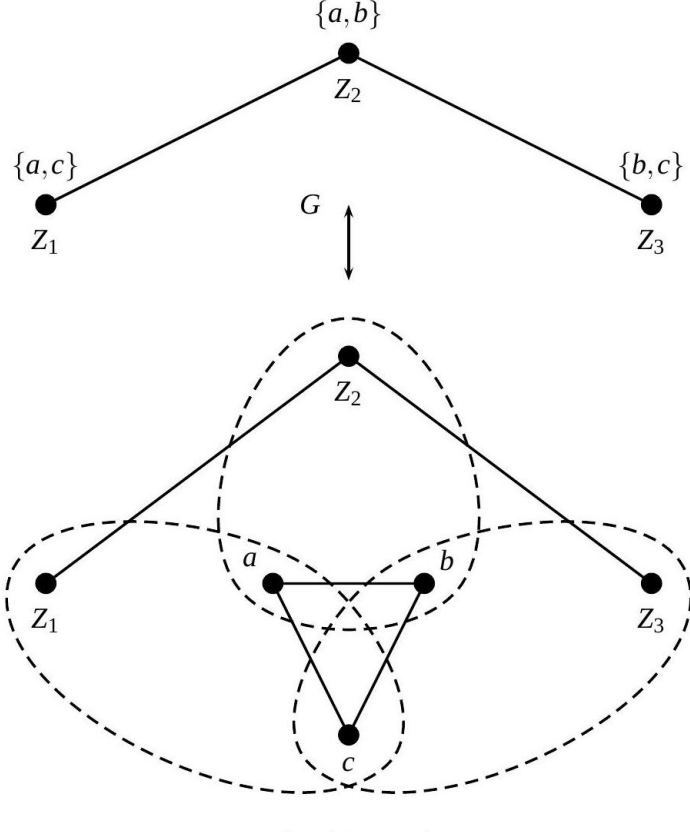
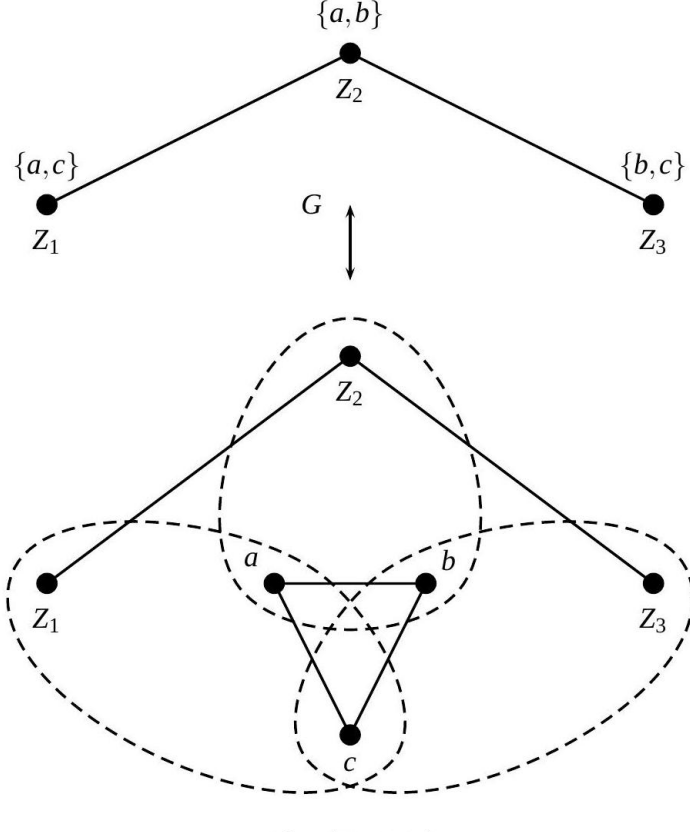
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voloshin-hypergraph_DIA_0254	242	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-242_518_836_1282_414.jpg			
voloshin-hypergraph_DIA_0254	244	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-244_386_1139_919_280.jpg			
voloshin-hypergraph_DIA_0255	250	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-250_606_1121_205_329.jpg			
voloshin-hypergraph_DIA_0255	251	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-251_492_1170_1129_311.jpg			

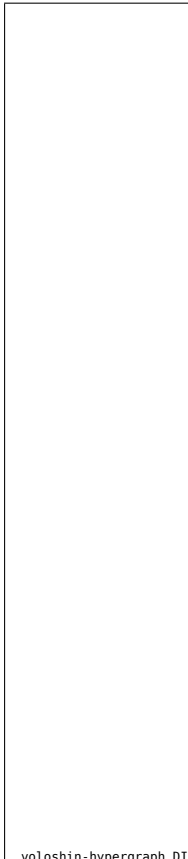
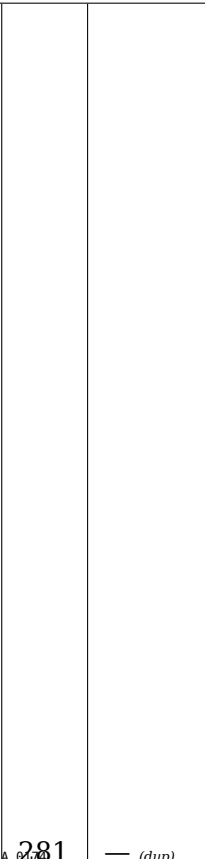
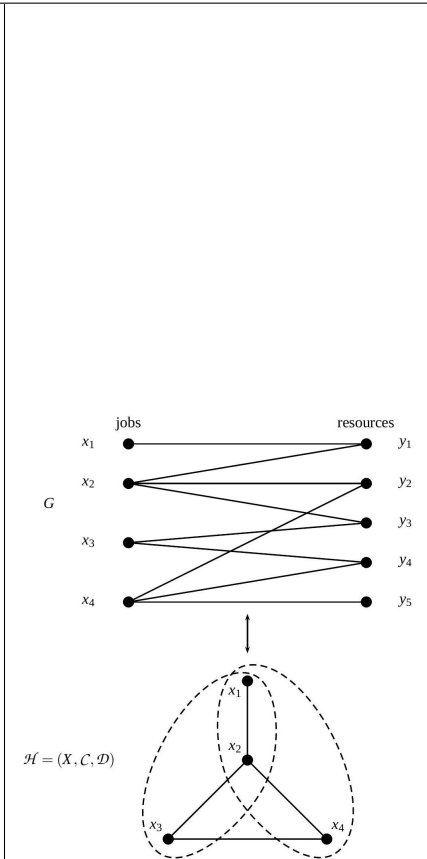
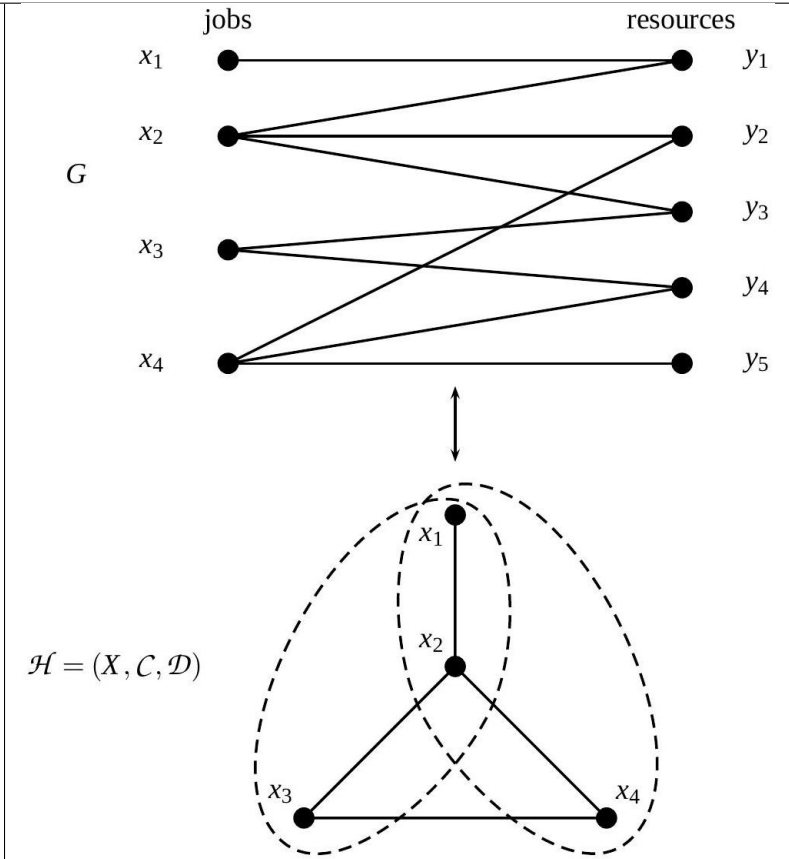
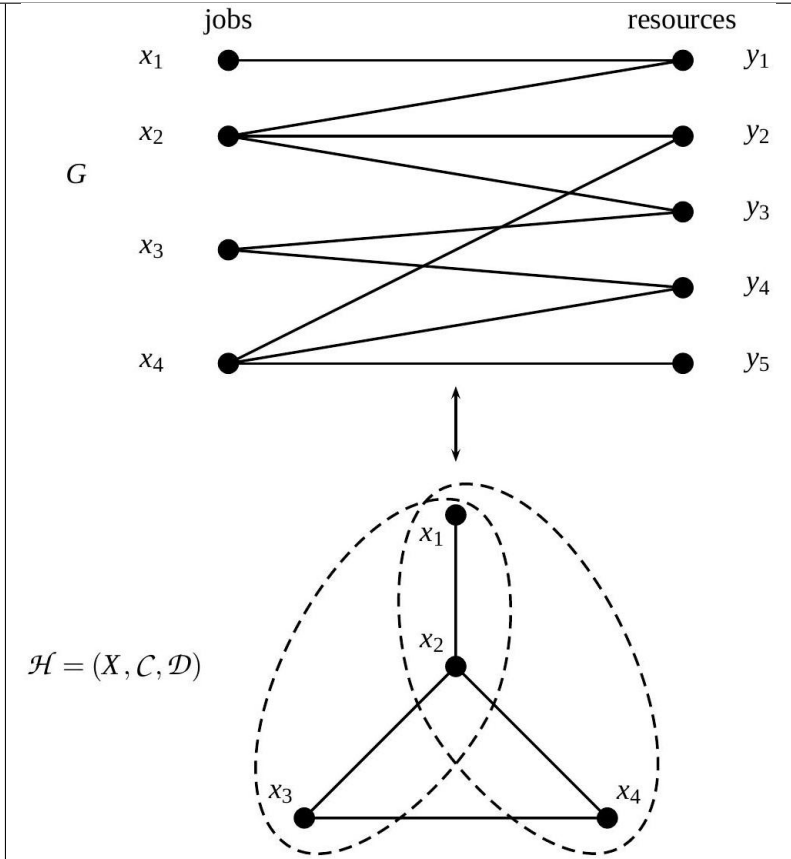
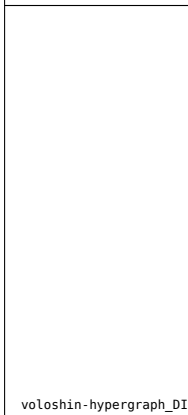
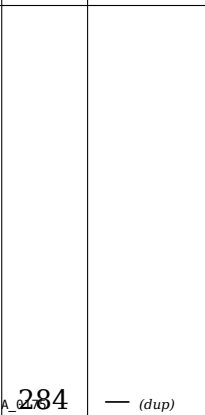
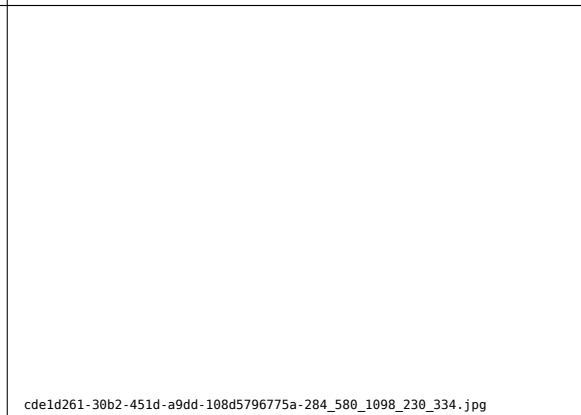
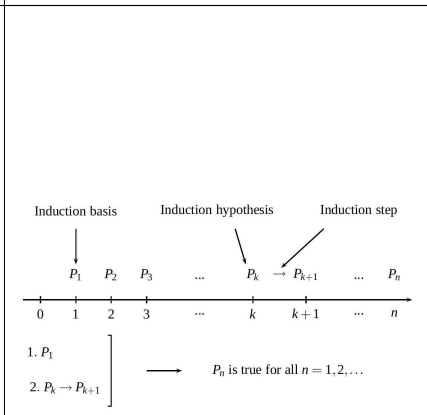
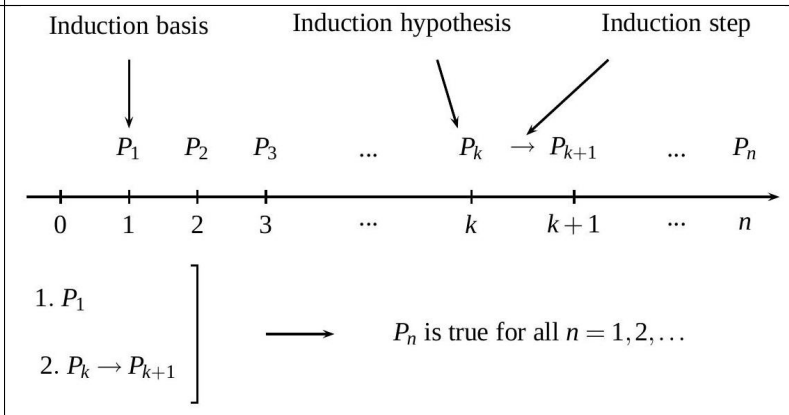
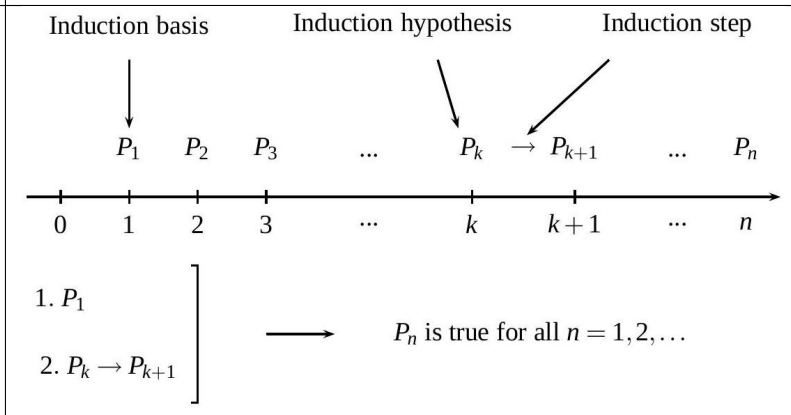
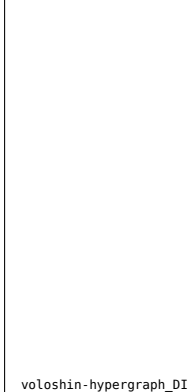
voloshin-hypergraph_DIA_0260	253	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-253_549_645_241_567.jpg			
voloshin-hypergraph_DIA_0265	257	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-257_640_1098_233_321.jpg			
voloshin-hypergraph_DIA_0266	258	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-258_557_1175_267_295.jpg			
voloshin-hypergraph_DIA_0269	259	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-259_484_1139_986_295.jpg			

voloshin-hypergraph_DIA_0261	261	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-261_834_1059_220_324.jpg			
voloshin-hypergraph_DIA_0262	262	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-262_844_1041_230_316.jpg			
voloshin-hypergraph_DIA_0265	265	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-265_818_1111_210_334.jpg			

voloshin-hypergraph_DIA_267	267	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-267_702_1152_207_303.jpg		
voloshin-hypergraph_DIA_269	269	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-269_541_1020_1059_365.jpg		
voloshin-hypergraph_DIA_271	271	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-271_544_702_199_531.jpg		

voloshin-hypergraph_DIA_0275	275 (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-275_1631_1201_233_300.jpg	 Diagram illustrating the decomposition of a K_4 hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4 into three 2-factors. The bottom part shows the colorings for each factor: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.	 Diagram illustrating the decomposition of a K_4^* hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4^* into three 2-factors. The bottom part shows the colorings for each factor: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.	 Diagram illustrating the decomposition of a K_4^* hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4^* into three 2-factors. The bottom part shows the colorings for each factor: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.
voloshin-hypergraph_DIA_0276	276 (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-276_917_1219_999_272.jpg	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.

voloshin-hypergraph_DIA_277	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-277_430_707_1491_528.jpg			
voloshin-hypergraph_DIA_280	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-280_1142_914_228_427.jpg			

	281	— (dup)					
	284	— (dup)					
	284	— (dup)	